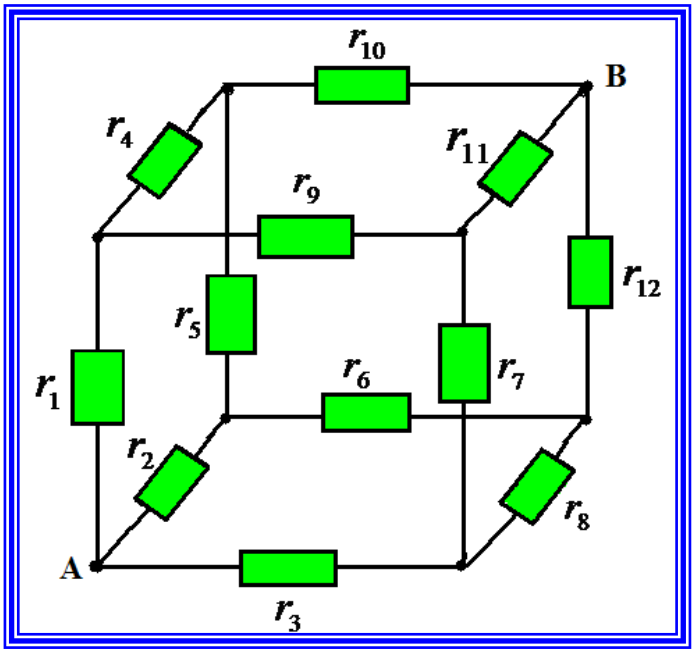
ELEKTR ZANJIR ELEMENTLARIGA DOIR MASALALAR YECHISH



O’ZBEKISTON RESPUBLIKASI OLIY VA O’RTA MAXSUS, KASB-HUNAR TA’LIMI VAZIRLIGI
O’RTA MAXSUS, KASB-HUNAR TA’LIMI MARKAZI
BUXORO VILOYATI HOKIMLIGI O’RTA MAXSUS, KASB-HUNAR TA’LIMI BOSHQARMASI
BUXORO DAVLAT TIBBIYOT INSTITUTI QOSHIDAGI 1-SON AKADEMIK LITSEY
V.Sh.Fayziyev, N.M.Narzullayev

ELEKTR ZANJIR ELEMENTLARIGA DOIR MASALALAR YECHISH

USLUBIY QO’LLANMA

I-QISM

Buxoro- 2014

Tuzuvchilar: Buxoro Davlat Tibbiyot Instituti qoshidagi
1-son akademik litsey fizika fani o’qituvchilari
N.M.Narzullayev, V.Sh.Fayziyev

Taqrizchilar: BuxDU, filologiya fanlari doktori
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BuxDU «Fizika kafedrasi»
dotsenti M.R .Jumayev

Buxoro Davlat Tibbiyot Instituti qoshidagi
1-son akademik litsey fizika fani o’qituvchisi
Ro’ziyev O’

Buxoro Davlat Tibbiyot Instituti qoshidagi 1-son akademik litseyi pedagogik kengashining № 6 - son qarori bilan nashrga tavsiya qilingan.

Kirish

Ushbu uslubiy qo’llanma II qismdan iborat bo’lib, akademik – litsey va kasb-hunar kollejlari hamda umum ta’lim maktab o’quvchi va o’qituvchilari uchun mo’ljallangan.

Mazkur uslubiy qo’llanmada DTM “Axborotnoma” sining 1999-2003 yillardagi sonlarida e’lon qilingan test topshiriqlari va turli xil masalalar yechish kitoblaridan ko’pgina sxemali masalalar tanlab olingan.

Uslubiy qo’llanmada masalalarning to’liq yechimlari keltirilgan bo’lib, masalalarni yechishda har xil usullardan foydalanilgan, ya’ni masalalarni yechishda ularni sxemalarini murakkab ko’rinishdan oddiy ko’rinishga aylantirib, masalalar yechilgan.

Uslubiy qo’llanma yordamida o’quvchilar o’tkazgichlarni ketma-ket va paralel ulash usullarini hamda turli xil sxemalarni ulashni o’rganishlari mumkin.

Ushbu uslubiy qo’llanmaning II-qismida nazariy usul bilan bir qatorda amaliy(laboratoriya)usulida yechib ko’rilgan masalalar ham joy olgan.

Ushbu uslubiy qo’llanmadan o’rta talim maktablari o’qituvchilari, ‘Bilimlar bellashuvi’, ‘Olimpiada’ hamda akademik litsey va kollejlar uchun olimpiadalarga tayyorlanayotgan o’quvchilar hamda oliy o’quv yurtlariga kiruvchilar foydalanishlari mumkin.

KONDENSATORLARNI ARALASH ULASHGA DOIR MASALALAR

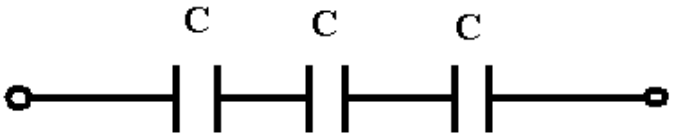
KONDENSATORLARNI KETMA-KET ULASH FORMULASI:

$$\frac{1}{C_{Um}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots \frac{1}{C_n}$$

KONDENSATORLARNI PARALLEL ULASH FORMULASI:

$$C_{Um} = C_1 + C_2 + C_3 + \ldots C_n$$

1.Sxemaning umumiy sig'imini hisoblang.

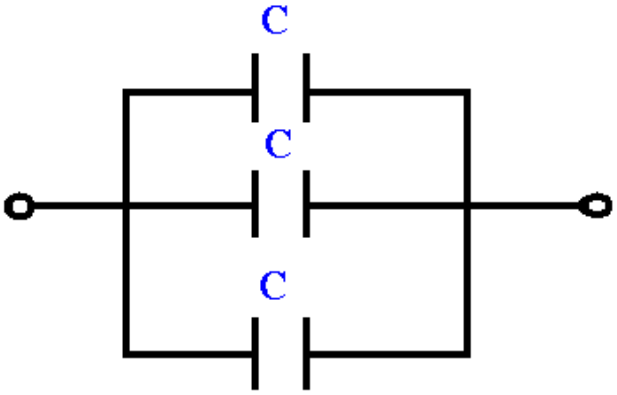


Har bir sig'imlar ketma-ket ulanganligi uchun.Quyidagicha hisoblanadi Kondensatorlarni ketma-ket ulashda quyidagi formuladan foydalanamiz.

$$\frac{1}{C_{Um}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots \frac{1}{C_n}$$

$$\frac{1}{C_{Um}} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C} = \frac{3}{C}; \qquad C_{Um} = \frac{C}{3}$$

2.Sxemaning umumiy sig'imini hisoblang

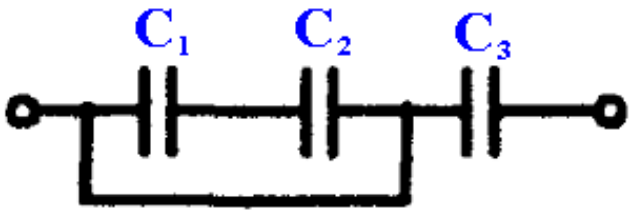


Har bir sig'imlar paralel ulanganligi uchun.Quyidagicha hisoblanadi kondensatorlarni parallel ulashda quyidagi formuladan foydalanamiz.

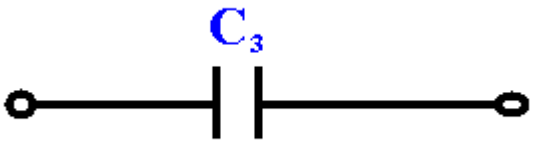
$$C_{Um} = C_1 + C_2 + C_3 + \ldots C_n$$

$$C_{Um} = C + C + C = 3C; \qquad C_{Um} = 3C$$

3.Sxemaning umumiy sig'imini hisoblang.($C_1 = C_2 = C_3 = C$)

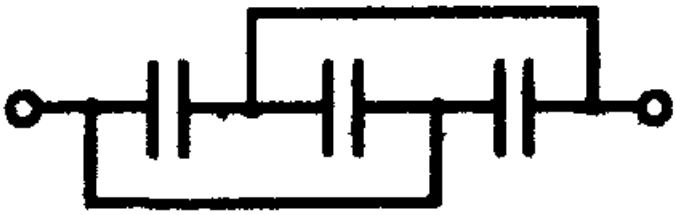


Sxemaning C_1 va C_2 sig'imlar orqali tok o'tmaydi,shuning uchun bu sig'imlarni sxemadan olib tashlaymiz va quyidagi ko'rinishga keladi.

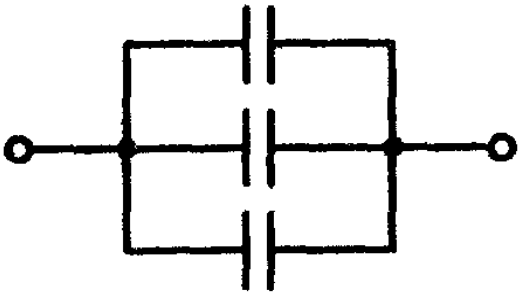


$$C_{Um} = C_3 = C$$

4. Sxemaning umumiy sig'imini hisoblang.($C_1 = C_2 = C_3 = C$)

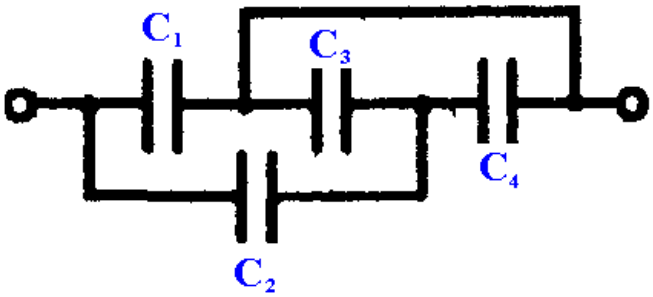


Sxemani quyidagi sodda ko'rinishda ifodalaymiz

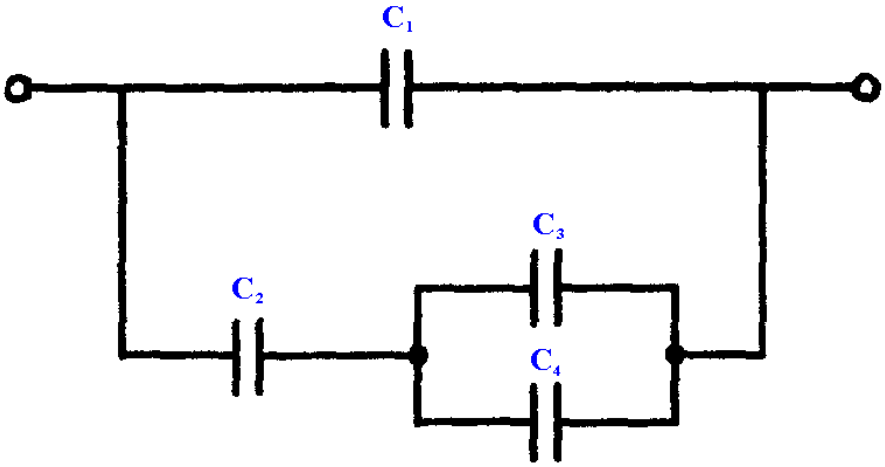


$$C_{Um} = C + C + C = 3C; \qquad C_{Um} = 3C$$

5.Sxemaning umumiy sig'imini hisoblang.($C_1 = C_2 = C_3 = C_4 = C$)



Sxemani quyidagi sodda ko’rinishda ifodalaymiz

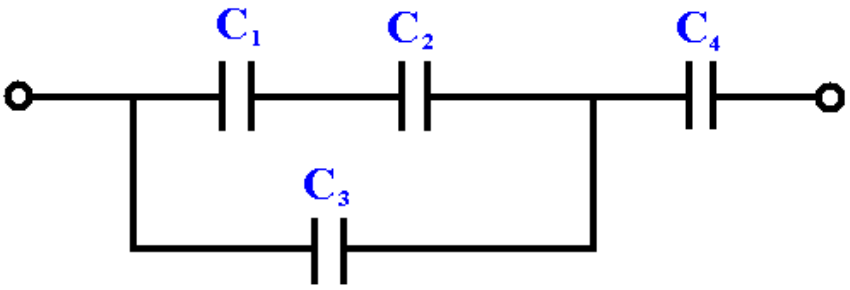


$$C_{34} = C_3 + C_4 = C + C = 2C$$

$$\frac{1}{C_{234}} = \frac{1}{C_2} + \frac{1}{C_{34}} = \frac{1}{C} + \frac{1}{2C} = \frac{3}{2C}; \quad C_{234} = \frac{2}{3}C$$

$$C_{Um} = C_1 + C_{234} = C + \frac{2}{3}C = \frac{5}{3}C; \qquad C_{Um} = \frac{5}{3}C$$

6.Sxemaning umumiy sig’imini hisoblang.($C_1 = C_2 = C_3 = C_4 = C$)



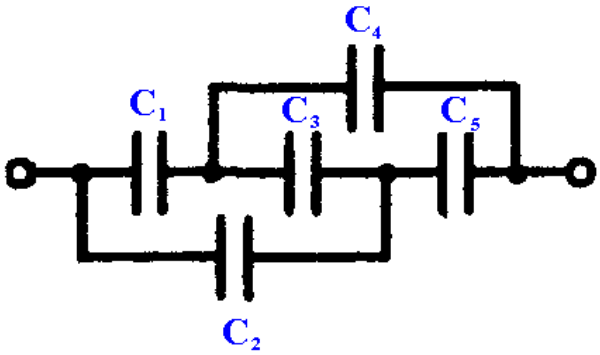
$$\frac{1}{C_{12}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{C} + \frac{1}{C} = \frac{2}{C}; \quad C_{12} = \frac{C}{2}.$$

$$C_{123} = C_{12} + C_3 = \frac{C}{2} + C = \frac{3}{2}C$$

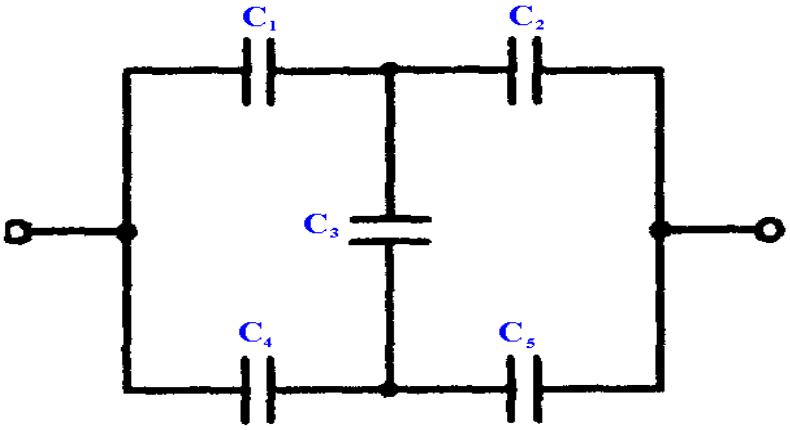
$$\frac{1}{C_{Um}} = \frac{1}{C_4} + \frac{1}{C_{123}} = \frac{1}{C} + \frac{2}{3C} = \frac{5}{3C}$$

$$C_{Um} = \frac{3}{5}C$$

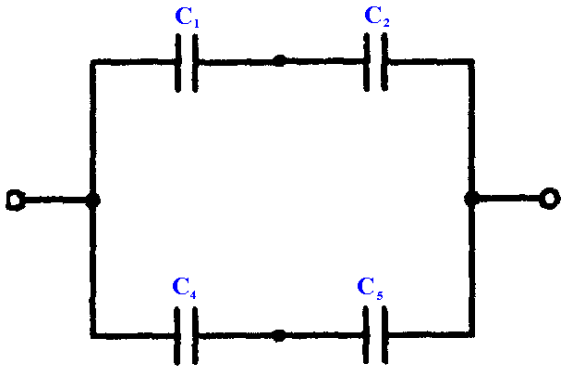
7.Sxemaning umumiy sig'imini hisoblang.($C_1 = C_2 = C_3 = C_4 = C_5 = C$)



Sxemani quyidagi sodda ko'rinishda ifodalaymiz



Shunga o'xshash sxemalar uchun quyidagi formula doimo o'rinlidir.Agar $\frac{C_1}{C_4} = \frac{C_2}{C_5}$ Nisbat teng bo'lgan hollarda C_3 sig'imdan tok o'tmaydi shuning uchun bu sig'imni sxemadan olib tashlaymiz.

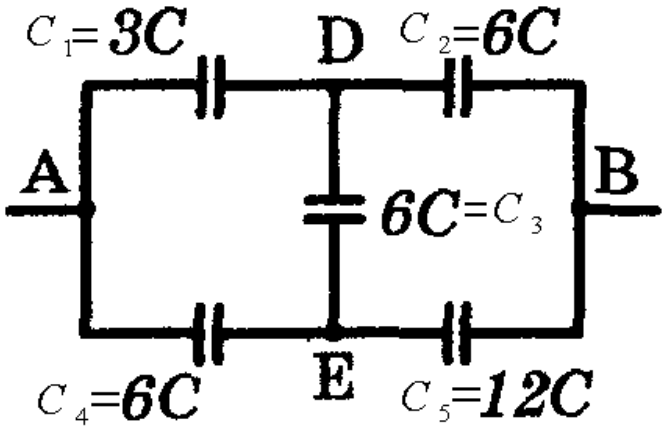


$$\frac{1}{C_{12}}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{1}{C}+\frac{1}{C}=\frac{2}{C};\quad C_{12}=\frac{C}{2}:$$

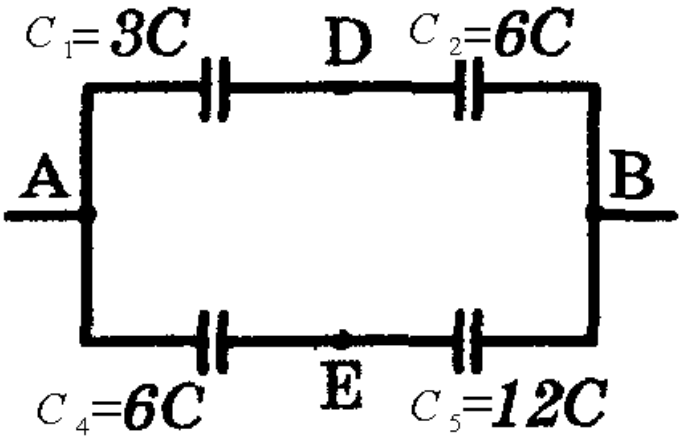
$$\frac{1}{C_{45}}=\frac{1}{C_4}+\frac{1}{C_5}=\frac{1}{C}+\frac{1}{C}=\frac{2}{C};\quad C_{45}=\frac{C}{2}:$$

$$C_{Um}=C_{12}+C_{45}=\frac{C}{2}+\frac{C}{2}=C;\quad C_{Um}=C$$

8.Sxemaning umumiy sig'imini hisoblang.



Oldingi masalada aytganimizdek. $\frac{C_1}{C_4}=\frac{C_2}{C_5}$ Nisbat teng bo'lgan hollarda C_3 sig'imdan tok o'tmaydi shuning uchun bu sig'imni sxemadan olib tashlaymiz.

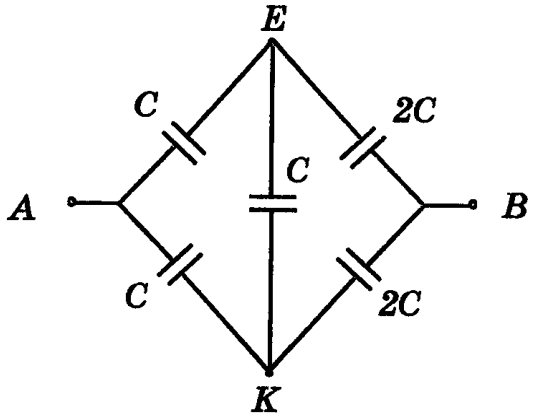


$$\frac{1}{C_{12}}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{1}{3C}+\frac{1}{6C}=\frac{3}{6C};\quad C_{12}=2C:$$

$$\frac{1}{C_{45}}=\frac{1}{C_4}+\frac{1}{C_5}=\frac{1}{6C}+\frac{1}{12C}=\frac{3}{12C};\quad C_{45}=4C:$$

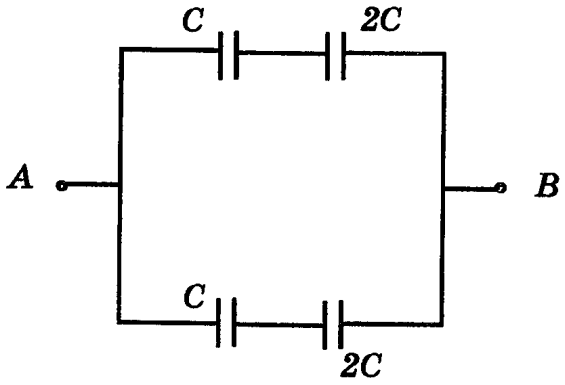
$$C_{Um}=C_{12}+C_{45}=2C+4C=6C;\quad C_{Um}=6C$$

9.Sxemaning umumiy sig’imini toping.



K va E nuqtalarda potensiallari teng potensiallar farqi 0 ga tengligi uchun KE yo’nalishda tok oqmaydi bu kondensator zaryadlanmaydi shuning uchun bu kondensatorni sxemadan olib tashlaymiz

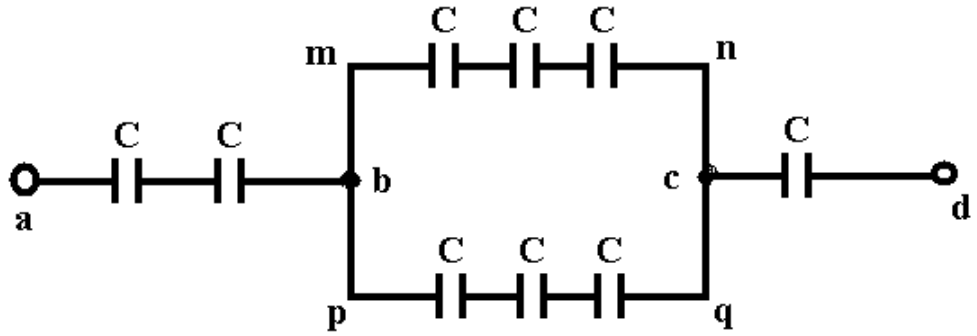
($\frac{C}{C}=\frac{2C}{2C}$ teng bo’lgani uchun ham yuqoridagi shart o’rinlidir)



$$\frac{1}{C_{12}}=\frac{1}{C}+\frac{1}{2C}=\frac{3}{2C};\quad C_{12}=\frac{2C}{3}:\qquad \frac{1}{C_{45}}=\frac{1}{C}+\frac{1}{2C}=\frac{3}{2C};\quad C_{45}=\frac{2C}{3}:$$

$$C_{Um}=C_{12}+C_{45}=\frac{2C}{3}+\frac{2C}{3}=\frac{4C}{3};\qquad C_{Um}=\frac{4C}{3}$$

10. ad nuqta orasidagi umumiy sig’imni toping.

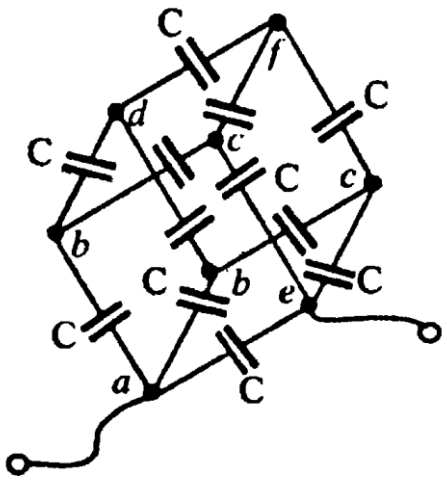


$$\frac{1}{C_{mn}}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C};\quad C_{mn}=\frac{C}{3}:\quad \frac{1}{C_{pq}}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C};\quad C_{pq}=\frac{C}{3}:$$

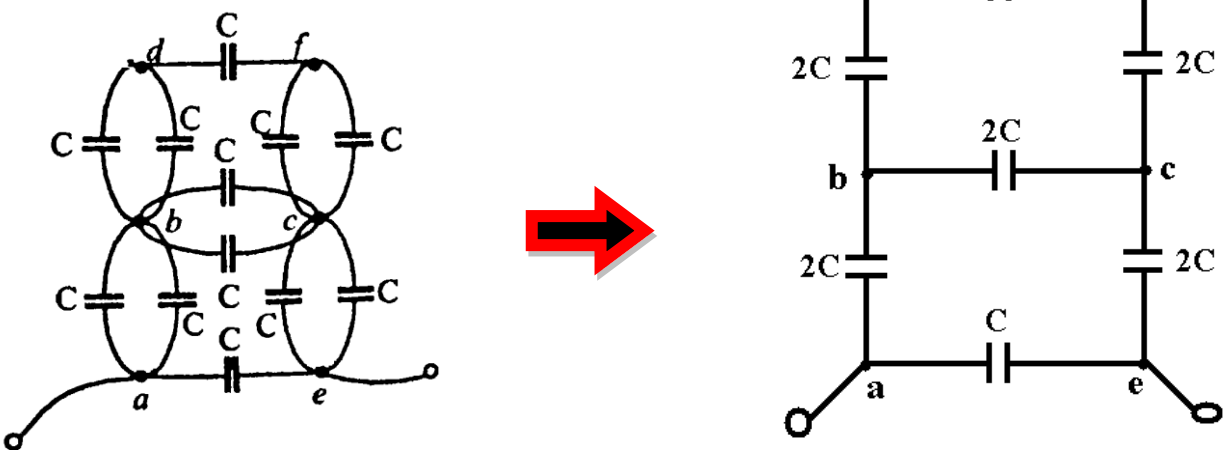
$$C_{bc}=C_{mn}+C_{pq}=\frac{C}{3}+\frac{C}{3}=\frac{2C}{3};\quad \frac{1}{C_{ab}}=\frac{1}{C}+\frac{1}{C}=\frac{2}{C};\quad C_{ab}=\frac{C}{2}:$$

$$\frac{1}{C_{ad}}=\frac{1}{C_{ab}}+\frac{1}{C_{bc}}+\frac{1}{C_{cd}}=\frac{2}{C}+\frac{3}{2C}+\frac{1}{C}=\frac{9}{2C};\quad C_{ad}=\frac{2C}{9}:$$

11. Sxemaning umumiy sig'imini toping.



Sxemani quyidagicha soddalashtiramiz, ya'ni b va c nuqtalarni tutashtiramiz.



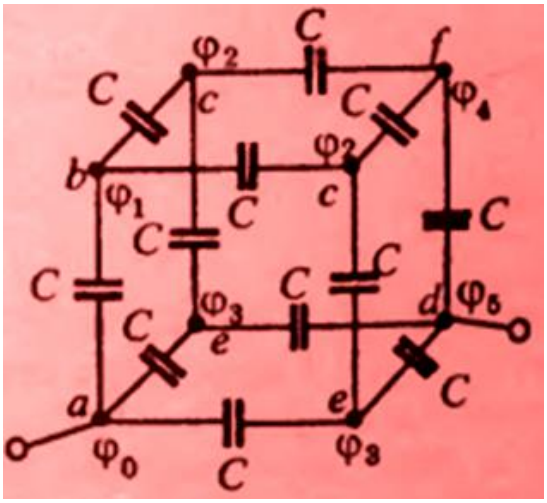
$$\frac{1}{C_{bdfc}}=\frac{1}{2C}+\frac{1}{C}+\frac{1}{2C}=\frac{4}{2C};\quad C_{bdfc}=\frac{C}{2}:$$

$$C_{bc}=C_{bdfc}+2C=\frac{C}{2}+2C=\frac{5C}{2}$$

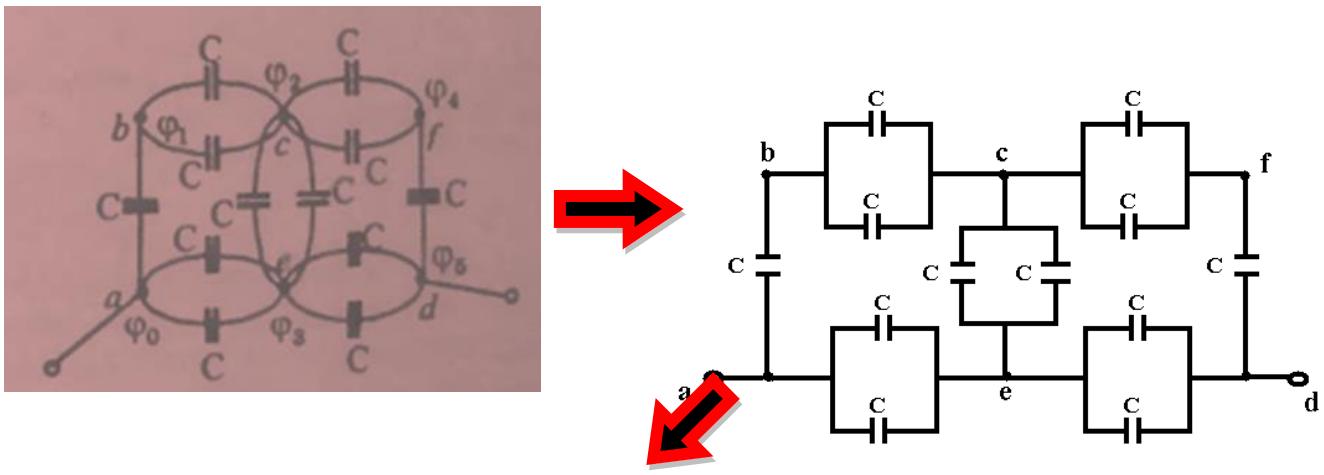
$$\frac{1}{C_{abce}}=\frac{1}{C_{ab}}+\frac{1}{C_{bc}}+\frac{1}{C_{ce}}=\frac{1}{2C}+\frac{2}{5C}+\frac{1}{2C};\quad C_{abce}=\frac{10C}{14}=\frac{5C}{7}:$$

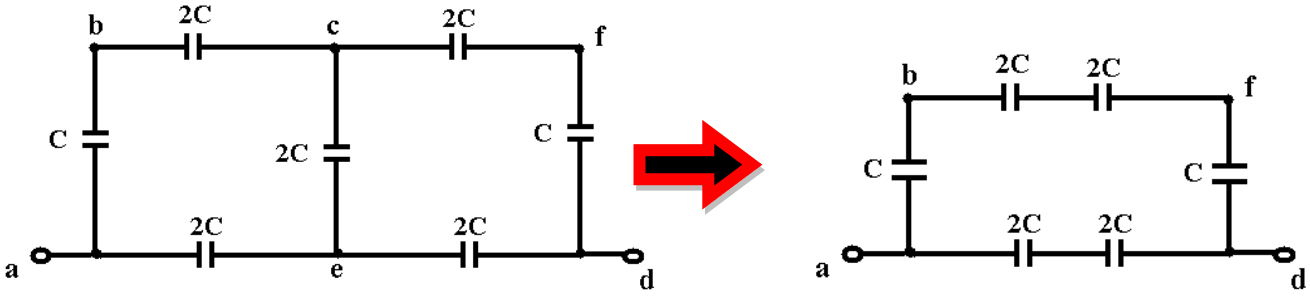
$$C_{ae}=C_{abce}+C=\frac{5C}{7}+C=\frac{12C}{7}$$

12. Sxemaning umumiy sig'imini toping.



Sxemani quyidagicha soddalashtiramiz, ya'ni e va c nuqtalarni tutashtiramiz.



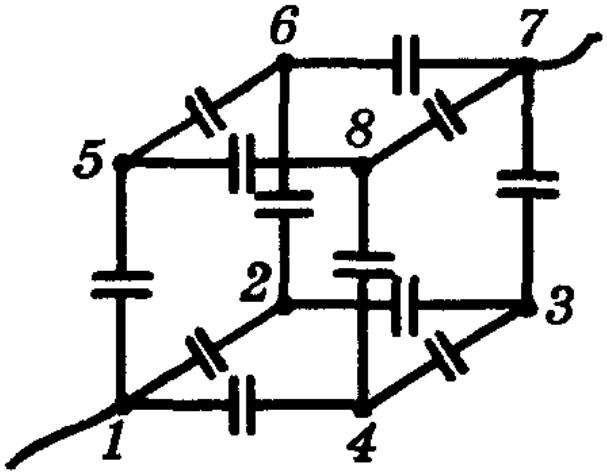


$\frac{2C}{2C}=\frac{2C}{2C}$ nisbat teng bo'lganligi uchun c va e nuqtalar potensiallari bir xil , potensiallar farqi 0 ga tengligi uchun ce nuqtalar orasidagi sig'im orqali tok o'tmaydi ya'ni bu sig'im zaryadlanmaydi shuning uchun uni sxemadan olib tashladik.

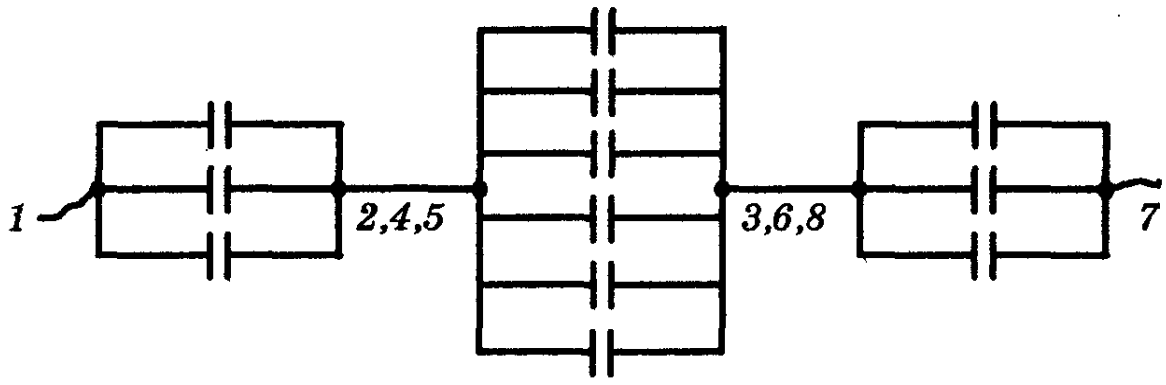
$$\frac{1}{C_{bf}}=\frac{1}{2C}+\frac{1}{2C}=\frac{2}{2C};\quad C_{bf}=C;\quad \frac{1}{C_{abfd}}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C};\quad C_{abfd}=\frac{C}{3}$$

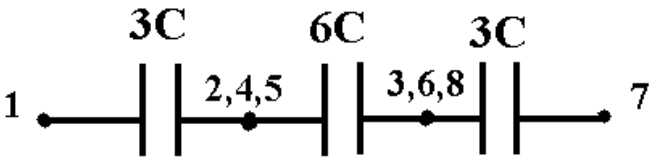
$$\frac{1}{C_{ad}}=\frac{1}{2C}+\frac{1}{2C}=\frac{2}{2C};\quad C_{ad}=C;\quad C_{um}=C_{abfd}+C_{ad}=\frac{C}{3}+C=\frac{4}{3}C,\quad C_{um}=\frac{4}{3}C.$$

13.Sxemaning umumiy sig'imini hisoblang.



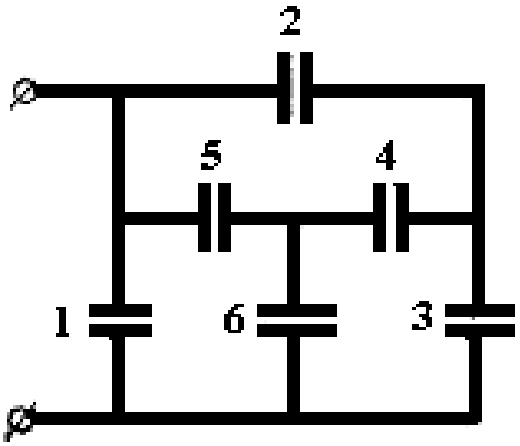
Sxemani sodda ko'rinishda ifodalaymiz ya'ni 2,4,5 va 3,6,8 nuqtalarni tutashtiramiz



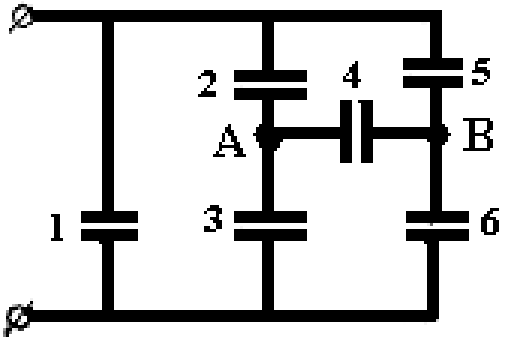


$$\frac{1}{C_{Um}}=\frac{1}{3C}+\frac{1}{6C}+\frac{1}{3C}=\frac{5}{6C};\qquad C_{Um}=\frac{6}{5}C=1{,}2C.$$

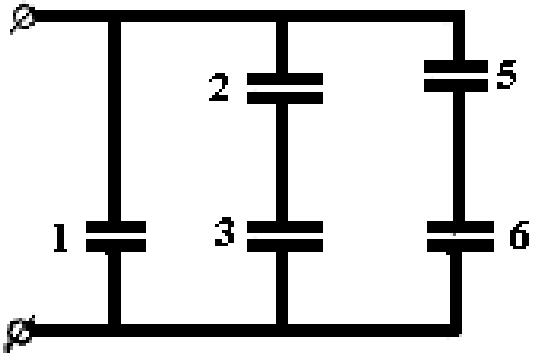
14. Rasmda tasvirlangan sxemaning umumiy sig'imini toping. Har bir kondensatorning sig'imi C ga teng.



Sxemani quyidagicha o'zgartiramiz



Hamma kondensatorlarning sig'imi birday bo'lgani tufayli A va B nuqtalar orasidagi potensiallar farqi nolga teng, C_4 kondensator hamma vaqt zaryadlanmagan bo'ladi va sxema ancha soddalashadi.

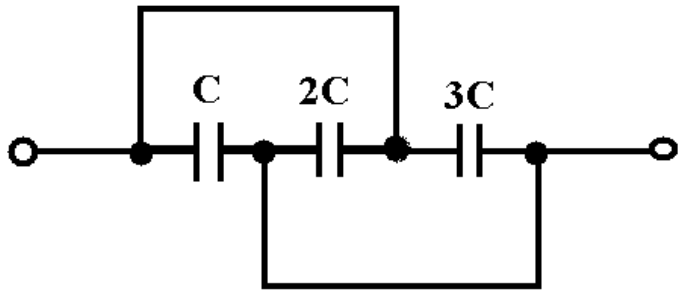


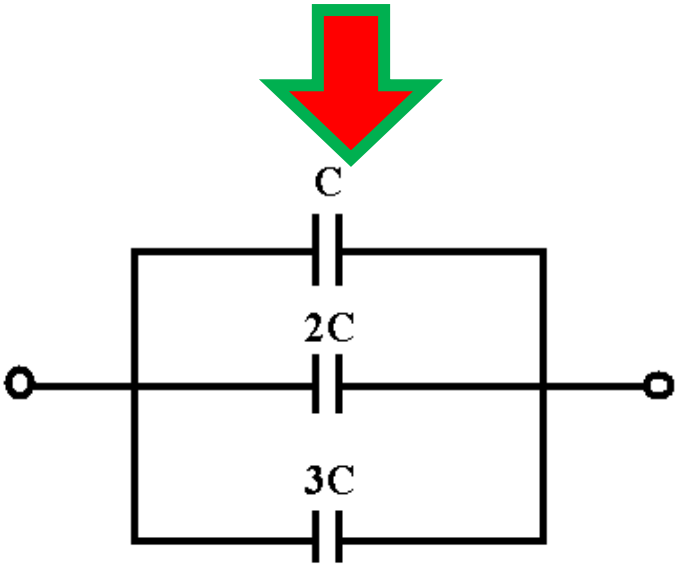
$$C_{23}=\frac{C_2\cdot C_3}{C_2+C_3}=\frac{C}{2};\qquad C_{56}=\frac{C_5\cdot C_6}{C_5+C_6}=\frac{C}{2};$$

$$C_{Umum}=C_1+C_{23}+C_{56}=C+\frac{C}{2}+\frac{C}{2}=2C$$

15.Sxemalarning umumiy sig'imini toping.

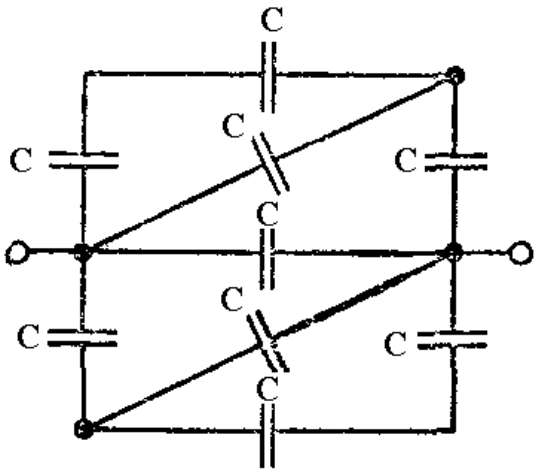
a) Rasm

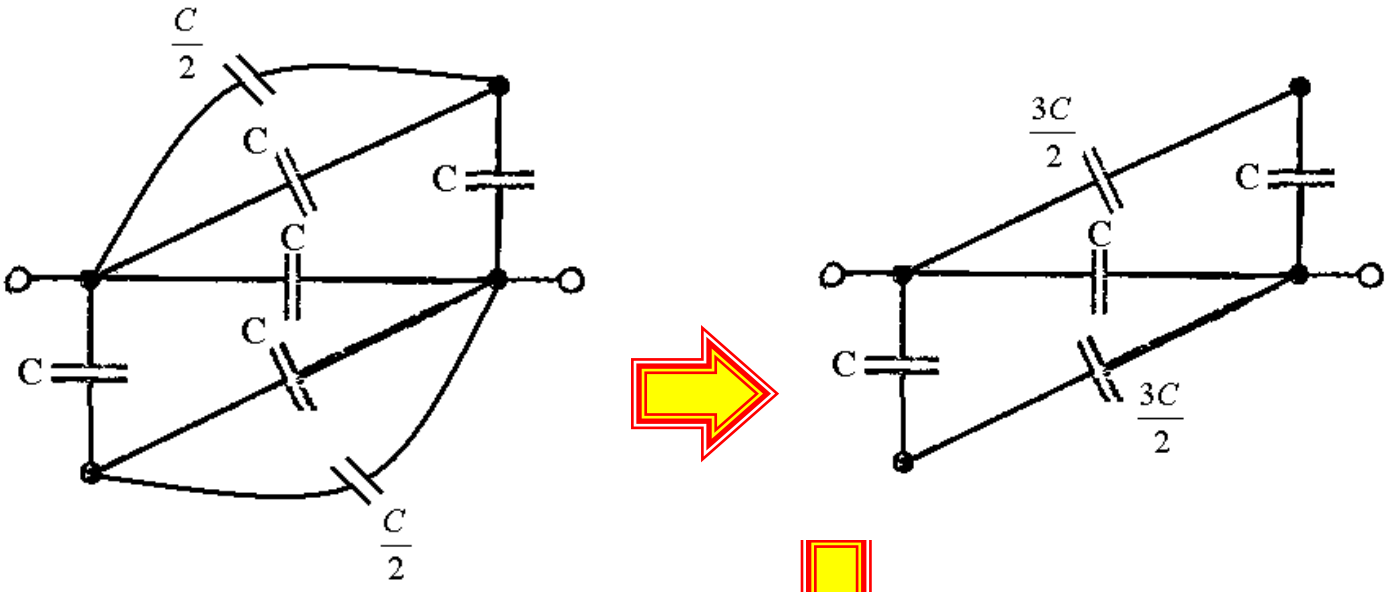


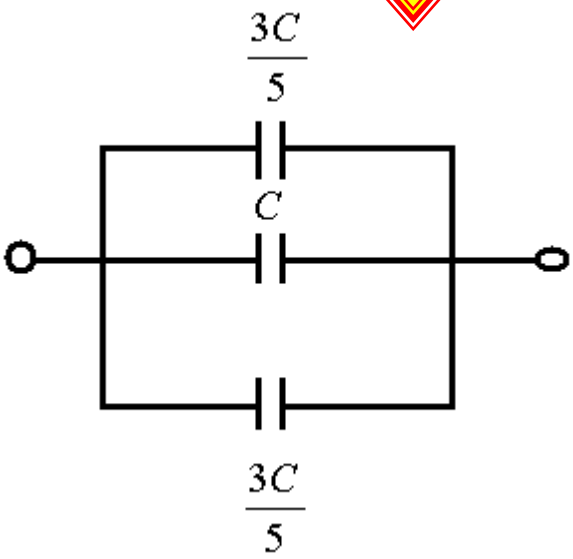


$$C_{Um}=C+2C+3C=6C$$

b) Rasm

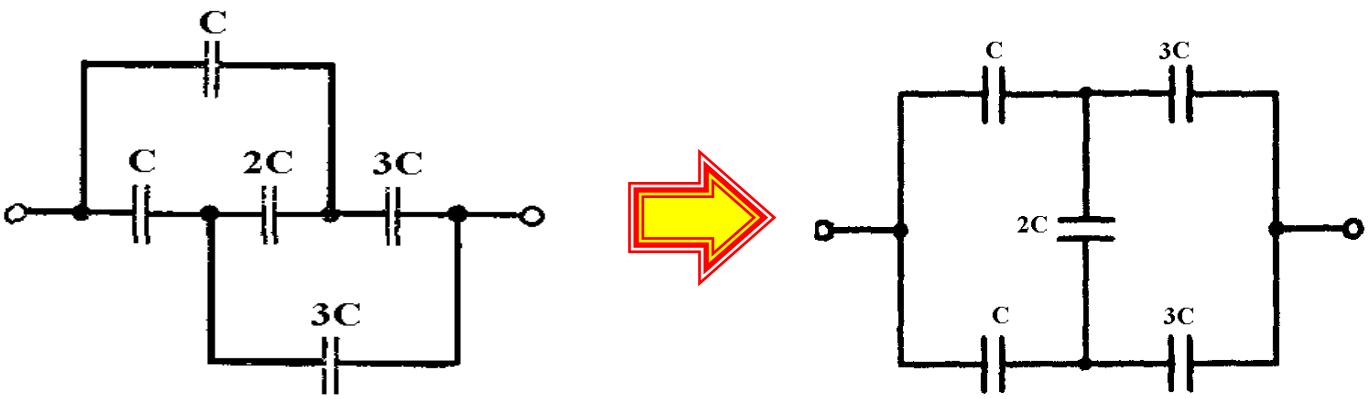




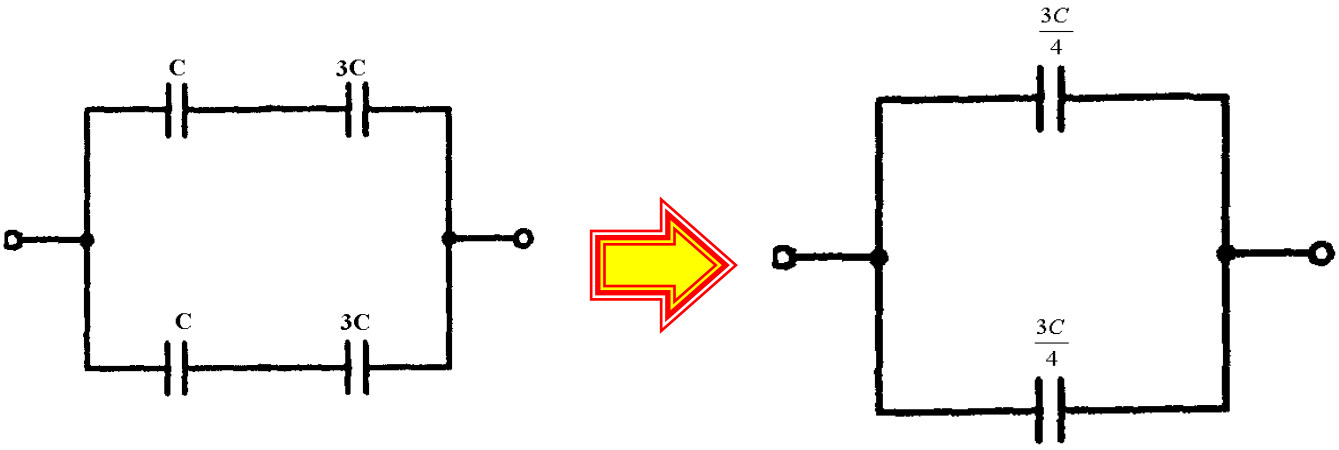


$$C_{Um} = \frac{3C}{5} + C + \frac{3C}{5} = \frac{11C}{5}$$

v) Rasm.

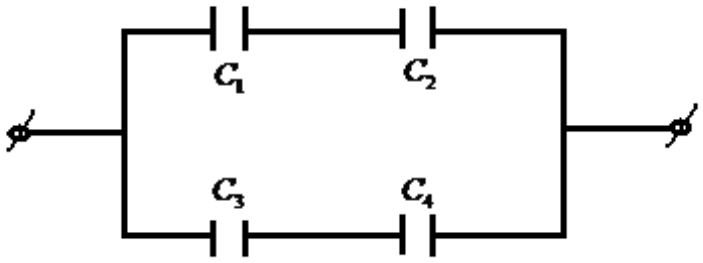


$$\frac{C}{C}=\frac{3C}{3C}$$ Nisbatlar tengligi uchun 2C kondensator zaryadlanmaydi shuning uchun uni sxemadan olib tashlaymiz.



$$C_{Um}=\frac{3C}{4}+\frac{3C}{4}=\frac{6C}{4}=1{,}5C$$

16.Sig'imlari $C_1\ 3mkF, C_2=5mkF, C_3=6mkF, va\, C_4=5mkF$ bo'lgan to'rtta kondensator rasmga ko'rsatilgan sxema bo'yicha ulangan. Hosil bo'lgan kondensatorlar batareyasining sag'imini anaqlang.

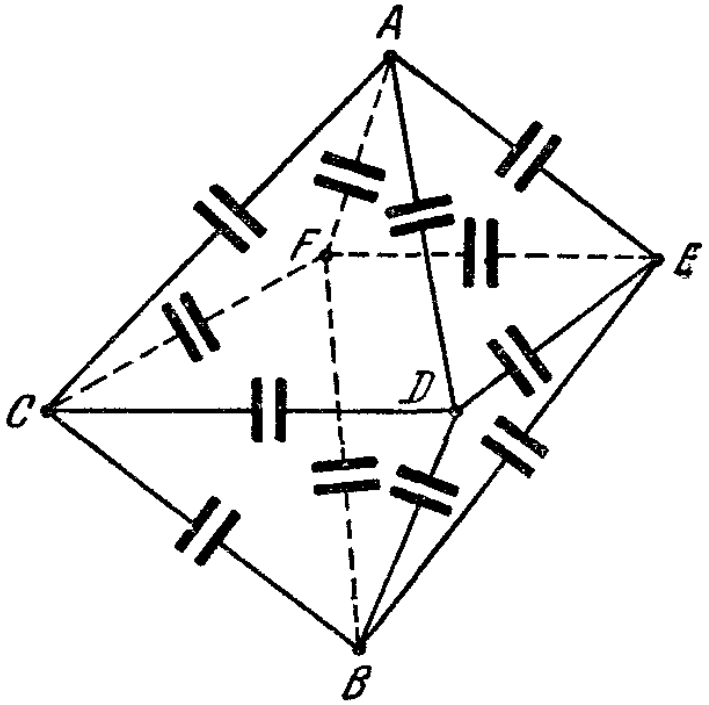


$$\frac{1}{C_{1.2}}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{1}{3}+\frac{1}{6}=\frac{3}{6};\qquad \frac{1}{C_{1.2}}=\frac{3}{6}=\frac{1}{2}\qquad C_{1.2}=2\,mkF$$

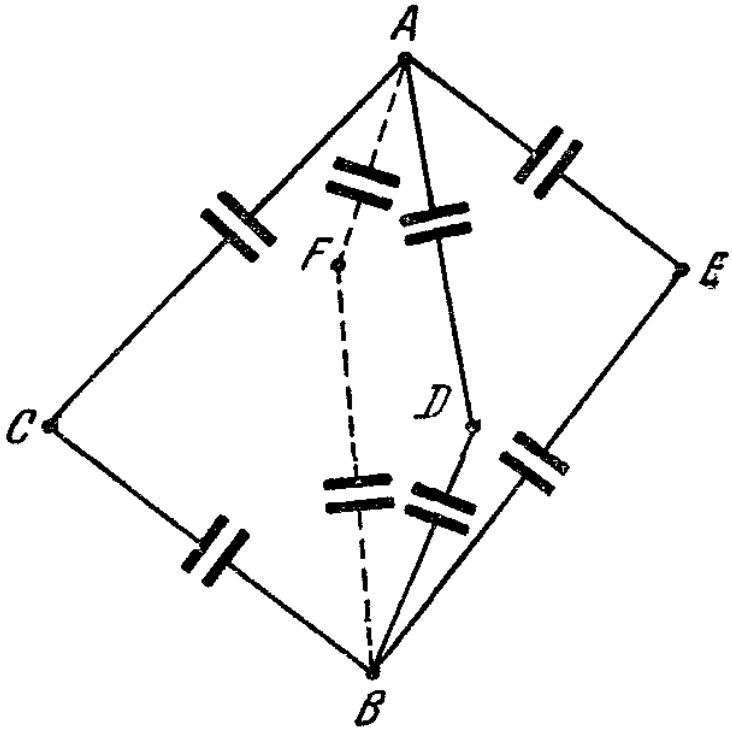
$$\frac{1}{C_{3.4}}=\frac{1}{C_3}+\frac{1}{C_4}=\frac{1}{5}+\frac{1}{10}=\frac{3}{10};\qquad \frac{1}{C_{3.4}}=\frac{3}{10}\qquad C_{3.4}=\frac{10}{3}\,mkF$$

$$C_{Um}=C_{1.2}+C_{3.4}=2+\frac{10}{3}=\frac{16}{3}\,mkF;\qquad Javob:\ C_{Um}=\frac{16}{3}\,mkF$$

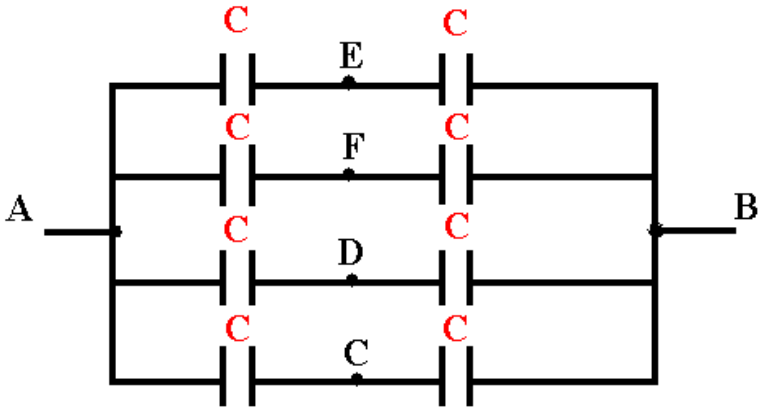
17. Har birining sig’imi C bo’lgan o’n ikkita bir xil kondensator ACD EFB sakkiz yoq ko’rinishida batareya qilib yig’ilgan.Bu kondensatorlar batareyasining A va B nuqtalari orasidagi sig’im qanday?

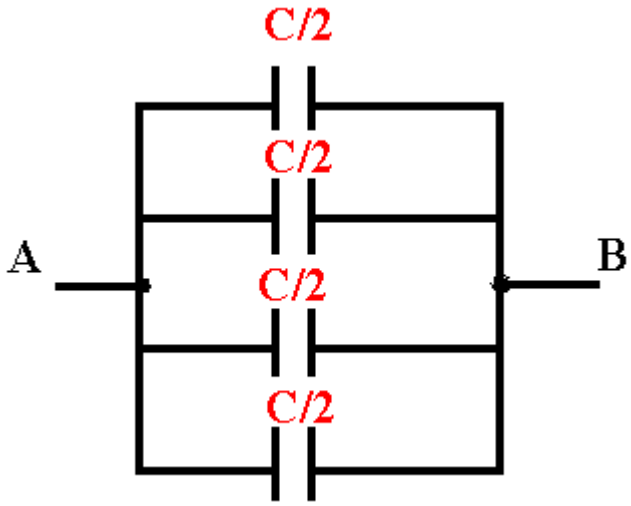


C, D, E va F nuqtalarning potensiallari tengligi batareyaning simmetrikligidan ma’lum. Binobarin, bu nuqtalar orasiga ulangan kondensatorlar zaryadlanmagan. Batareya sig’imini hisoblashda bu kondensatorlarni hisobga olmasa bo’ladi. Unda batareya quyidagi ko’rinishga keladi.



Bu sxemani quyidagi sodda ko’rinishda ifodalaymiz

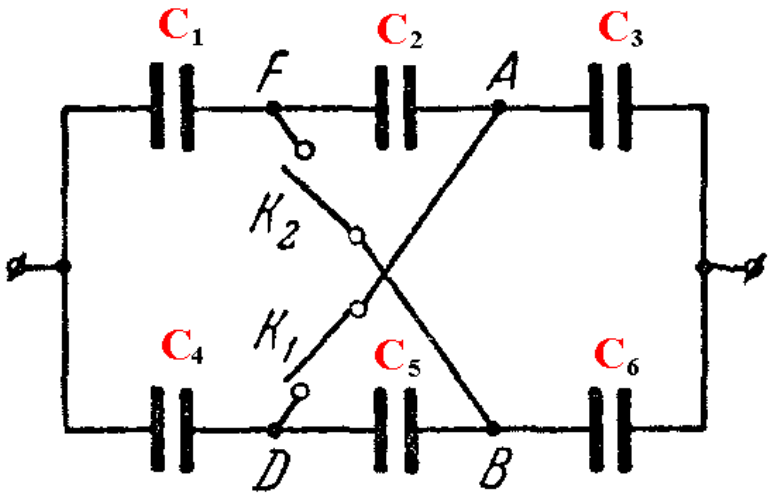




$$C_{AB} = \frac{C}{2} + \frac{C}{2} + \frac{C}{2} + \frac{C}{2} = 2C; \qquad C_{AB} = 2C$$

18. Quyidagi sxemada tasvirlangan kondensatorlar batareyasi bir xil sig'imli kondensatorlardan tuzilgan. Agar K_1 va K_2 kalitlar ulansa, batareyaning umumiy sig'imi necha marta o'zgaradi?

$$C_1 = C_2 = C_3 = C_4 = C_5 = C_6 = C$$



Kalit ulangunga qadar batareyaning umumiy sig'imi

$$\frac{1}{C_{123}}=\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C};\ C_{123}=\frac{C}{3}:$$

$$\frac{1}{C_{456}}=\frac{1}{C_4}+\frac{1}{C_5}+\frac{1}{C_6}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C};\ C_{456}=\frac{C}{3}:$$

$$C_{Um}=\frac{C}{3}+\frac{C}{3}=\frac{2C}{3}:$$

Kalitlar ulangunga qadar batareyaning sig'imi $C_{Um}=\frac{2C}{3}:$ ni tashkil qiladi.

Kalitlar ulangandan keyin hosil bo'lgan sxemani ko'ramiz. Simmetriya mulohazalaridan A va B nuqtalarning potensiallari teng ekanligi tushunarli; F va D nuqtalar haqida ham shuni aytish mumkin. Ikkinchidan, A va D shuningdek, B va F nuqtalarning potensiallari teng, chunki ular o'tkazgichlar bilan ulangan. Binobarin,o'rtadagi kondensatorlar zaryadlanmaydi. Shuning uchun C_2 va C_5 kondensatorlarni hisobga olmaymiz.

$$\frac{1}{C_{13}}=\frac{1}{C_1}+\frac{1}{C_3}=\frac{1}{C}+\frac{1}{C}=\frac{2}{C};\ C_{13}=\frac{C}{2}:$$

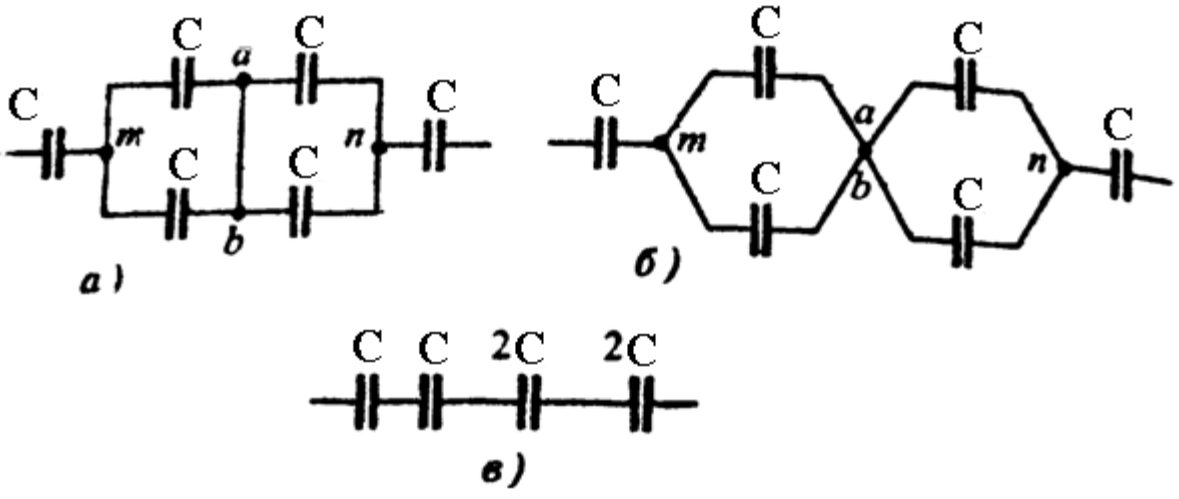
$$\frac{1}{C_{46}}=\frac{1}{C_4}+\frac{1}{C_6}=\frac{1}{C}+\frac{1}{C}=\frac{2}{C};\ C_{46}=\frac{C}{2}:$$

$$C_{Um}=\frac{C}{2}+\frac{C}{2}=C:$$

Javob: 1,5marta ortadi

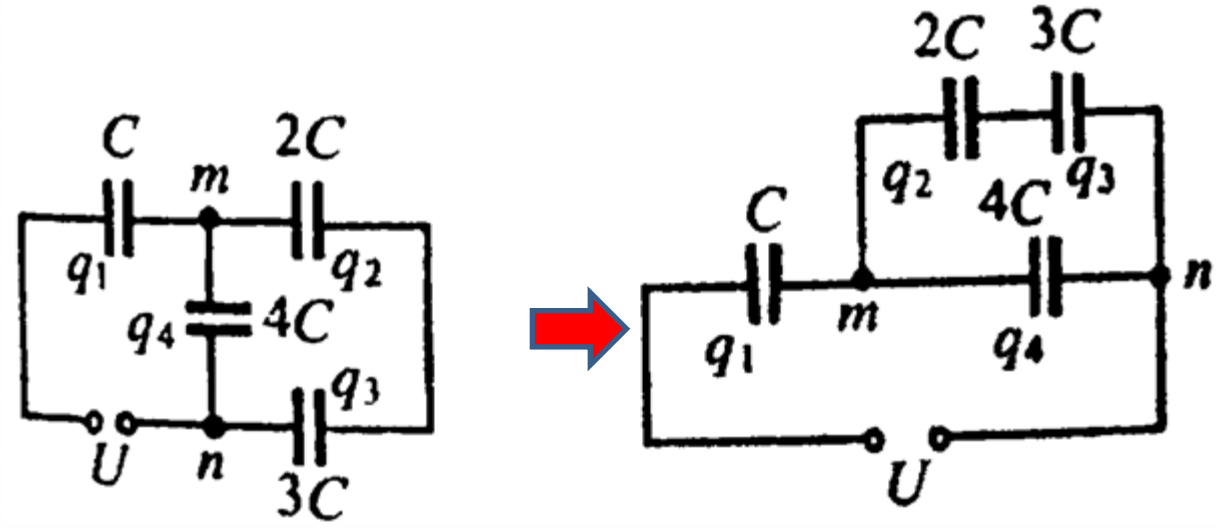
19.Sxemaning umumiy sig'imini toping.

Masalani yechish uchun a rasmdagi sxemani b va v rasmdagi sxemalar kabi soddalashtiramiz.



$$\frac{1}{C_{Um}}=\frac{1}{C}+\frac{1}{C}+\frac{1}{2C}+\frac{1}{2C}=\frac{6}{2C}=\frac{3}{C};\quad C_{Um}=\frac{C}{3}$$

20. Berilgan sxemada C=0,2nF; manbaning kuchlanishi 200V bo'lsa har bir kondensatorda to'planadigan zaryadni aniqlang?



Sxemani quyidagicha soddalashtiramiz.2-sxemadan ko'rinib turibdiki 2C va 3C sig'imli kondensatorlar o'zaro ketma-ketligi uchun ulardagi zaryadlar bir-xil $q_2=q_3$ birinchi kondensator manbaga ketma-ket ulanganligi uchun

$$q_1=C_{Um}\cdot U=q_2+q_4$$

$$C_{23}=\frac{2C\cdot 3C}{2C+3C}=\frac{6C}{5};\qquad C_{234}=C_{23}+C_4=\frac{6C}{5}+4C=\frac{26C}{5};$$

$$\frac{1}{C_{Um}}=\frac{1}{C_{234}}+\frac{1}{C_1}=\frac{5}{26C}+\frac{1}{C}=\frac{31}{26C};\quad C_{Um}=\frac{26C}{31};$$

$$q_1=C_{Um}\cdot U=\frac{26}{31}\cdot 0{,}2\cdot 10^{-9}\cdot 200=\frac{1040}{31}\cdot 10^{-9}=34\cdot 10^{-9}\,kl=34\,nkl$$

Paralel ulanganda kuchlanishlar tengligidan foydalanamiz

$$U_{23}=U_4;\quad U_{23}=\frac{q_2}{C_{23}};\ (q_2=q_3)\quad U_4=\frac{q_3}{C_4}$$

$$\frac{q_2}{C_{23}}=\frac{q_4}{C_4};\qquad \frac{5q_2}{6C}=\frac{q_4}{4C};\qquad \frac{5}{6}q_2=\frac{q_4}{4};\qquad q_4=\frac{10}{3}q_2$$

$q_1=C_{Um}\cdot U=q_2+q_4$ ushbu formulaga asosan

$$q_1=C_{Um}\cdot U=q_2+q_4=q_2+\frac{10}{3}q_2=\frac{13}{3}q_2;\qquad q_2=\frac{3}{13}q_1$$

$$q_2=\frac{3}{13}\cdot 34\cdot 10^{-9}=7{,}8\cdot 10^{-9}\,kl;\quad q_3=q_2=7{,}8\cdot 10^{-9}\,kl=7{,}8\,nkl$$

$$q_4=\frac{10}{3}q_2=\frac{10}{3}\cdot 7{,}8\cdot 10^{-9}=26\cdot 10^{-9}\,kl=26\,nkl$$

REZISTORLARNI ARALASH ULASHGA DOIR MASALALAR

REZISTORLARNI KETMA-KET ULASH FORMULASI:

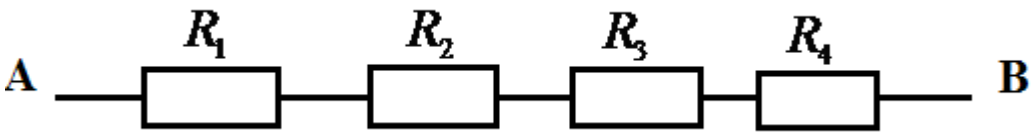
$$R_{Um} = R_1 + R_2 + R_3 + \ldots R_n$$

REZISTORLARNI PARALLEL ULASH FORMULASI:

$$\frac{1}{R_{Um}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_n}$$

1.AB nuqta orasidagi umumiy qarshilikni toping.

1-HOL



Agar qarshiliklar har-xil bo'lsa

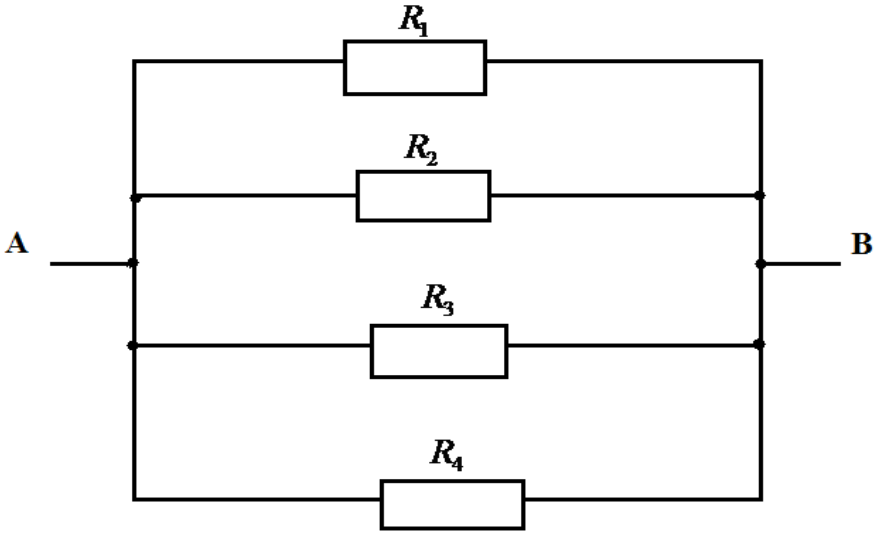
$$R_{AB} = R_1 + R_2 + R_3 + R_4$$

Agar qarshiliklar bir xil bo'lsa

$$R_1 = R_2 = R_3 = R_4 = R$$

$$R_{AB} = R_1 + R_2 + R_3 + R_4 = R + R + R + R = 4R$$

2. AB nuqta orasidagi umumiy qarshilikni toping.



Agar

qarshiliklar har-xil bo'lsa

$$\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

$$R_{AB} = \frac{R_1 \cdot R_2 \cdot R_3 \cdot R_4}{R_2 \cdot R_3 \cdot R_4 + R_1 \cdot R_3 \cdot R_4 + R_1 \cdot R_2 \cdot R_4 + R_1 \cdot R_2 \cdot R_3}$$

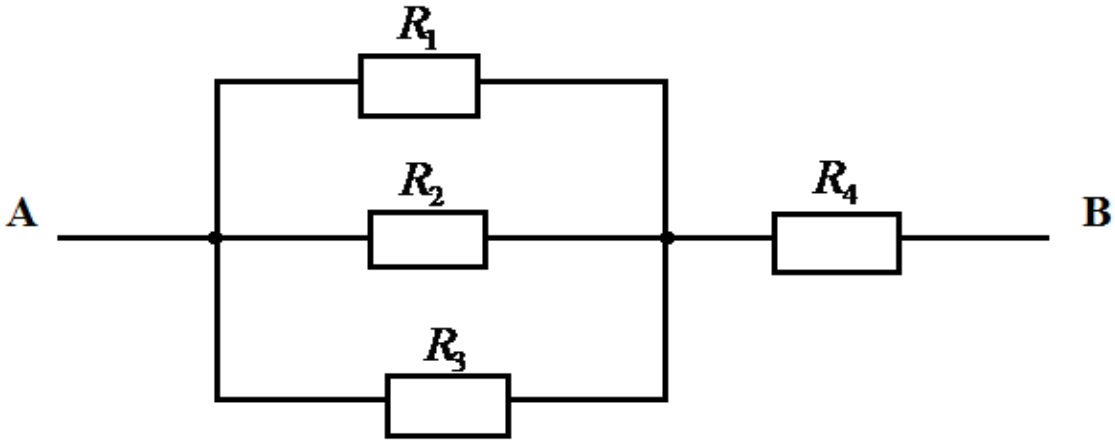
Agar qarshiliklar bir xil bo'lsa

$$R_1 = R_2 = R_3 = R_4 = R$$

$$\frac{1}{R_{AB}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{4}{R}$$

$$R_{AB} = \frac{R}{4}$$

3. AB nuqta orasidagi umumiy qarshilikni toping.



Agar qarshiliklar har-xil bo'lsa

$$\frac{1}{R_{123}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{R_2 \cdot R_3}{R_1 \cdot R_2 \cdot R_3} + \frac{R_1 \cdot R_3}{R_1 \cdot R_2 \cdot R_3} + \frac{R_1 \cdot R_2}{R_1 \cdot R_2 \cdot R_3} = \frac{R_2 \cdot R_3 + R_1 \cdot R_3 + R_1 \cdot R_2}{R_1 \cdot R_2 \cdot R_3}$$

$$R_{123} = \frac{R_1 \cdot R_2 \cdot R}{R_2 \cdot R_3 + R_1 \cdot R_3 + R_1 \cdot R_2}; \qquad R_{AB} = R_{123} + R_4$$

$$R_{AB} = \frac{R_1 \cdot R_2 \cdot R}{R_2 \cdot R_3 + R_1 \cdot R_3 + R_1 \cdot R_2} + R_4$$

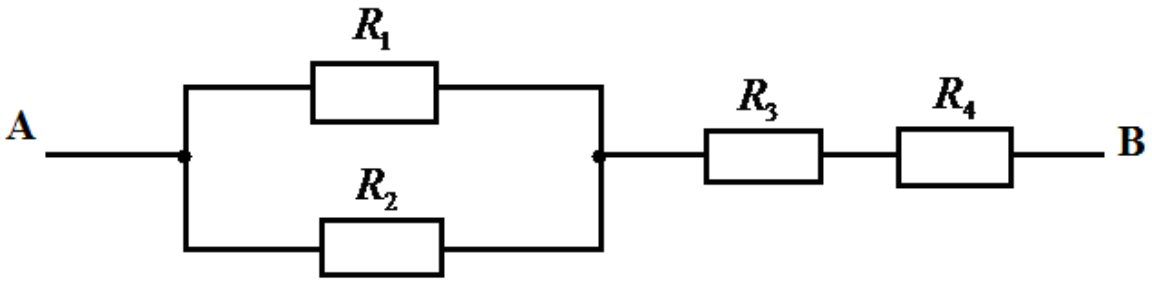
Agar qarshiliklar bir xil bo'lsa

$$R_1 = R_2 = R_3 = R_4 = R$$

$$\frac{1}{R_{123}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}; \qquad R_{123} = \frac{R}{3}$$

$$R_{AB} = R_{123} + R_4 = \frac{R}{3} + R = \frac{4}{3}R$$

4. AB nuqta orasidagi umumiy qarshilikni toping.



Agar qarshiliklar har-xil bo’lsa

$$\frac{1}{R_{12}}=\frac{1}{R_1}+\frac{1}{R_2}=\frac{R_1+R_2}{R_1\cdot R_2};\qquad R_{12}=\frac{R_1\cdot R_2}{R_1+R_2}$$

$$R_{34}=R_3+R_4;\qquad R_{AB}=R_{12}+R_{34}$$

$$R_{AB}=\frac{R_1\cdot R_2}{R_1+R_2}+R_3+R_4$$

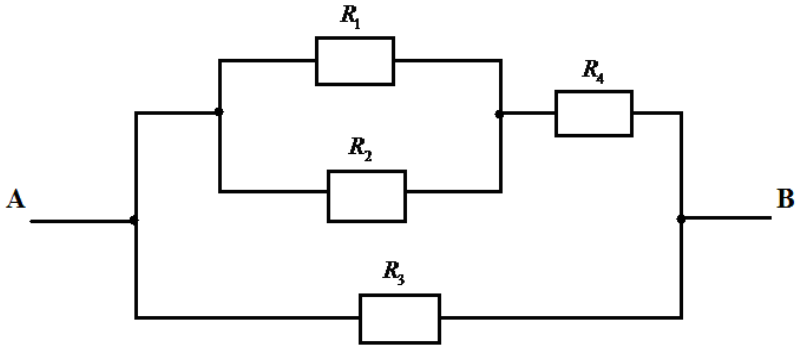
Agar qarshiliklar bir xil bo’lsa

$$R_1=R_2=R_3=R_4=R$$

$$\frac{1}{R_{12}}=\frac{1}{R_1}+\frac{1}{R_2}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R};\qquad R_{12}=\frac{R}{2}$$

$$R_{34}=R_3+R_4=R+R=2R;\qquad R_{AB}=R_{12}+R_{34}=\frac{R}{2}+2R=\frac{5}{2}R$$

5. AB nuqta orasidagi umumiy qarshilikni toping.



Agar qarshiliklar har-xil bo’lsa

$$\frac{1}{R_{12}}=\frac{1}{R_1}+\frac{1}{R_2}=\frac{R_1+R_2}{R_1\cdot R_2};\qquad R_{12}=\frac{R_1\cdot R_2}{R_1+R_2}$$

$$R_{124}=R_{12}+R_4;$$

$$\frac{1}{R_{AB}}=\frac{1}{R_{124}}+\frac{1}{R_3}=\frac{R_3+R_{124}}{R_{124}\cdot R_3};\qquad R_{AB}=\frac{R_{124}\cdot R_3}{R_3+R_{124}}$$

$$R_{AB}=\frac{\left(\frac{R_1\cdot R_2}{R_1+R_2}+R_4\right)\cdot R_3}{R_3+\frac{R_1\cdot R_2}{R_1+R_2}+R_4}$$

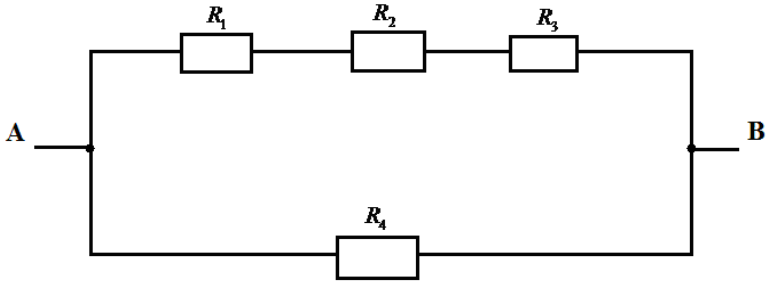
Agar qarshiliklar bir xil bo'lsa

$$R_1=R_2=R_3=R_4=R$$

$$\frac{1}{R_{12}}=\frac{1}{R_1}+\frac{1}{R_2}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R}\quad;\quad R_{12}=\frac{R}{2};\quad R_{124}=R_{12}+R_4=\frac{R}{2}+R=\frac{3}{2}R;$$

$$\frac{1}{R_{AB}}=\frac{1}{R_{124}}+\frac{1}{R_3}=\frac{2}{3R}+\frac{1}{R}=\frac{5}{3R};\qquad R_{AB}=\frac{3}{5}R$$

6. AB nuqta orasidagi umumiy qarshilikni toping.



Agar qarshiliklar har-xil bo'lsa, $R_1;R_2;R_3$ qarshiliklar ketma-ket ulanganligi uchun $R_{123}=R_1+R_2+R_3$ chiqqan natija esa R_4 ga parallel ulanganligi uchun umumiy qarshilik quyidagicha topiladi

$$\frac{1}{R_{AB}}=\frac{1}{R_{123}}+\frac{1}{R_4}=\frac{R_4+R_{123}}{R_{123}\cdot R_4}$$

$$R_{AB}=\frac{R_{123}\cdot R_4}{R_4+R_{123}}$$

Agar qarshiliklar bir xil bo'lsa

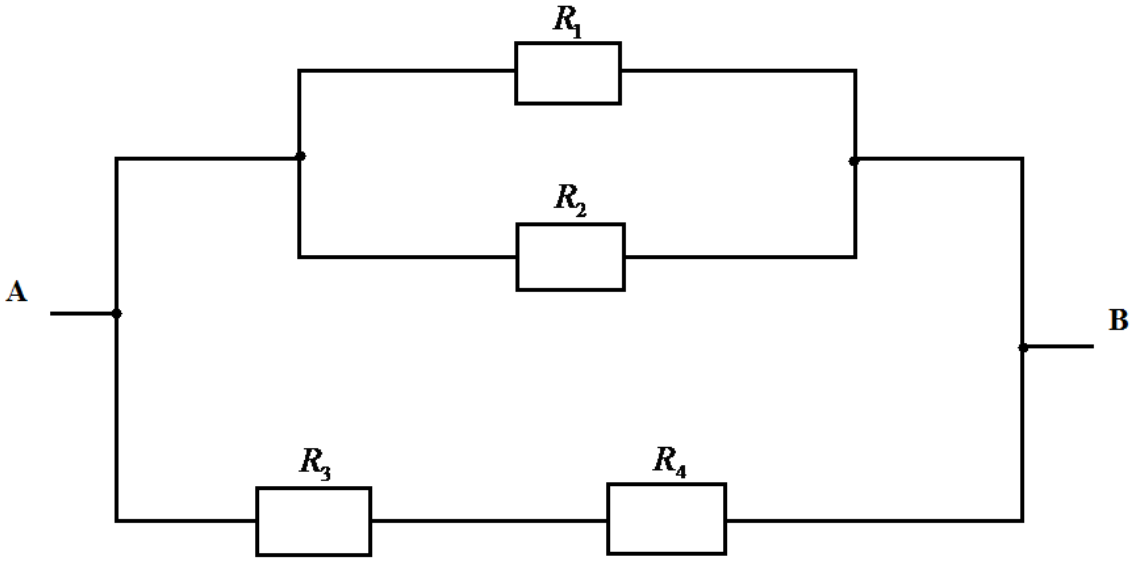
$$R_1=R_2=R_3=R_4=R$$

$$R_{123}=R_1+R_2+R_3=R+R+R=3R$$

$$\frac{1}{R_{AB}}=\frac{1}{R_{123}}+\frac{1}{R_4}=\frac{1}{3R}+\frac{1}{R}=\frac{4}{3R}$$

$$R_{AB}=\frac{3R}{4}$$

7. AB nuqta orasidagi umumiy qarshilikni toping.



Agar qarshiliklar har-xil bo’lsa

R_1 va R_2 qarshiliklar parallel ulangan, R_3 va R_4 qarshiliklar ketma-ket ulanganligi uchun ularni quyidagicha hisoblaymiz.

$$\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_2 + R_1}{R_1 \cdot R_2};$$

$$R_{12} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$R_{34} = R_3 + R_4$$

Chiqqan natijalarni parallel ulaymiz

$$\frac{1}{R_{AB}} = \frac{1}{R_{12}} + \frac{1}{R_{34}} = \frac{R_{34} + R_{12}}{R_{12} \cdot R_{34}}; \qquad R_{AB} = \frac{R_{12} \cdot R_{34}}{R_{34} + R_{12}}$$

Agar qarshiliklar bir xil bo’lsa

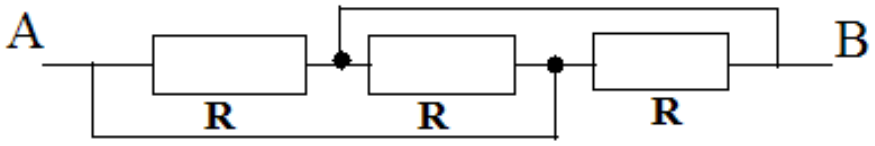
$$R_1 = R_2 = R_3 = R_4 = R$$

$$\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R}; \qquad R_{12} = \frac{R}{2}$$

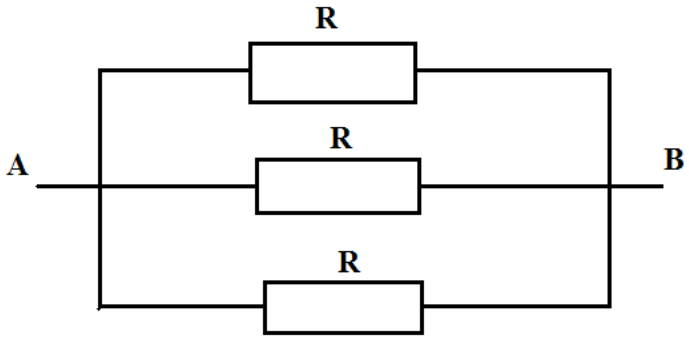
$$R_{34} = R_3 + R_4 = R + R = 2R$$

$$\frac{1}{R_{AB}} = \frac{1}{R_{12}} + \frac{1}{R_{34}} = \frac{1}{\frac{R}{2}} + \frac{1}{2R} = \frac{2}{R} + \frac{1}{2R} = \frac{5}{2R}; \qquad R_{AB} = \frac{2R}{5}$$

8.Rasmda tasvirlangan sxemaning umumiy qarshiligini toping.

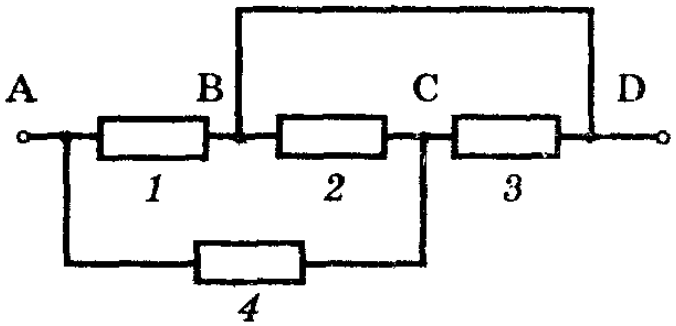


Bu zanjirni soddalashtiramiz

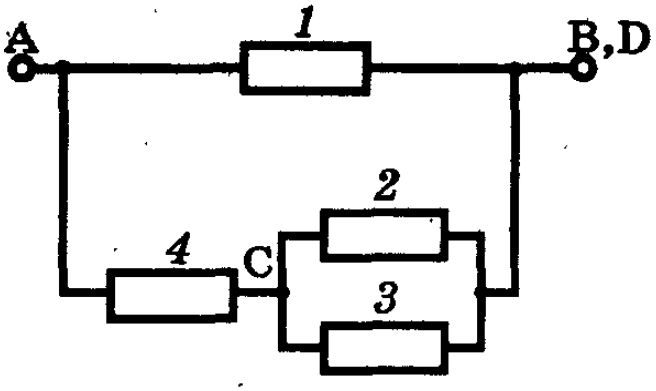


$$\frac{1}{R_{AB}}=\frac{1}{R}+\frac{1}{R}+\frac{1}{R}=\frac{3}{R};\quad R_{AB}=\frac{R}{3};\quad Javob: R_{AB}=\frac{R}{3}$$

9.Rasmda tasvirlangan sxemaning umumiy qarshiligini toping.(Barcha rezistorlarning qarshiligi R ga teng)



Sxemani soddalashtiramiz

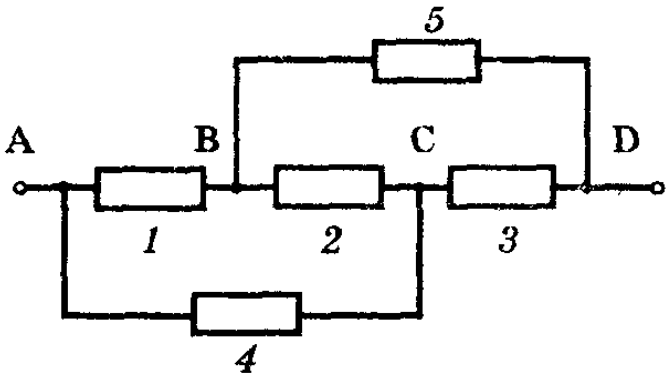


$$\frac{1}{R_{23}}=\frac{1}{R_2}+\frac{1}{R_3}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R};\qquad R_{23}=\frac{R}{2}:$$

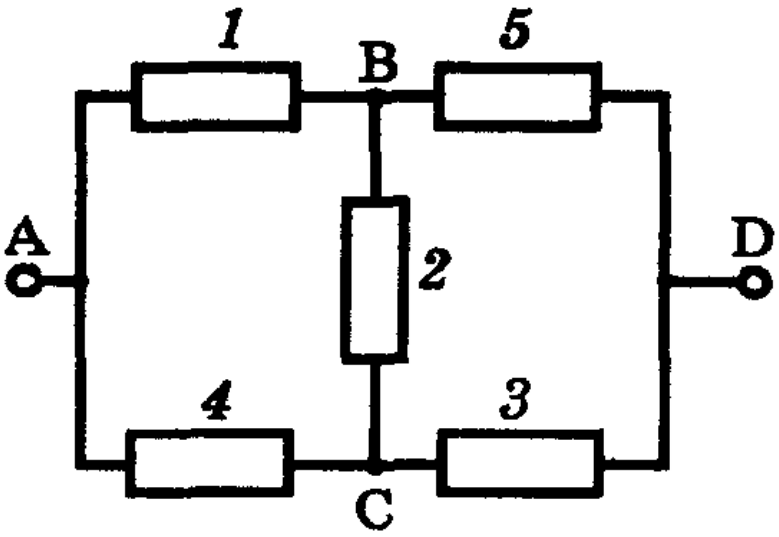
$$R_{234}=R_{23}+R_4=\frac{R}{2}+R=\frac{3}{2}R;\qquad R_{234}=\frac{3}{2}R:$$

$$\frac{1}{R_{um}}=\frac{1}{R_{234}}+\frac{1}{R_1}=\frac{2}{3R}+\frac{1}{R}=\frac{5}{3R};\quad R_{Um}=\frac{3}{5}R:$$

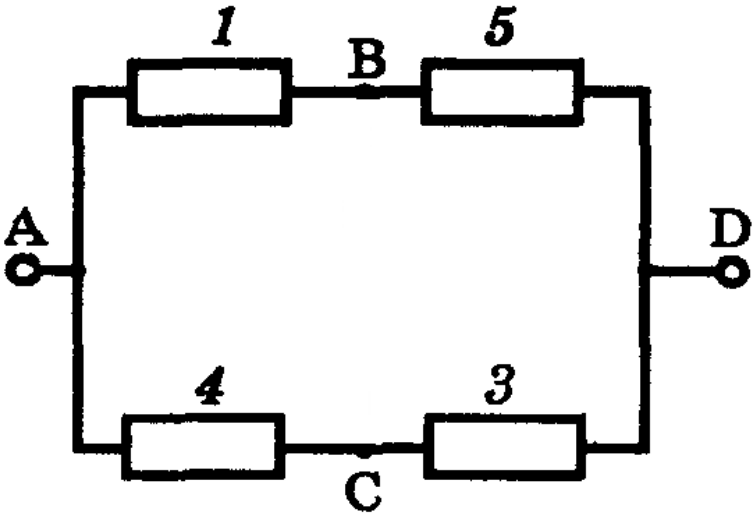
10.Rasmda tasvirlangan sxemaning umumiy qarshiligini toping.
(Barcha rezistorlarning qarshiligi R ga teng)



Sxemani soddalashtiramiz



$\frac{R_1}{R_4}=\frac{R_5}{R_3}$ Nisbatlar tengligi uchun B va C nuqtalarning potensiallar farqi nolga teng, shuning uchun 2-rezistordan tok o’tmayni . Tok o’tmaganligi uchun 2-rezistorni sxemadan olib tashlaymiz.

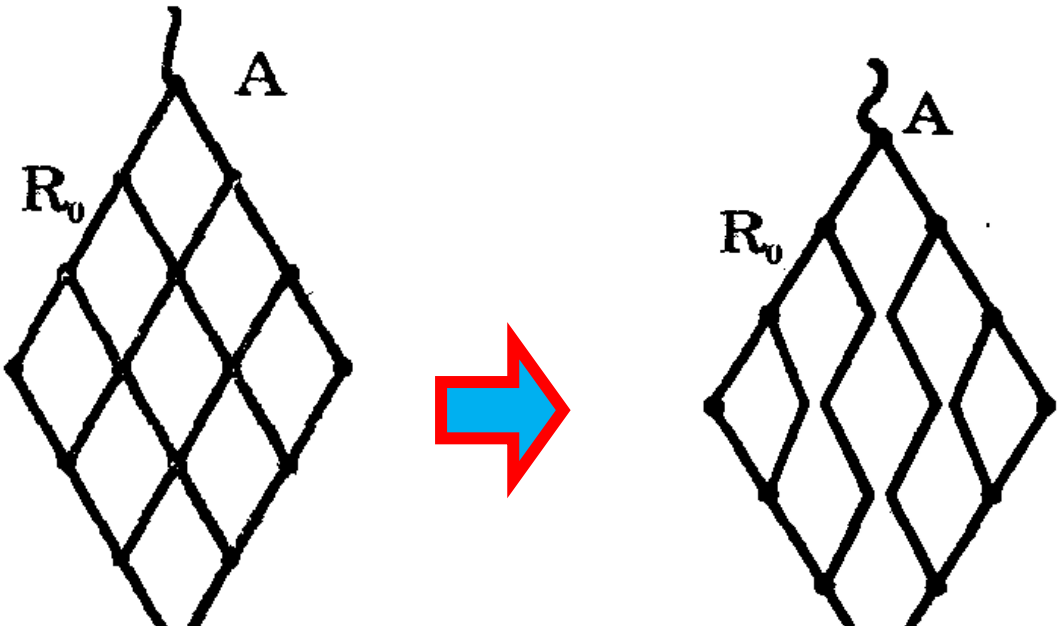


$$R_{15} = R_1 + R_5 = R + R = 2R; \qquad R_{15} = 2R:$$

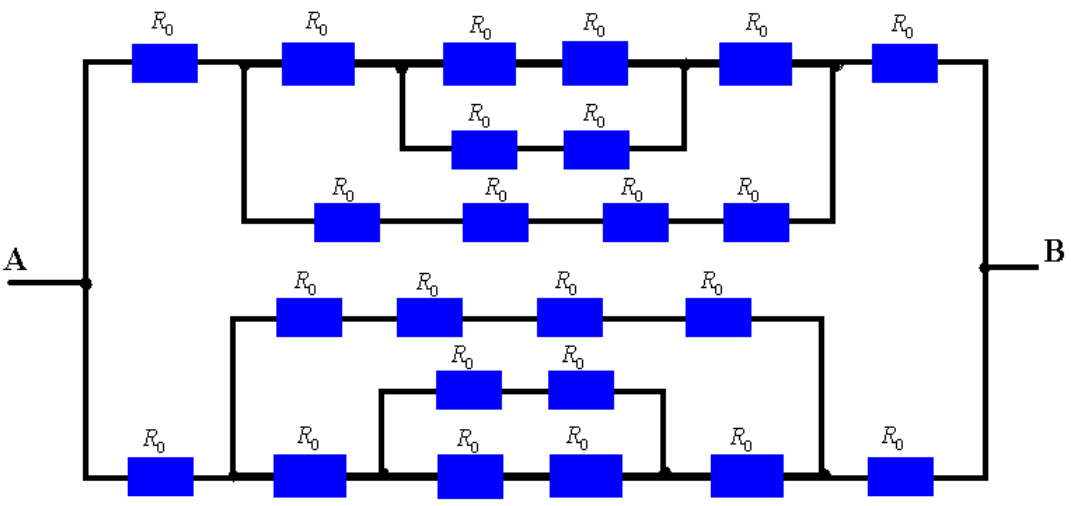
$$R_{43} = R_4 + R_3 = R + R = 2R; \qquad R_{43} = 2R:$$

$$\frac{1}{R_{AB}} = \frac{1}{R_{15}} + \frac{1}{R_{43}} = \frac{1}{2R} + \frac{1}{2R} = \frac{2}{2R}; \quad R_{AB} = R:$$

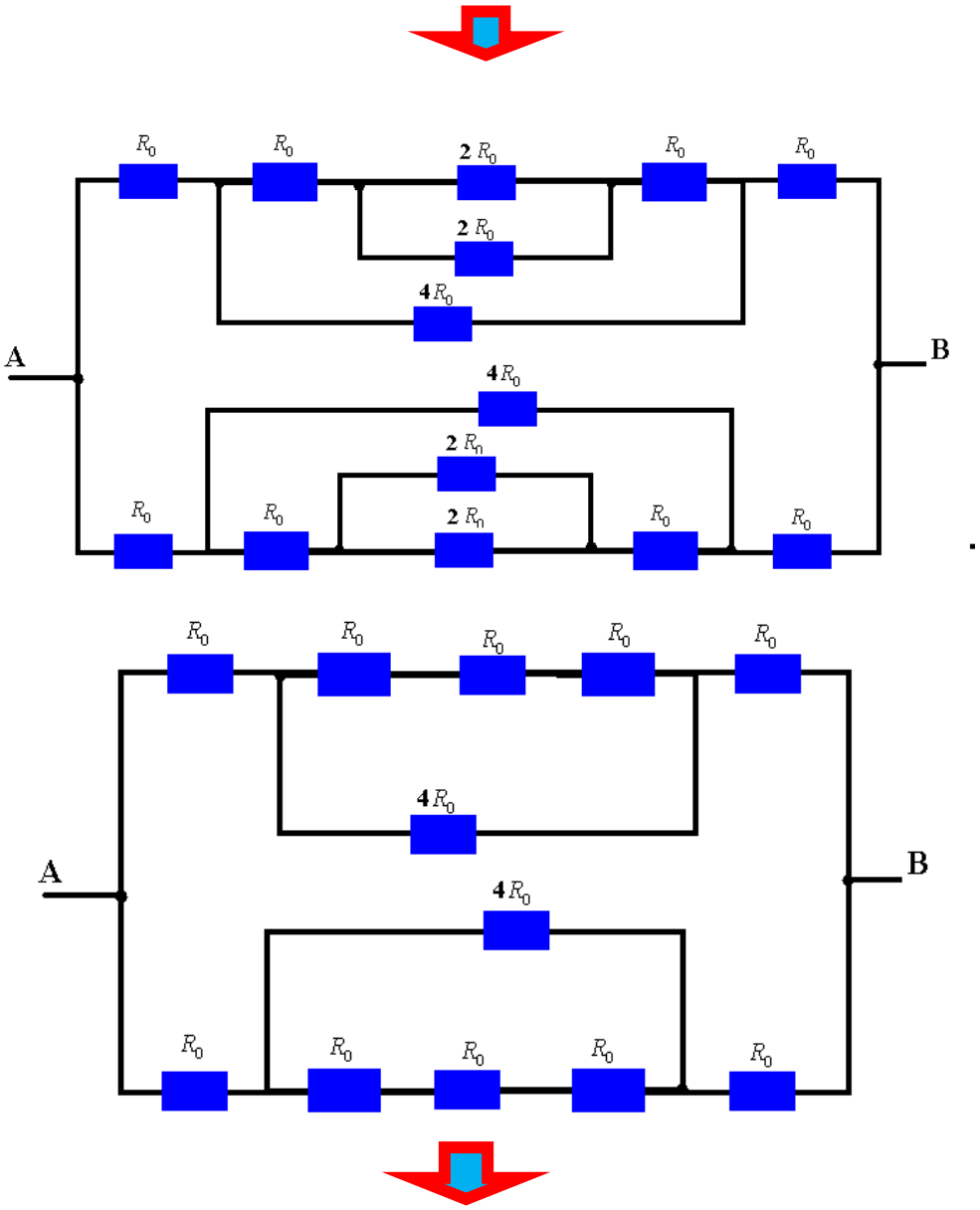
11.AB nuqta orasidagi umumiy qarshilikni toping.
(Barcha qismlarning qarshiligi R ga teng)

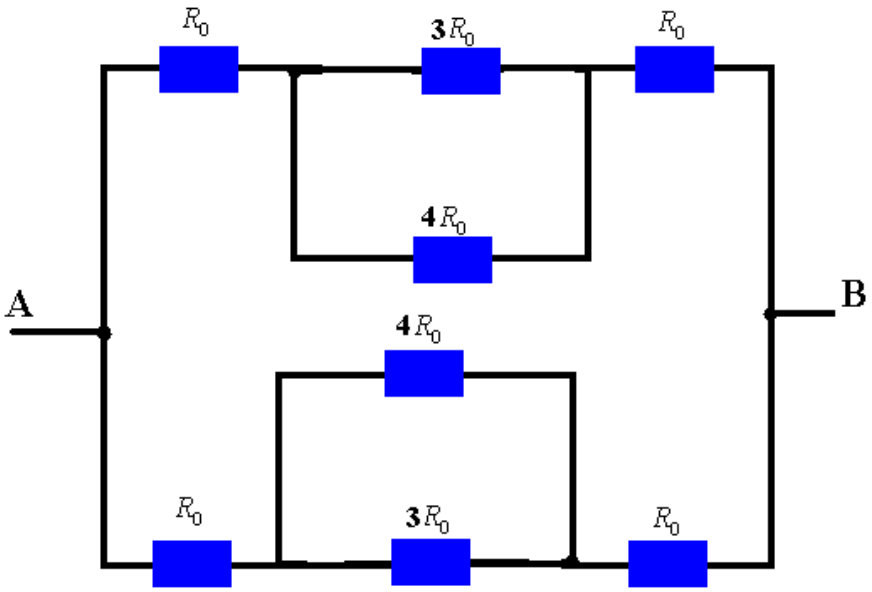


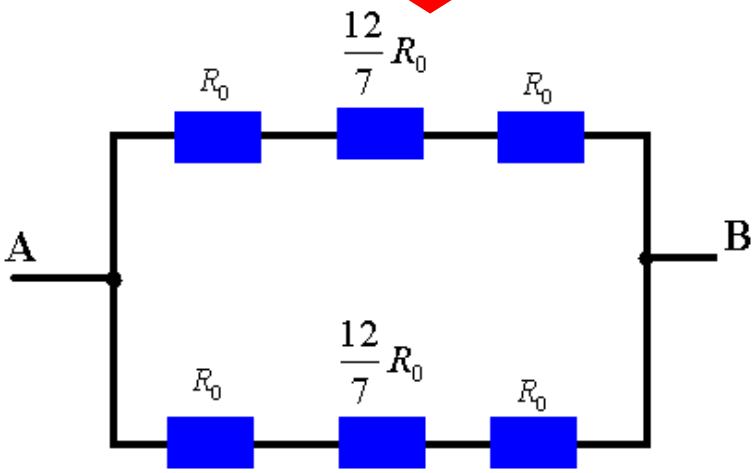
Sxemani quyidagi soda

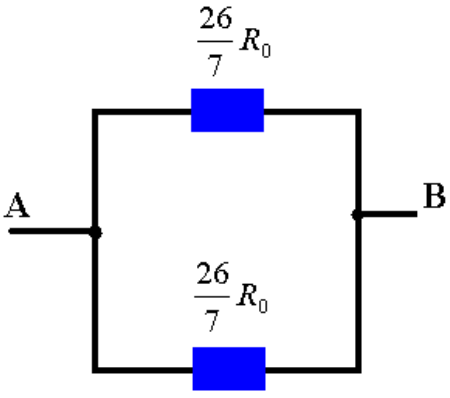


ko’rinishda ifodalaymiz.







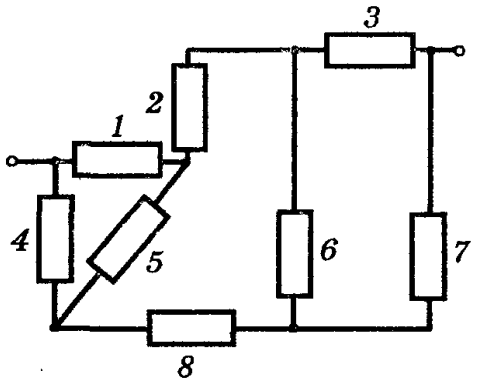


$$\frac{1}{R_{AB}} = \frac{7}{26R_0} + \frac{7}{26R_0} = \frac{14}{26R_0} = \frac{7}{13R_0};$$

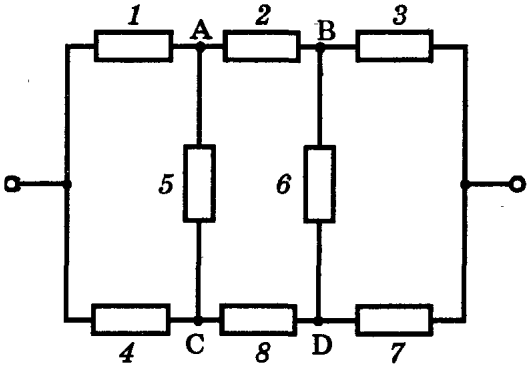
$$R_{AB} = \frac{13}{7}R_0$$

12. Zanjirning umumiy qarshiligini, Har-bir rezistordan o’tadigan tok kuchini toping.

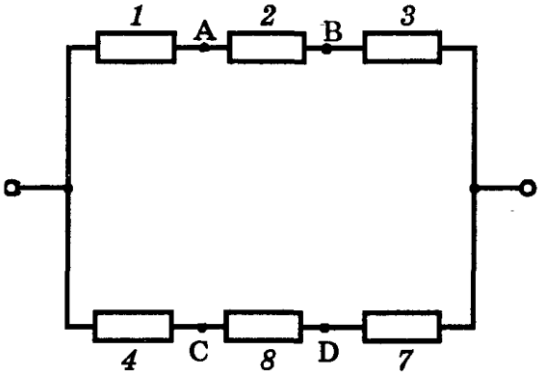
$U = 84v; R_1 = R_5 = R_8 = 12\Omega;\ R_2 = R_6 = R_7 = 6\Omega;\ R_4 = 24\Omega; R_3 = 3\Omega;$



Sxemani soddalashtiramiz



$R_1 : R_2 : R_3 = R_4 : R_8 : R_7,$ qarshiliklar nisbati tengligi uchun A va C ;B va D nuqtalarda potensiallar teng, potensiallar farqi o ga teng.Shuning uchun R_5 va R_6 rezistor orqali tok o’tmaydi va sxemamiz quyidagicha soddalashadi



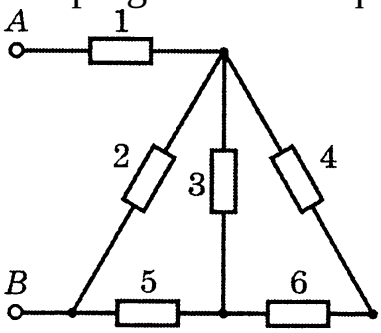
$$\frac{1}{R_{Um}} = \frac{1}{21} + \frac{1}{42} = \frac{3}{42};\quad R_{Um} = 14\Omega$$

$$I_5 = 0;\ I_6 = 0$$

$$I_1 = I_2 = I_3 = \frac{U}{R_1 + R_2 + R_3} = 4\ \text{A},$$

$$I_4 = I_7 = I_8 = \frac{U}{R_4 + R_7 + R_8} = 2\ \text{A}.$$

13.AB nuqta orasidagi umumiy qarshilikni va har bir rezistor orqali o’tadigan tok kuchini toping. A va B nuqta potensiallar farqi 12 v



$R_1 = 4$ Ом

$R_2 = 4$ Ом

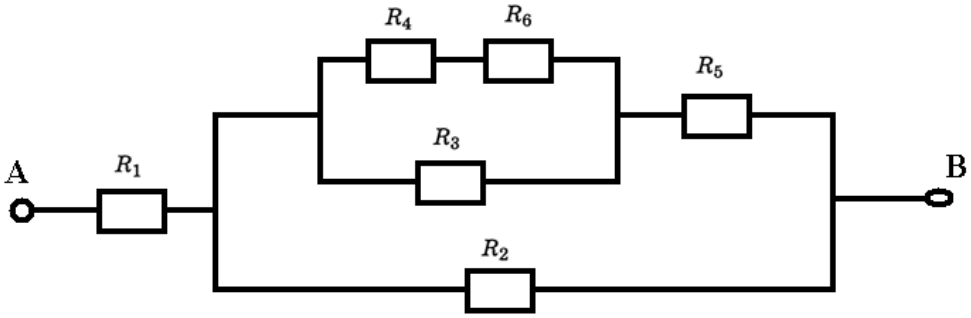
$R_3 = 5$ Ом

$R_4 = 3$ Ом

$R_5 = 1{,}5$ Ом

$R_6 = 2$ Ом

Sxemani quyidagicha soddalashtiramiz.



$$R_{46} = R_4 + R_6 = 3 + 2 = 5\Omega;$$

$$\frac{1}{R_{346}} = \frac{1}{R_3} + \frac{1}{R_{46}} = \frac{1}{5} + \frac{1}{5} = \frac{2}{5};\ R_{346} = 2{,}5\Omega$$

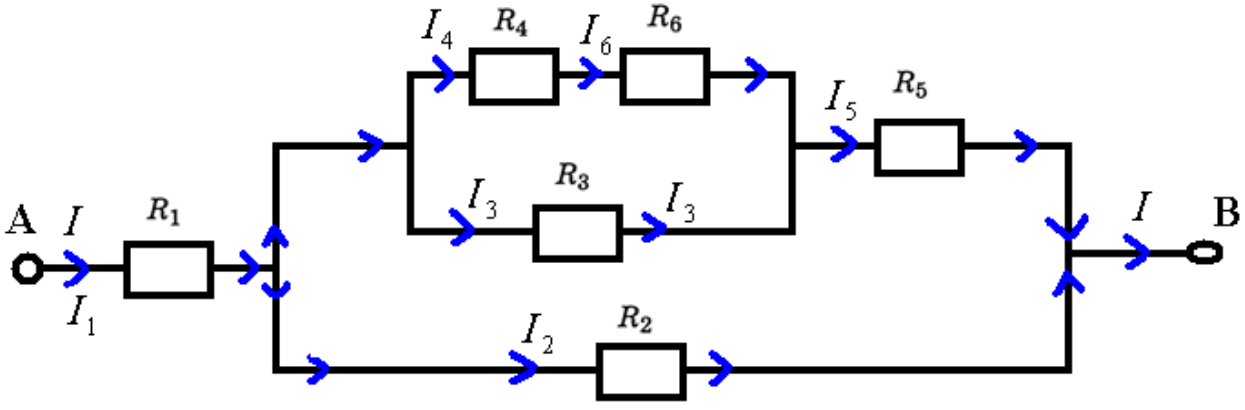
$$R_{3465} = R_{346} + R_5 = 2{,}5 + 1{,}5 = 4\Omega$$

$$\frac{1}{R_{34652}} = \frac{1}{R_{3465}} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4};\ R_{34652} = 2\Omega$$

$$R_{AB} = R_{34652} + R_1 = 2 + 4 = 6\Omega$$

$$I = \frac{U}{R_{AB}} = \frac{12}{6} = 2A$$

Demak A nuqtadan I tok kuchi kirib B nuqtadan chiqib ketadi deb sxemani quyidagicha o’zgartiramiz.



Krixgov 1-qoidasiga ko’ra $I = I_1 = I_2 + I_5$

$I_5 = I_3 + I_6$

Ketma ketligi uchun $I_4 = I_6$

$I = I_1 = I_2 + I_5 = 2A;$

$U_5 = U_2$ *bundan*

$I_5 \cdot R_5 = I_2 \cdot R_2$

$I_5 \cdot 1{,}5 = I_2 \cdot 4$

$I_5 = I_2 \cdot \frac{4}{1{,}5} = \frac{8}{3} \cdot I_2$

$I_2 + I_5 = 2$

$I_2 + \frac{8}{3} I_2 = 2$

$\frac{11}{3} I_2 = 2;\ I_2 = \frac{6}{11}$

$I_5 = \frac{8}{3} \cdot I_2 = \frac{8}{3} \cdot \frac{6}{11} = \frac{16}{11}$

$U_4 + U_6 = U_3$ *bundan*

$I_4 \cdot R_4 + I_6 \cdot R_6 = I_3 \cdot R_3$

$I_4 \cdot 3 + I_6 \cdot 2 = I_3 \cdot 5;\ (I_4 = I_6)$

$5 \cdot I_6 = I_3 \cdot 5;\quad I_6 = I_3$

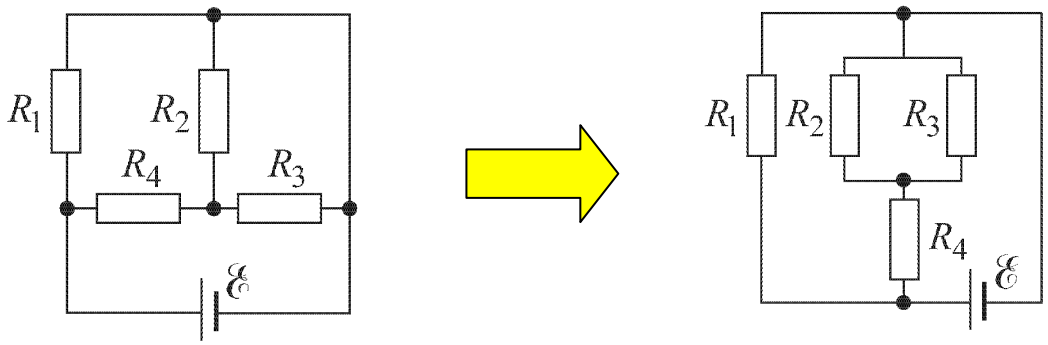
$I_3 + I_6 = I_5 = \frac{16}{11}$ *tenglikdan*

$2I_3 = \frac{16}{11};\quad I_3 = \frac{8}{11}$

$I_3 = I_6 = I_4 = \frac{8}{11} A$

14. Manbaning tok kuchini va R_1 rezistordagi tok kuchini aniqlang.

($\varepsilon = 1{,}5V;\ r = 0{,}5\Omega; R_1 = 5\Omega; R_2 = 9\Omega; R_3 = 3\Omega;\ R_4 = 2{,}75\Omega$)



$R' = \frac{R_2 R_3}{R_2 + R_3}.\qquad R'' = R' + R_4 = \frac{R_2 R_3}{R_2 + R_3} + R_4 = \left(\frac{9{,}0 \cdot 3{,}0}{12} + 2{,}75\right)$ Ом $= 5{,}0$ Ом.

$R = \frac{R_1 R''}{R_1 + R''} = \frac{R_1}{2} = 2{,}5$ Ом. $\qquad I = \frac{\mathscr{E}}{R + r} = \frac{1{,}5}{2{,}5 + 0{,}50} \frac{\text{В}}{\text{Ом}} = 0{,}50$ А.

$U_1 = U_{234}$ *tengligidan*

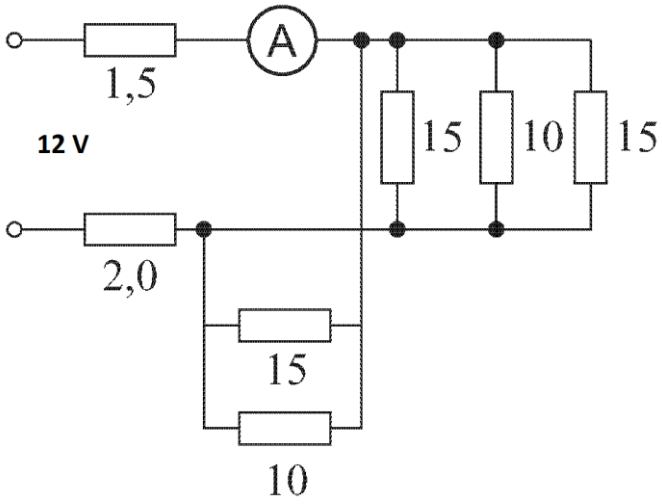
$I_1 \cdot R_1 = I'' \cdot R''$

$I_1 \cdot 5 = I'' \cdot 5;\quad I_1 = I''$

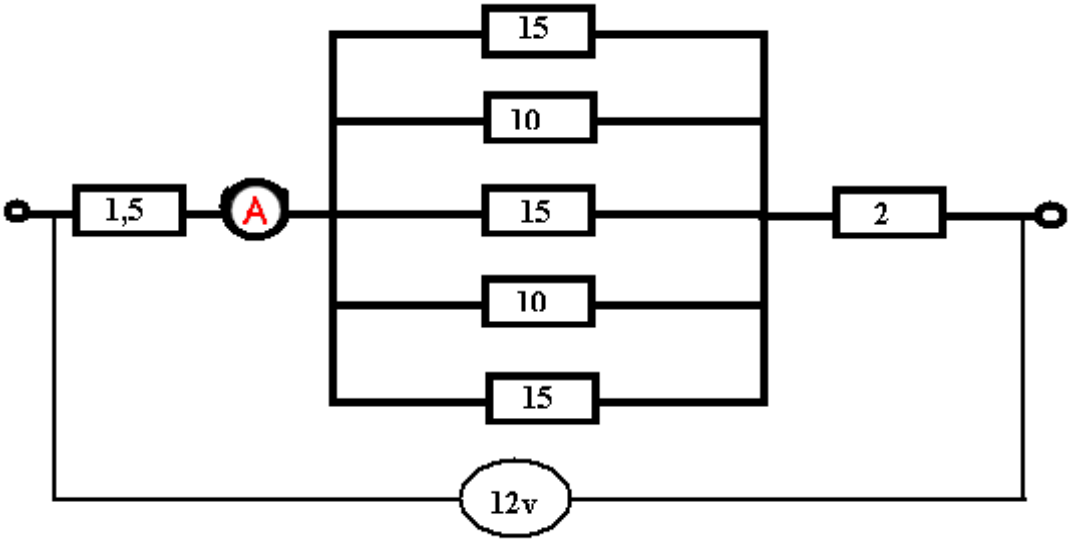
$I = I_1 + I'' = 0{,}5A$

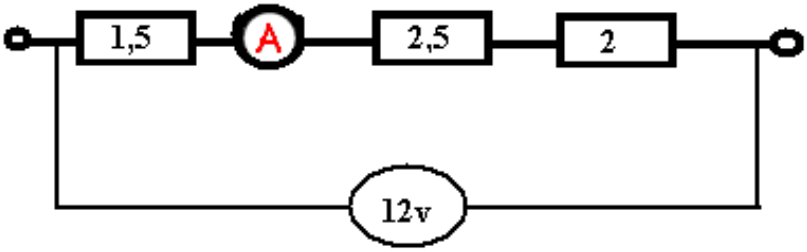
$I_1 = I'' = \frac{I}{2} = \frac{0{,}5}{2} = 0{,}25A$

15.Zanjirdagi ampermetr necha amperni ko’rsatadi?



Sxemani soddalashtiramiz.



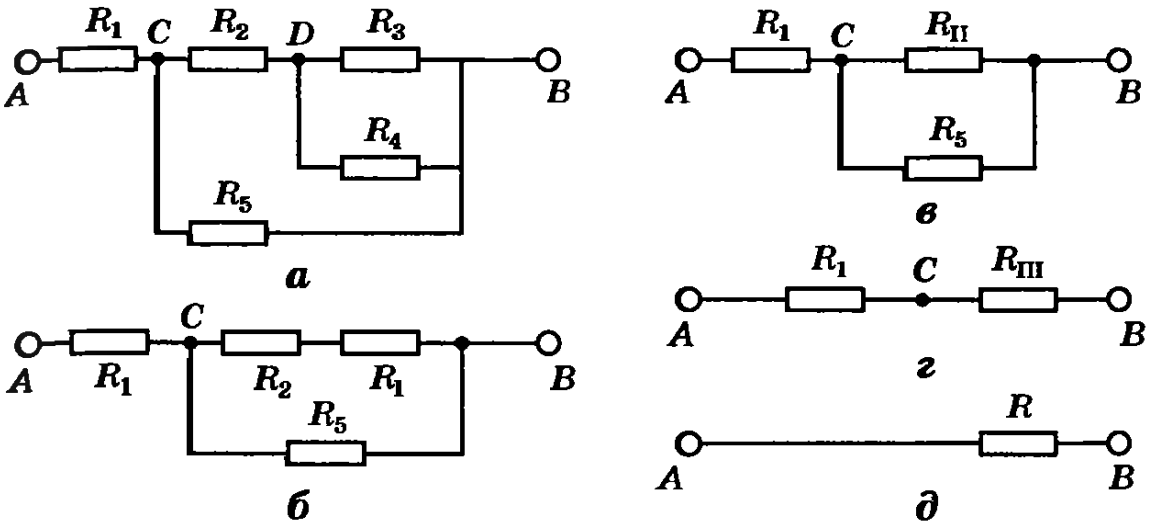


$$R_{Um} = 1{,}5 + 2{,}5 + 2 = 6\Omega$$

$$I = \frac{U}{R_{Um}} = \frac{12}{6} = 2A$$

16.Rasmda tasvirlangan har-bir rezistordan o’tadigan tok kuchini va umumiy qarshilikni toping. $R_1 = 6\Omega; R_2 = 5\Omega;\ R_3 = 4\Omega; R_4 = 12\Omega;\ R_5 = 8\Omega; U_{AB} = 20v$

Sxemani quyidagicha soddalashtiramiz va hisoblaymiz



$$R_{34}=\frac{R_3\cdot R_4}{R_3+R_4}=\frac{4\cdot 12}{4+12}=\frac{48}{16}=3\Omega;\qquad R_{234}=R_2+R_{34}=5+3=8\Omega;$$

$$R_{2345}=\frac{R_{234}\cdot R_5}{R_{234}+R_5}=\frac{8\cdot 8}{8+8}=\frac{64}{16}=4\Omega;\quad R_{AB}=R_1+R_{2345}=6+4=10\Omega;$$

$$I=I_1=\frac{U_{AB}}{R_{AB}}=\frac{20}{10}=2A;\quad U_{CB}=I_1\cdot R_{2345}=2\cdot 4=8v;\quad I_2=\frac{U_{CB}}{R_{234}}=\frac{8}{8}=1A;$$

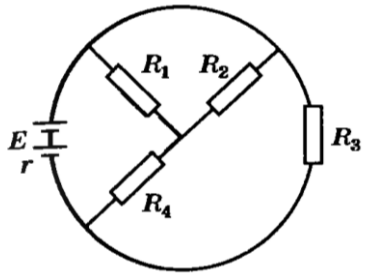
$$I_5=\frac{U_{CB}}{R_5}=\frac{8}{8}=1A;\qquad R_{234}=R_5\ tengligidan\ I_{234}=I_5=\frac{I}{2}=1A;$$

DB uchastka uchun $U_{DB}=I_3R_3=I_4R_4+I_2R_{34};\ U_{DB}=1\cdot 3=3v$

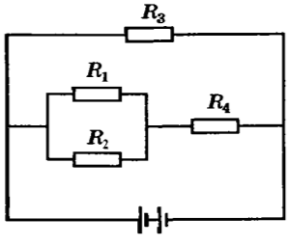
$$I_3=\frac{U_{DB}}{R_3}=\frac{3}{4}=0{,}75A,\quad I_4=\frac{U_{DB}}{R_4}=\frac{3}{12}=0{,}25A.$$

17. Rasmda tasvirlangan sxema manbasini tok kuchini toping.

($R_1=1\Omega;\ R_2=3\Omega;\ R_3=1{,}5\Omega;\ R_4=0{,}75\Omega;\ \varepsilon=3v;\ r=0{,}25\Omega$)



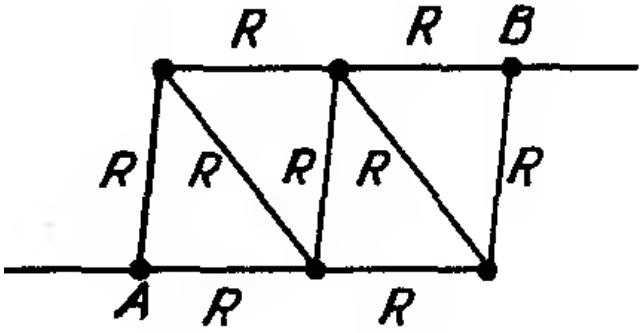
Sxemani quyidagicha soddalashtiramiz



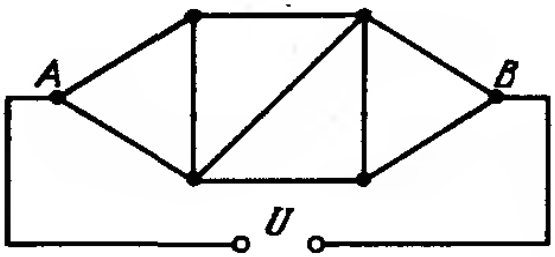
$$R_{12}=\frac{R_1\cdot R_2}{R_1+R_2}=\frac{1\cdot 3}{1+3}=\frac{3}{4}=0{,}75\Omega;\quad R_{124}=R_{12}+R_4=0{,}75+0{,}75=1{,}5\Omega$$

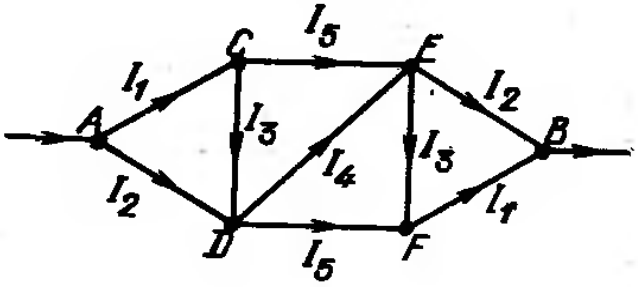
$$R_{Um}=\frac{R_{124}\cdot R_3}{R_{124}+R_3}=\frac{1{,}5\cdot 1{,}5}{1{,}5+1{,}5}=\frac{2{,}25}{3}=0{,}75\Omega;\quad I=\frac{\varepsilon}{R+r}=\frac{3}{0{,}75+0{,}25}=3A$$

18.AB nuqta orasidagi umumiy qarshilikni toping.



Sxemaning umumiy qarshiligini topish uchun uning A va B nuqtalari potensiallar farqini U bilian belgilaymiz(1-rasm), 2-rasmda esa sxemadagi rezistorlardagi toklarning harakat yo'nalishlari ko'rsatilgan.





Sxemadan ko'rinib turibdiki CE va DF parallel ulanganligi uchun va qarshiliklari bir-xil bo'lganligi uchun ularda harakatlanadigan toklar teng,bu fikrni CD va EF nuqtalar orasidagi toklar uchun ham shunday deyish mumkin.

Krixgovning 1-qoidasiga ko'ra

$$I=I_1+I_2;\qquad I_2+I_3=I_4+I_5;$$

Krixgovning 2-qoidasiga ko'ra

$$(I_2+I_5+I_1)R=U;\quad (I_3+I_4)R=I_5R;\quad (I_1+I_3)R=I_2R$$

Krixgovning 1-2-qoidalaridan chiqqan natijalardan

$$I_2=\frac{6}{5}I_1,\quad I_3=\frac{I_1}{5},\qquad I_4=\frac{3}{5}I_1,\qquad I_5=\frac{4}{5}I_1.\ \textit{Bulardan}$$

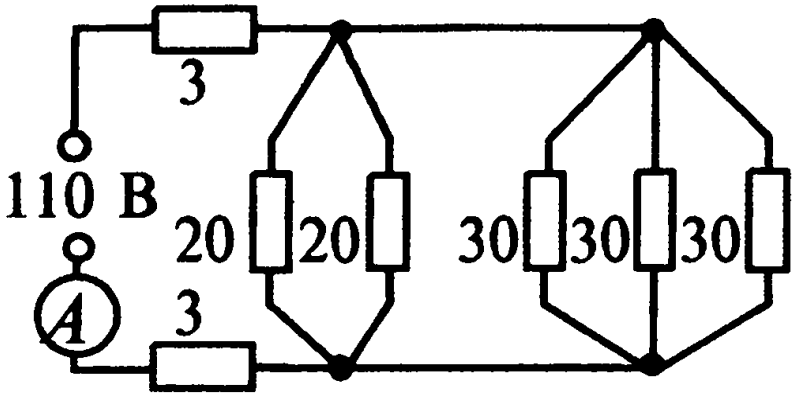
$$U=(I_1+\frac{4}{5}I_1+\frac{6}{5}I_1)R.\qquad \frac{U}{I_1}=3R.$$

Umumiy qarshilik

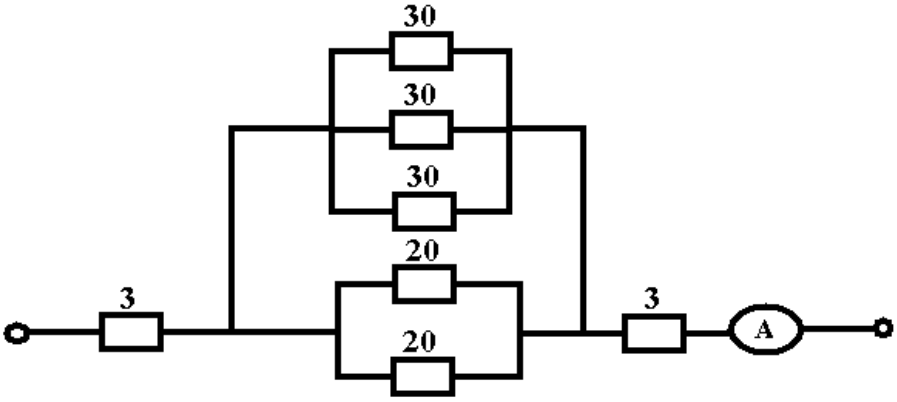
$$R_{AB}=\frac{U}{I}=\frac{U}{I_1+I_2}=\frac{U}{I_1+\frac{6}{5}I_1}=\frac{5}{11}\cdot\frac{U}{I_1}=\frac{5}{11}\cdot\frac{3\cdot R\cdot I_1}{I_1}=\frac{15}{11}R,$$

$$R_{AB}=\frac{15}{11}R$$

19. Sxemaning umumiy qarshiligini va Ampermetr ko'rsatgichini aniqlang.

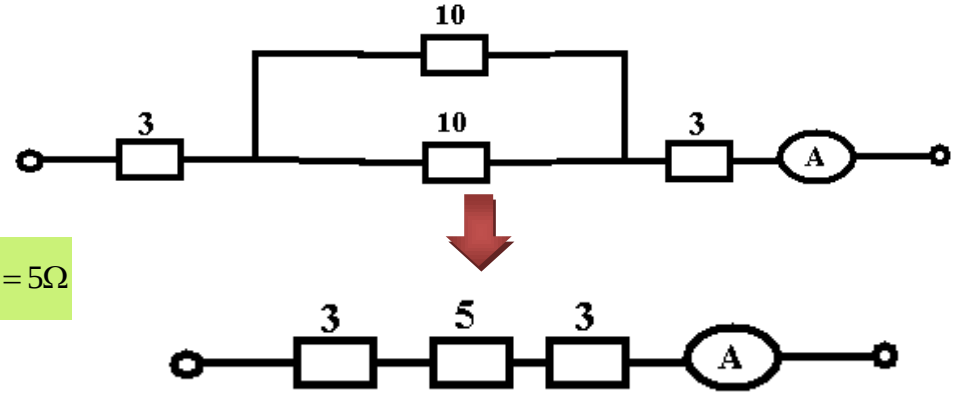


Sxemani quyidagicha soddalashtiramiz.



$$R_{p1}=\frac{30}{3}=10\Omega$$

$$R_{p2}=\frac{20}{2}=10\Omega$$

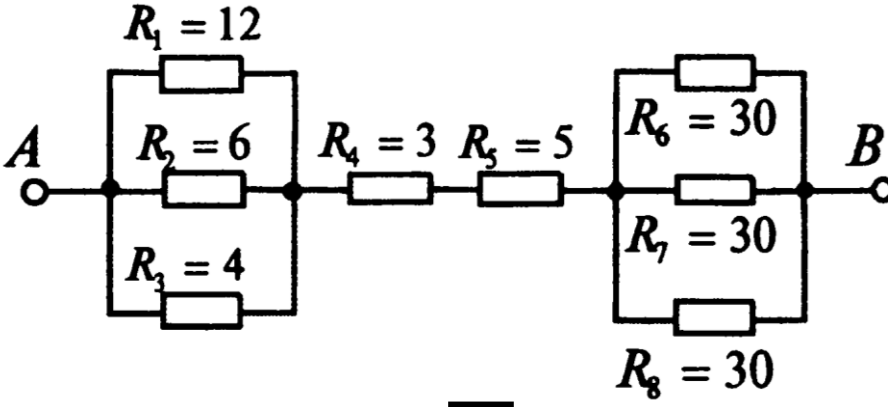


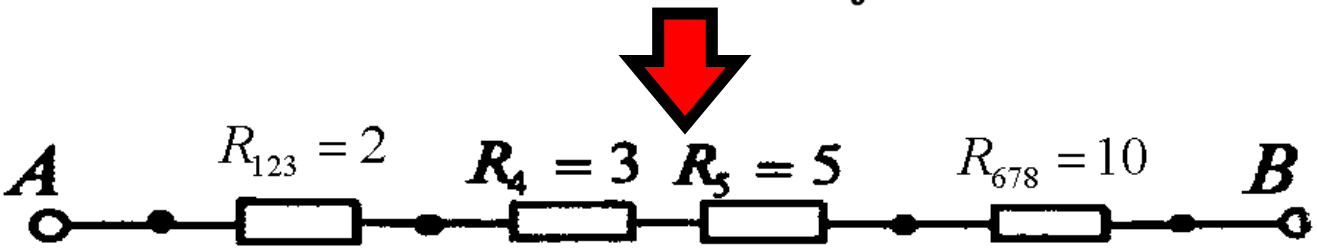
$$R_{p3}=\frac{10}{2}=5\Omega$$

$$R_{Um} = 3+5+3 = 11\Omega$$

$$I = \frac{U}{R_{Um}} = \frac{110}{11} = 10A$$

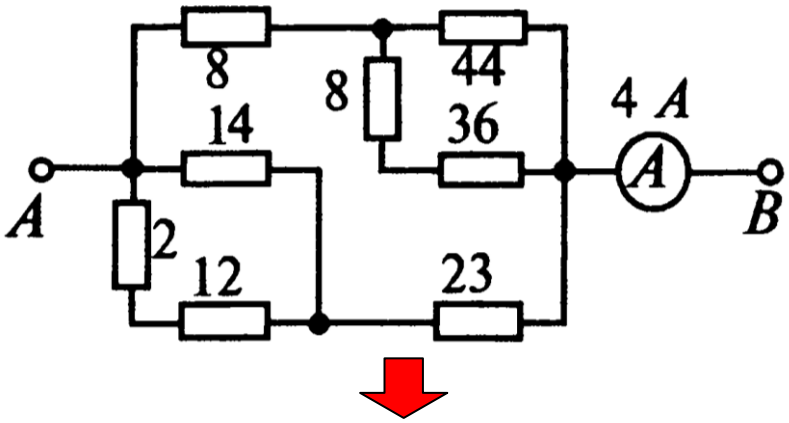
20.AB nuqta orasidagi umumiy qarshilikni toping.

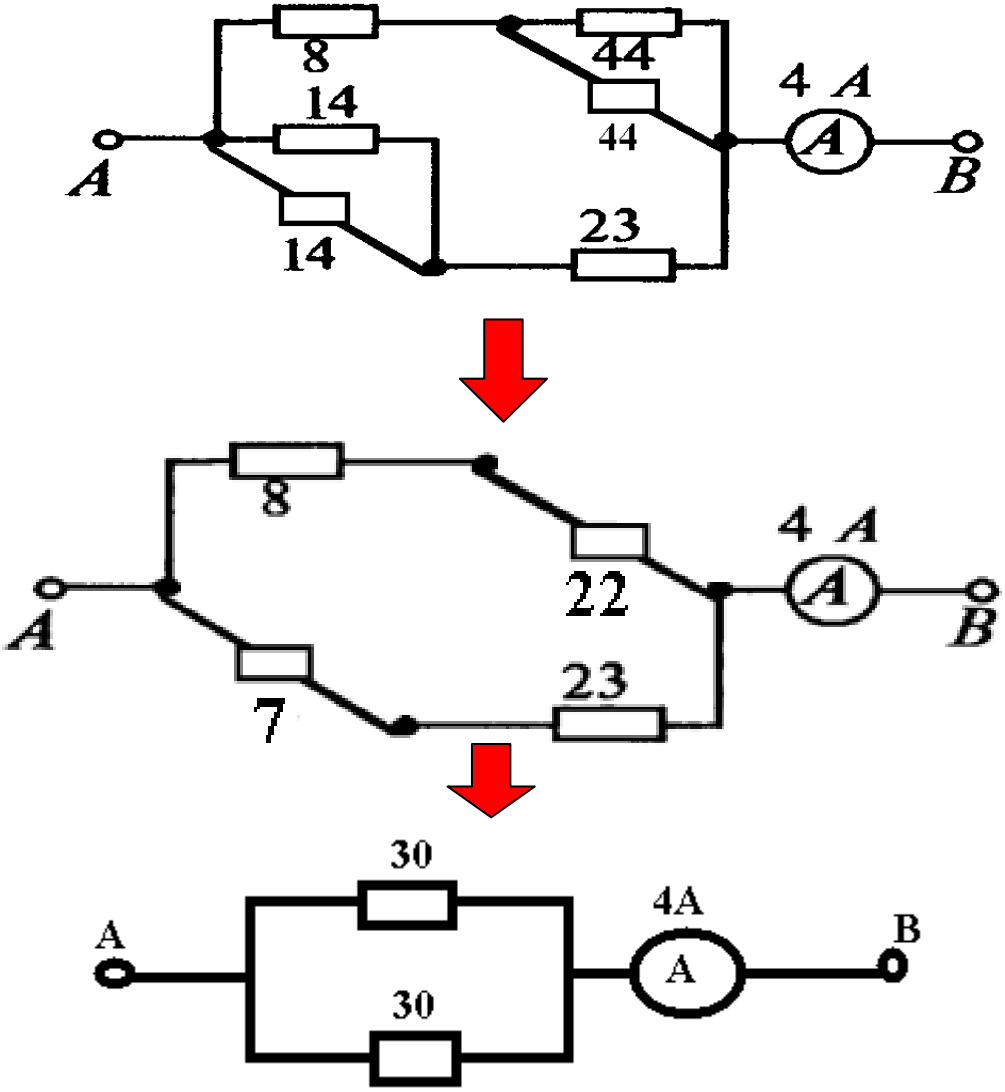




$$R_{Um} = R_{123} + R_4 + R_5 + R_{678} = 2+3+5+10 = 20\Omega$$

21.AB nuqta orasidagi umumiy qarshilikni va AB nuqta urasidagi potensiallar farqini toping.

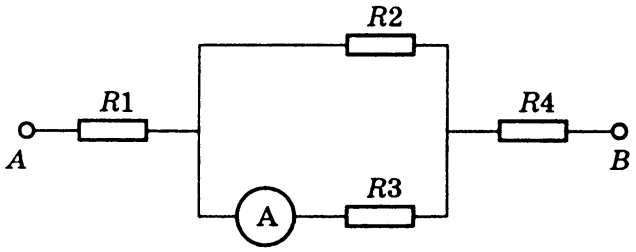




$$R_{Um} = \frac{30 \cdot 30}{30 + 30} = 15\Omega; \qquad U = I \cdot R_{Um} = 4 \cdot 15 = 60V$$

22.Rasmda tasvirlangan sxemaning barcha rezistorlaridagi tok kuchi va kuchlanishni toping. (3-rezistordan o'tadigan tok kuchi 2A)

$R_1 = 2\Omega; \quad R_2 = 10\Omega; \quad R_3 = 15\Omega; \quad R_4 = 4\Omega; \quad I_3 = 2A$



2- va 3- rezistorlar parallel ulanganligi uchun

$$U_2 = U_3; \qquad R_2 \cdot I_2 = R_3 \cdot I_3; \qquad I_2 = I_3 \frac{R_3}{R_2} = 2 \cdot \frac{15}{10} = 3A$$

$$U_3 = U_2 = R_2 \cdot I_2 = 10 \cdot 3 = 30V$$

Krixgovning 1-qoidasiga ko’ra

$$I = I_1 = I_4 = I_2 + I_3 = I_3 \frac{R_2 + R_3}{R_2} = 2\frac{10+15}{10} = 5A$$

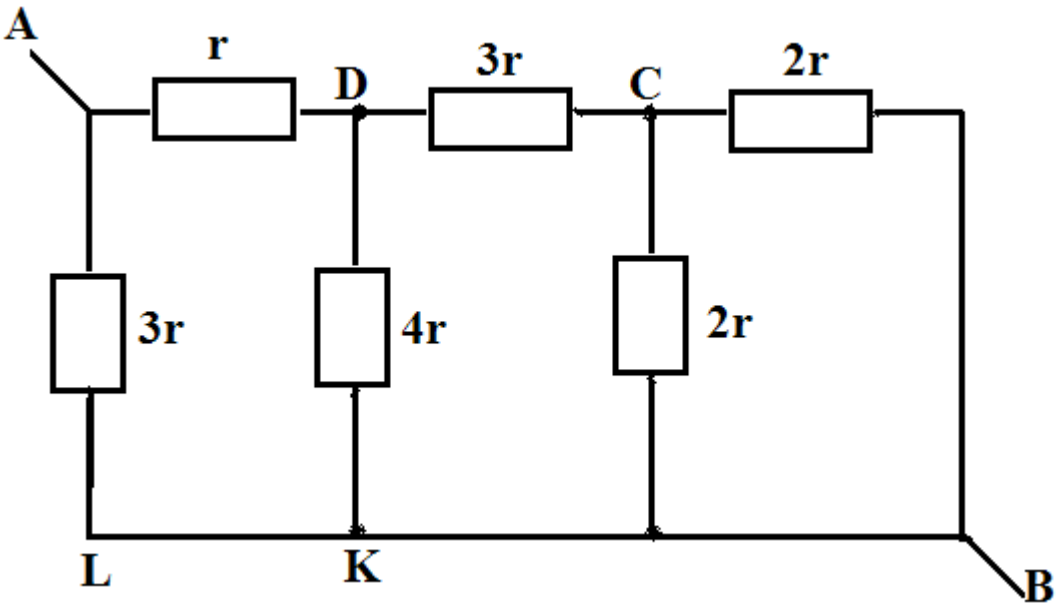
Om qonunidan kuchlanishni topamiz

$$U_1 = R_1 I_1 = I_3 \frac{R_1(R_2 + R_3)}{R_2} = 2\frac{2(10+15)}{10} = 10V$$

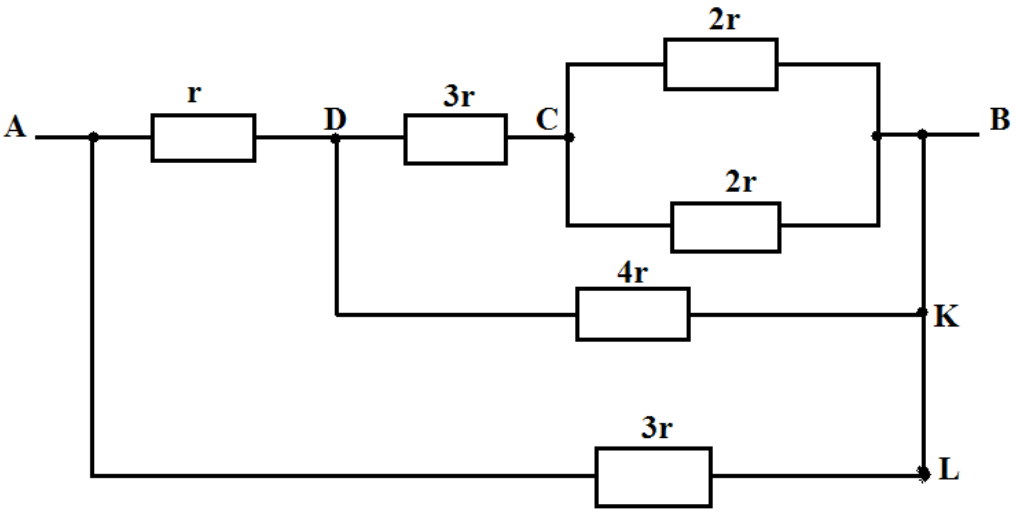
$$U_4 = R_4 I_4 = I_3 \frac{R_4(R_2 + R_3)}{R_2} = 2\frac{4(10+15)}{10} = 20V$$

$$U_{AB} = U_1 + U_4 + U_3 = 10 + 20 + 30 = 60V$$

23.Ava B nuqtalar orasidagi umumiy qarshilikni toping .



Bu zanjirni soddalashtiramiz



$$\frac{1}{R_{BS}}=\frac{1}{2r}+\frac{1}{2r}=\frac{1}{r};\qquad R_{BS}=r$$

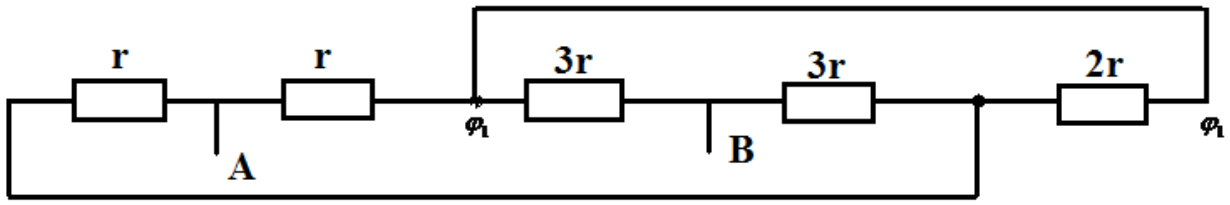
$$R_{BD}=r+3r=4r$$

$$\frac{1}{R_{BDK}}=\frac{1}{R_{KD}}+\frac{1}{R_{BD}}=\frac{1}{4r}+\frac{1}{4r}=\frac{2}{4r};\qquad R_{BDK}=2r$$

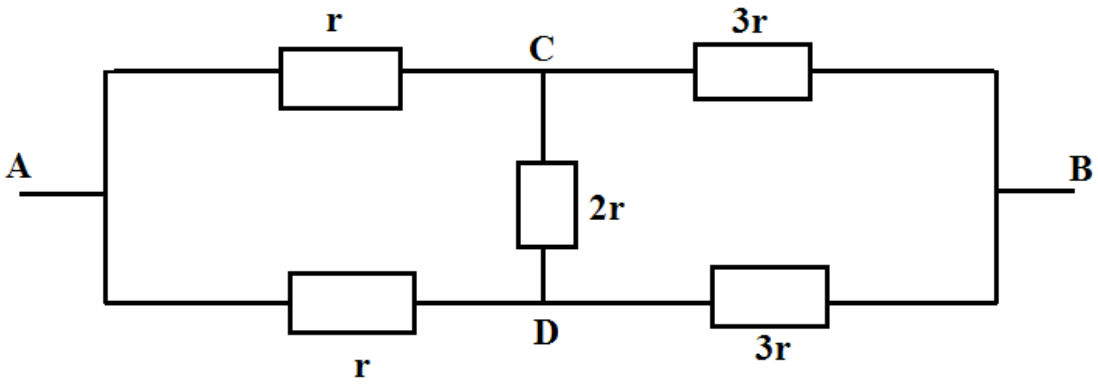
$$R_{BDKA}=R_{BDK}+R_{DA}=2r+r=3r$$

$$\frac{1}{R_{AB}}=\frac{1}{R_{BDKA}}+\frac{1}{R_{LA}}=\frac{1}{3r}+\frac{1}{3r}=\frac{2}{3r};\qquad Javob: R_{AB}=1{,}5r$$

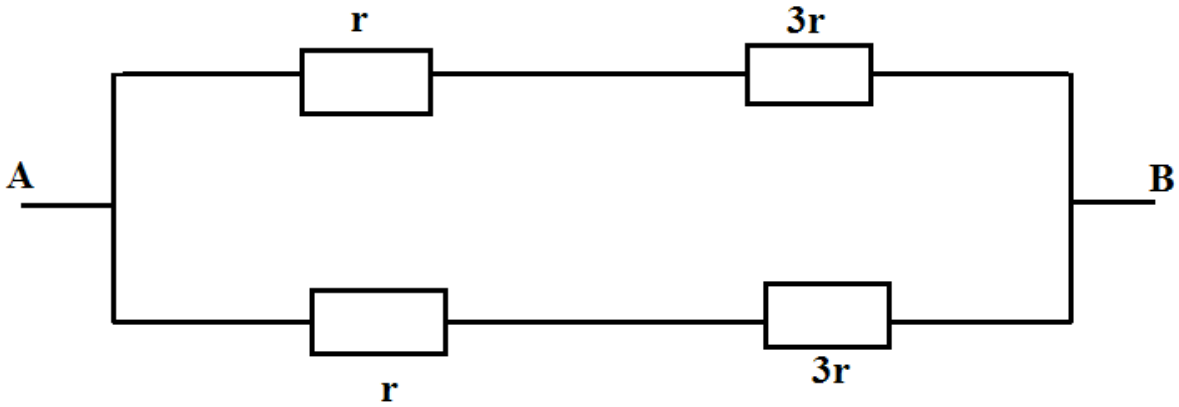
24.Rasmda tasvirlangan sxemaning umumiy qarshiligni hisobang?



Bu zanjirni soddalashtiramiz :



$\varphi_{CD}=O$ Shu sababdan 2R qarshilikdan tok o'tmaydi shuning uchun chizma o'zgaradi



$\varphi_{CD} = O$ *Shu sababdan* $2R$ *qarshilikdan tok o'tmaydi.*

$R' = r_1 + r_3 \qquad R'' = r_2 + r_4$

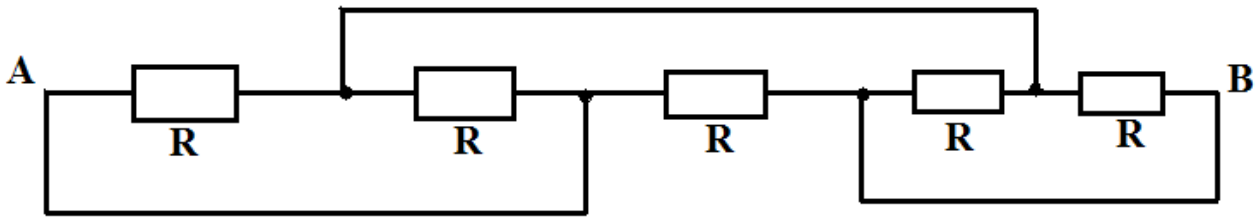
$$\frac{1}{R_{AB}} = \frac{1}{R'} + \frac{1}{R''} \rightarrow R_{AB} = \frac{R' \cdot R''}{R' + R''}$$

$R' = r + 3r = 4r \qquad R'' = r + 3r = 4r$

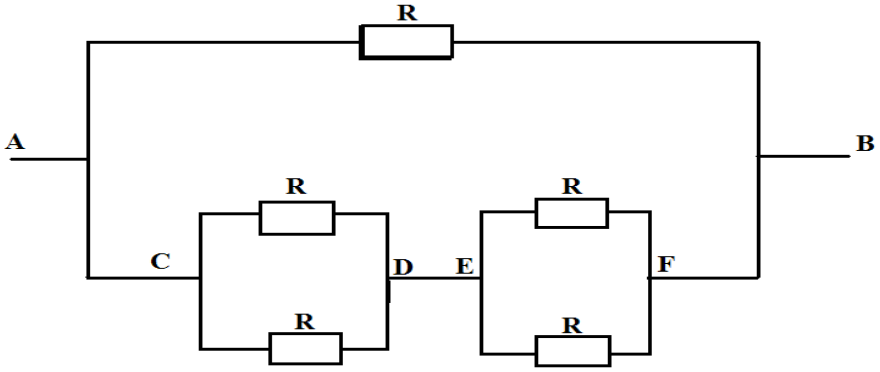
$$R_{AB} = \frac{4r \cdot 4r}{4r + 4r} = \frac{16r^2}{8r} = 2r$$

Javob: $R_{AB} = 2r$

25. Har bir rezistorning qarshiligi R ga teng bo'lsa, A va B nuqtalar orasidagi umumiy qarshilik qanday bo'ladi?



Bu zanjirni soddalashtiramiz :



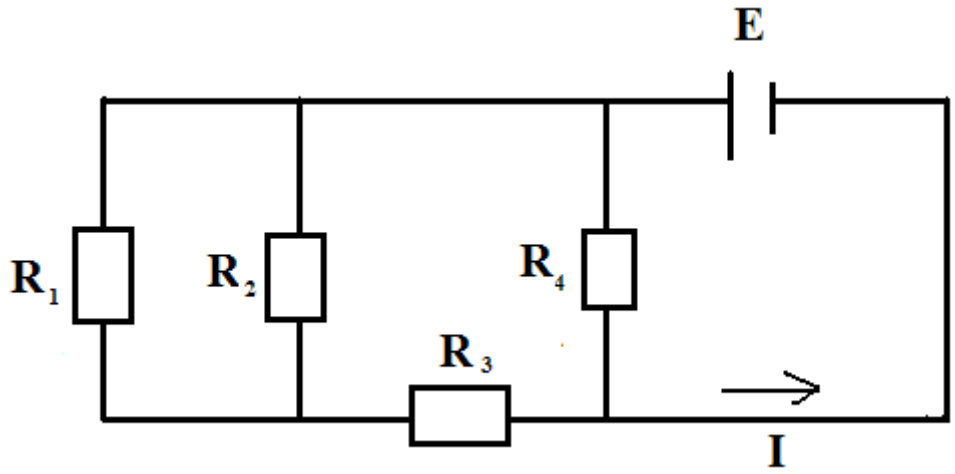
Yechish:

$$\frac{1}{R_{CD}}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R};\quad R_{CD}=\frac{R}{2};\quad \frac{1}{R_{EF}}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R}\qquad R_{EF}=\frac{R}{2}$$

$$R_{CF}=R_{CD}+R_{EF}=\frac{R}{2}+\frac{R}{2}=R;\quad \frac{1}{R_{AB}}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R}$$

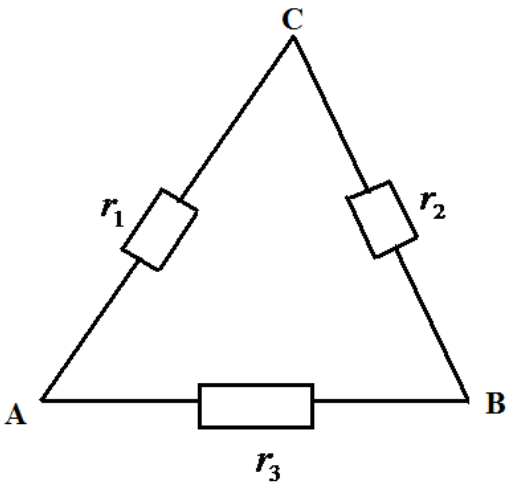
$$Javob:\quad R_{AB}=\frac{R}{2}$$

26. Quyidagi sxemadagi EYK i 12 V bo'lgan manbaning ichki qarshiligini toping. Barcha rezistorlarning qarshiligi 4 **Ω** dan, zanjirning tarmoqlanmagan qismidagi tokning kuchi 4 A ga teng.

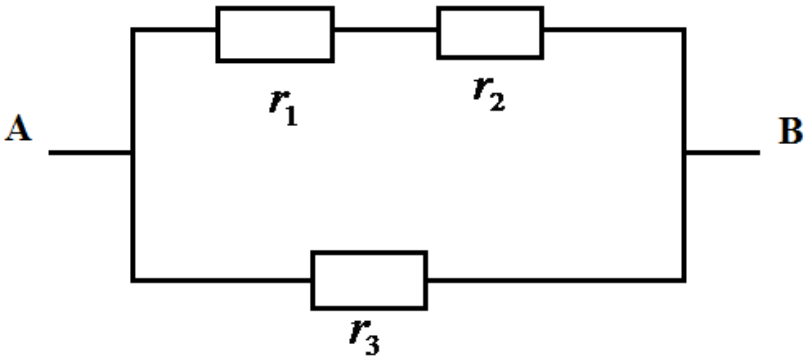


Berilgan:	Formula:	Yechilishi:
I=4A ε=12V R=4 **Ω**	$I=\frac{\varepsilon}{R_u+r}$ $IR_u+Ir=\varepsilon$ $r=\frac{\varepsilon}{I}-R_u$ $\frac{1}{R^*}=\frac{1}{R_1}+\frac{1}{R_2}$ $R^{**}=R^*+R_3$ $\frac{1}{R_u}=\frac{1}{R^{**}}+\frac{1}{R_4}$	$\frac{1}{R^*}=\frac{1}{4}+\frac{1}{4}=\frac{2}{4}=\frac{1}{2\Omega}$ $R^*=2\Omega$ $R^{**}=2\Omega+4\Omega=6\Omega$ $\frac{1}{R_u}=\frac{1}{6}+\frac{1}{4}=\frac{5}{12\Omega}$ $R_u=2{,}4\ \Omega$ $r=\frac{12V}{4A}-2{,}4\Omega=3\Omega-2{,}4\Omega=0{,}6\Omega$ $Javob:\quad r=0{,}6\Omega$
r-?		

27. Uchta qarshilik rasmda ko’rsatilgan sxema bo’yicha ulangan.Qarshiliklar zanjirning AB nuqtalariga ulangan bo’lsa,tutashmaning qarshiligini toping.(R_{AB}=?)

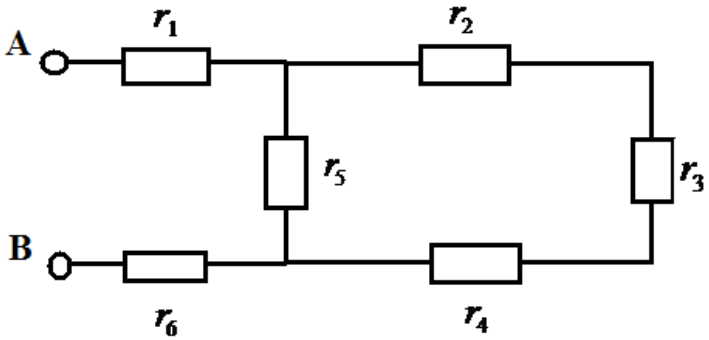


Bu zanjirni soddalashtiramiz:

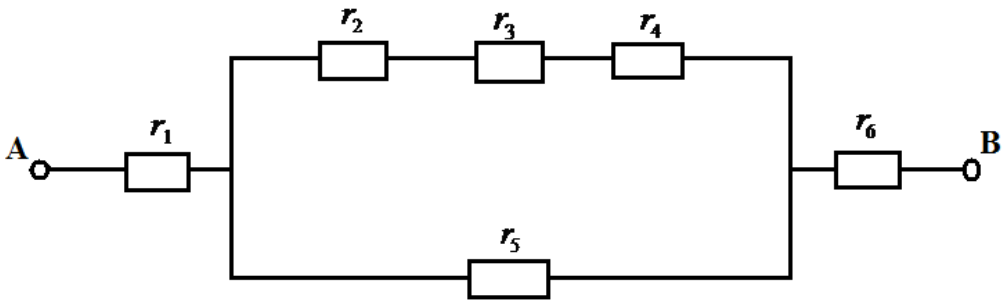


Berilgan:	Formula:	Yechilishi:
$r_1 = 20\,om$ $r_2 = 30\,om$ $r_3 = 50\,om$	*Qarshiliklarni parallel ulash*: $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_n}$ *Qarshiliklarni ketma – ket ulash*: $R_K = R_1 + R_2 + R_3 + \ldots R_n$	$R^* = r_1 + r_2 = 20 + 30 = 50\,om$ $\frac{1}{R_{AB}} = \frac{1}{R^*} + \frac{1}{r_3} = \frac{1}{50} + \frac{1}{50} = \frac{2}{50} = \frac{1}{25}$ *Javob*: $R_{AB} = 25\,om$
R_{AB}-?		

28. Rasmda ko’rsatilgan zanjirning umumiy qarshiligini hisoblang. (R_{AB}=?)

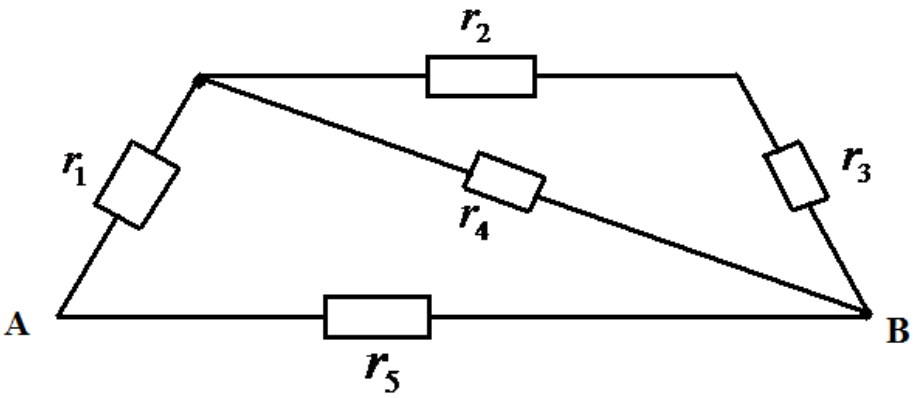


Bu zanjirni soddalashtiramiz:

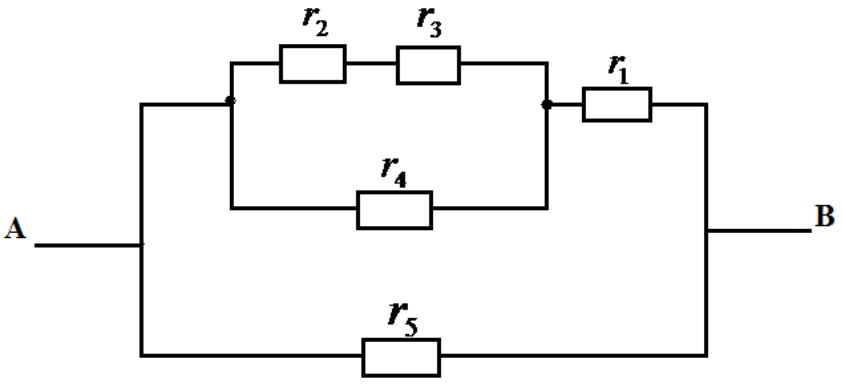


Berilgan:	Formula:	Yechilishi:
$r_1 = 4\Omega$ $r_2 = 5\Omega$ $r_3 = 5\Omega$ $r_4 = 5\Omega$ $r_5 = 10\Omega$ $r_6 = 10\Omega$	*Qarshiliklarni parallel ulash*: $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_n}$ *Qarshiliklarni ketma – ket ulash*: $R_K = R_1 + R_2 + R_3 + \ldots R_n$	$R^* = r_2 + r_3 + r_4 = 5+5+5 = 15\Omega$ $\frac{1}{R^{**}} = \frac{1}{R^*} + \frac{1}{r_5} = \frac{1}{15} + \frac{1}{10}$; $R^{**} = 6\Omega$ $R_{AB} = r_1 + R^{**} + r_6 = 6+4+10 =$ $= 20\Omega$ *Javob*: $R_{AB} = 20\Omega$
R_{AB}-?		

29. Rasmda ko’rsatilgan zanjirning umumiy qarshiligini hisoblang. (R_{AB}=?)

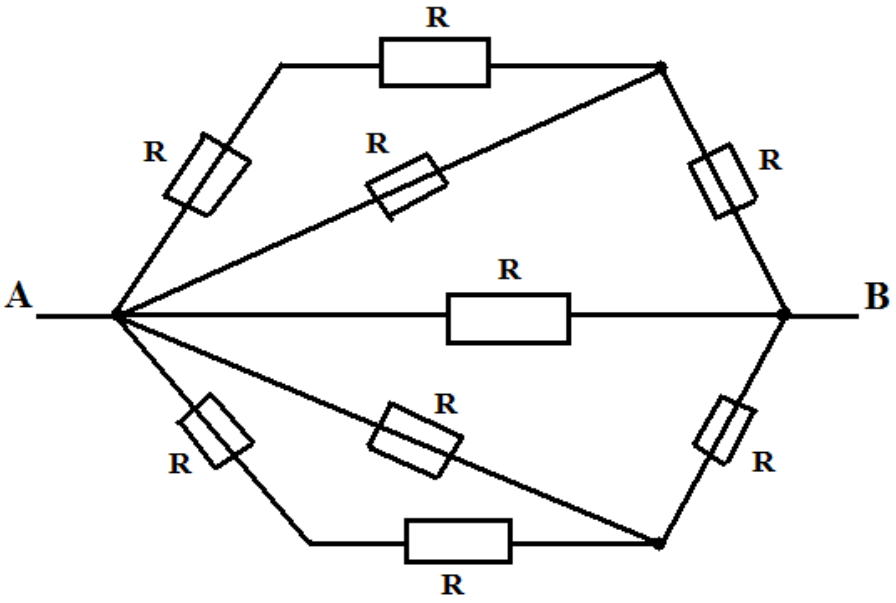


Bu zanjirni soddalashtiramiz :

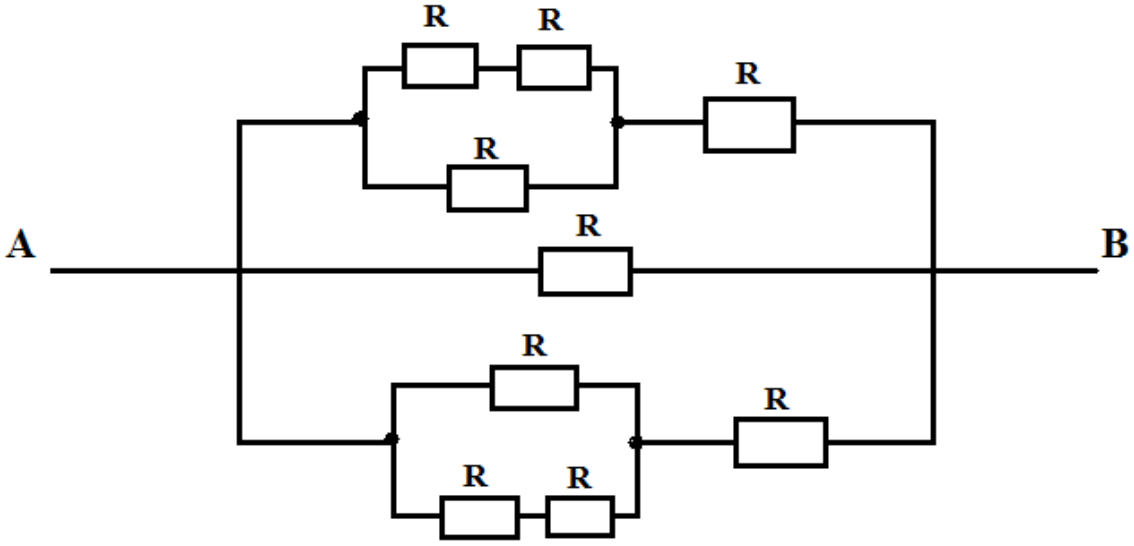


Berilgan:	Formula:	Yechilishi:
$r_1 = 2\Omega$ $r_2 = 3\Omega$ $r_3 = 5\Omega$ $r_4 = 8\Omega$ $r_5 = 6\Omega$ R_{AB}-?	*Qarshiliklarni parallel ulash*: $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_n}$ *Qarshiliklarni ketma – ket ulash*: $R_K = R_1 + R_2 + R_3 + \ldots R_n$	$R^* = r_2 + r_3 = 3 + 5 = 8\Omega$ $\frac{1}{R^{**}} = \frac{1}{R^*} + \frac{1}{r_4} = \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$; $R^{**} = 4\Omega$ $R^{***} = r_1 + R^{**} = 2 + 4 = 6\Omega$ $\frac{1}{R_{AB}} = \frac{1}{R^{***}} + \frac{1}{r_5}$ $R_{AB} = \frac{R^{***} \cdot r_5}{R^{***} + r_5} = \frac{6 \cdot 6}{6 + 6} = 3\ \Omega$ *Javob*: $R_{AB} = 3\Omega$

30.Agar har bir o’tkazgichning qarshiligi R bo’lsa. Zanjirning A va B nuqtalar orasidagi qismining umumiy qarshiligini toping.



Bu zanjirni soddalashtiramiz :



Yechilishi:

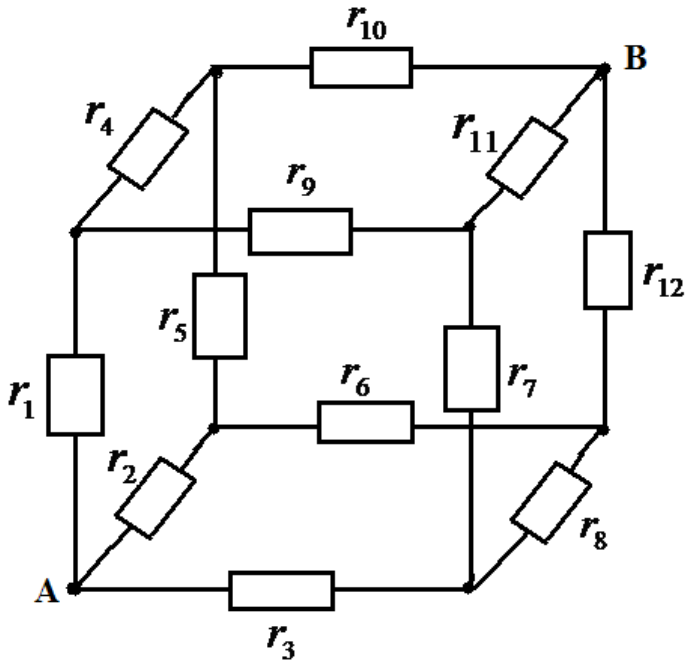
$$R_1 = R + R = 2R; \quad \frac{1}{R_2} = \frac{1}{R_1} + \frac{1}{R} \qquad R_2 = \frac{3}{2}R$$

$$R_3 = R_2 + R = \frac{3}{2}R + R = \frac{5}{2}R$$

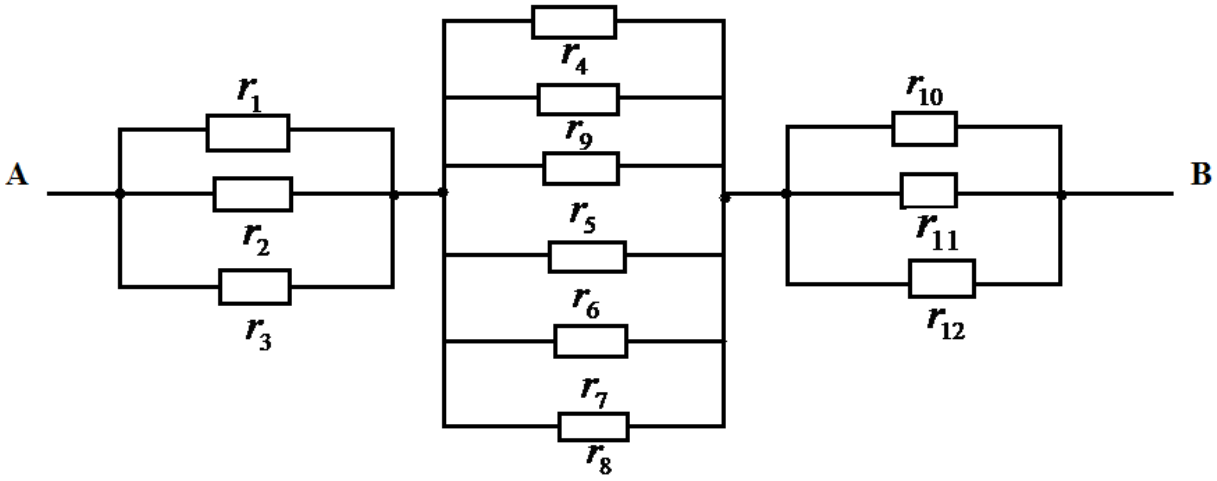
$$R_1 = R_4 \qquad R_2 = R_5 \qquad R_3 = R_6$$

$$\frac{1}{R_{AB}} = \frac{1}{R_3} + \frac{1}{R} + \frac{1}{R_6} \qquad R_{AB} = \frac{5}{11}R; \quad Javob: \quad R_{AB} = \frac{5}{11}R$$

31.Agar har bir o’tkazgichning qarshiligi R bo’lsa. Zanjirning A va B nuqtalar orasidagi qismining umumiy qarshiligini toping.



Bu zanjirni soddalashtiramiz :



Berilgan:

$$r_1 = r_2 = r_3 = r_4 = r_5 = r_6 =$$
$$= r_7 = r_8 = r_9 = r_{10} = r_{11} = r_{12} = R$$

Yechilishi:

$$\frac{1}{R_1} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}; \quad R_1 = \frac{R}{3}$$

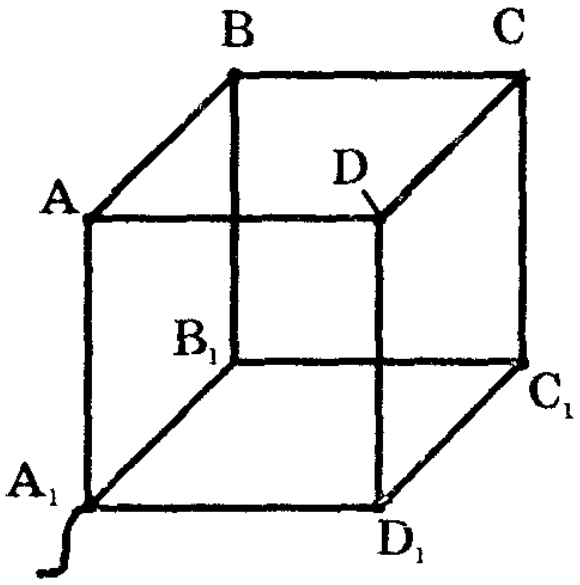
$$\frac{1}{R_2} = \frac{1}{r_4} + \frac{1}{r_9} + \frac{1}{r_5} + \frac{1}{r_6} + \frac{1}{r_7} + \frac{1}{r_8} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{6}{R}; \quad R_2 = \frac{R}{6}$$

$$\frac{1}{R_3} = \frac{1}{r_{10}} + \frac{1}{r_{11}} + \frac{1}{r_{12}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}; \quad R_3 = \frac{R}{3}$$

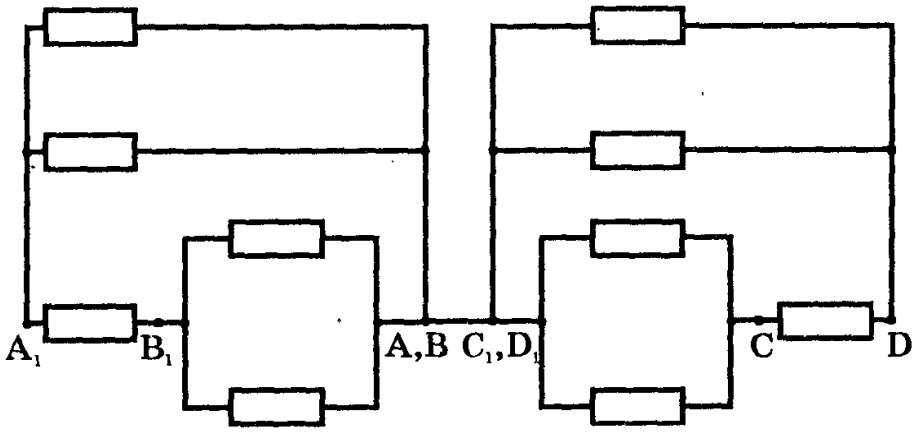
$$R_{AB} = R_1 + R_2 + R_3 = \frac{R}{3} + \frac{R}{6} + \frac{R}{3} = \frac{5R}{6}; \quad Javob: \quad R_{AB} = \frac{5}{6}R$$

32.Agar har bir o'tkazgichning qarshiligi R bo'lsa. Zanjirning A_1 va D nuqtalar orasidagi qismining umumiy qarshiligini toping.

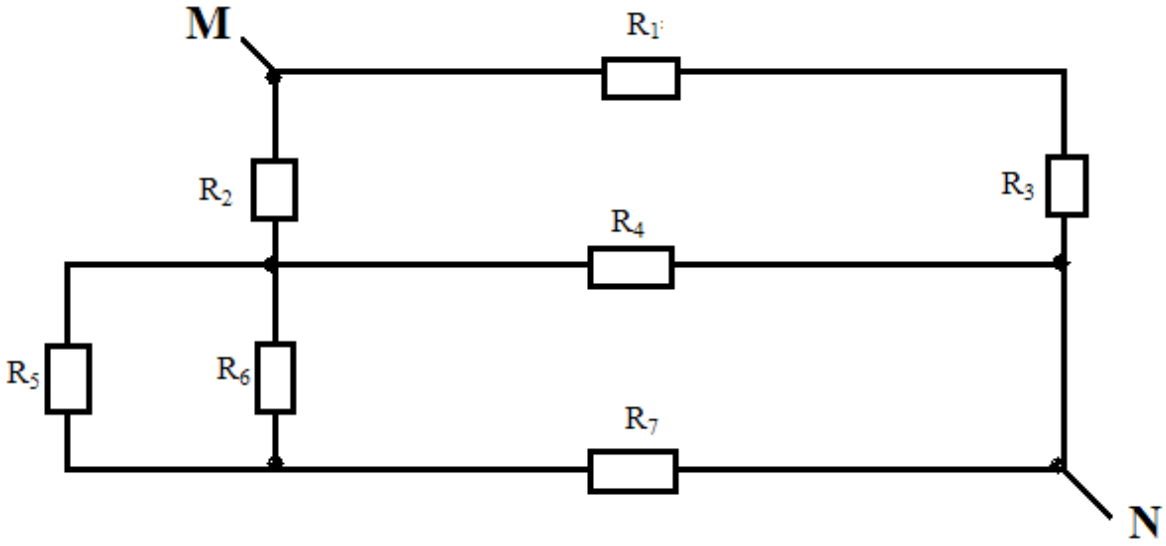
$$r_1 = r_2 = r_3 = r_4 = r_5 = r_6 =$$
$$= r_7 = r_8 = r_9 = r_{10} = r_{11} = r_{12} = R$$



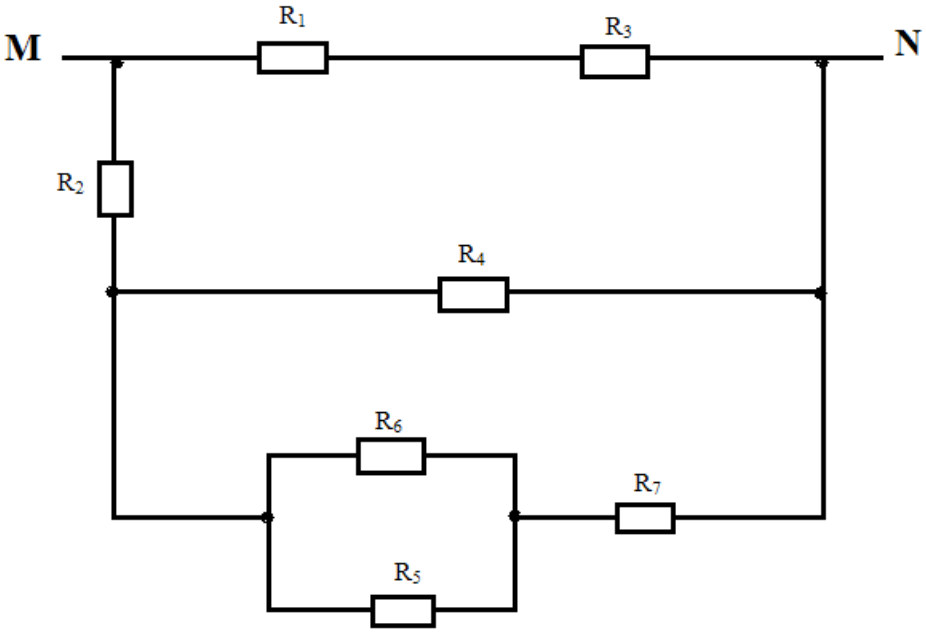
Bu zanjirni soddalashtiramiz:



33. Rasmda tasvirlangan zanjirning M va N nuqtalari orasidagi umumiy elektr qarshilikni aniqlang.

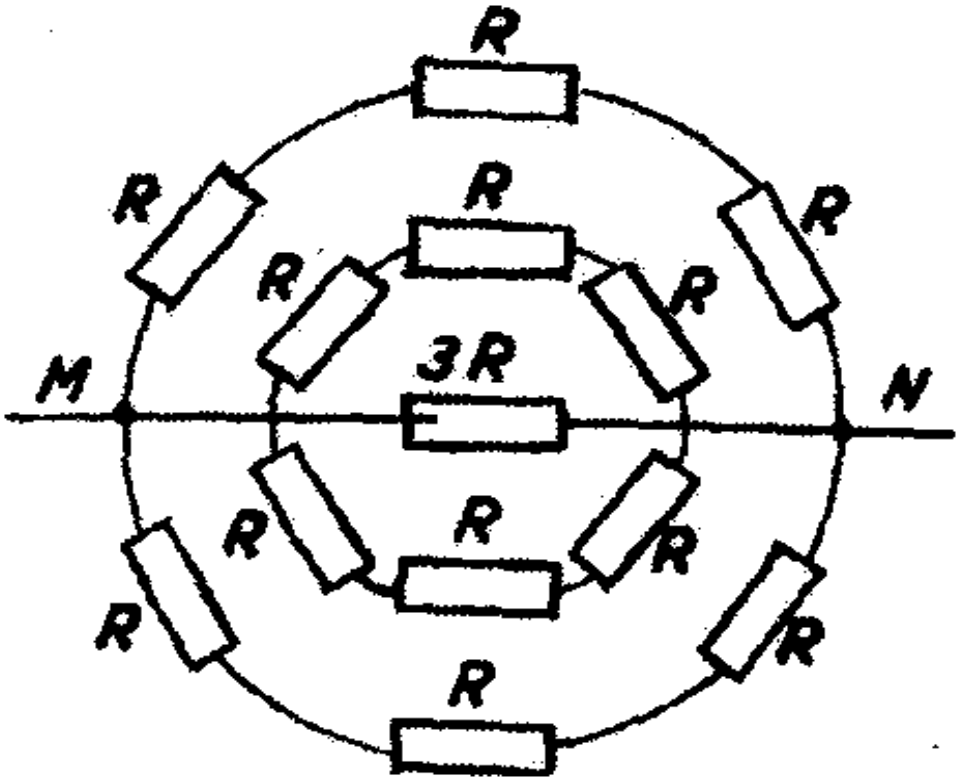


Bu zanjirni soddalashtiramiz:

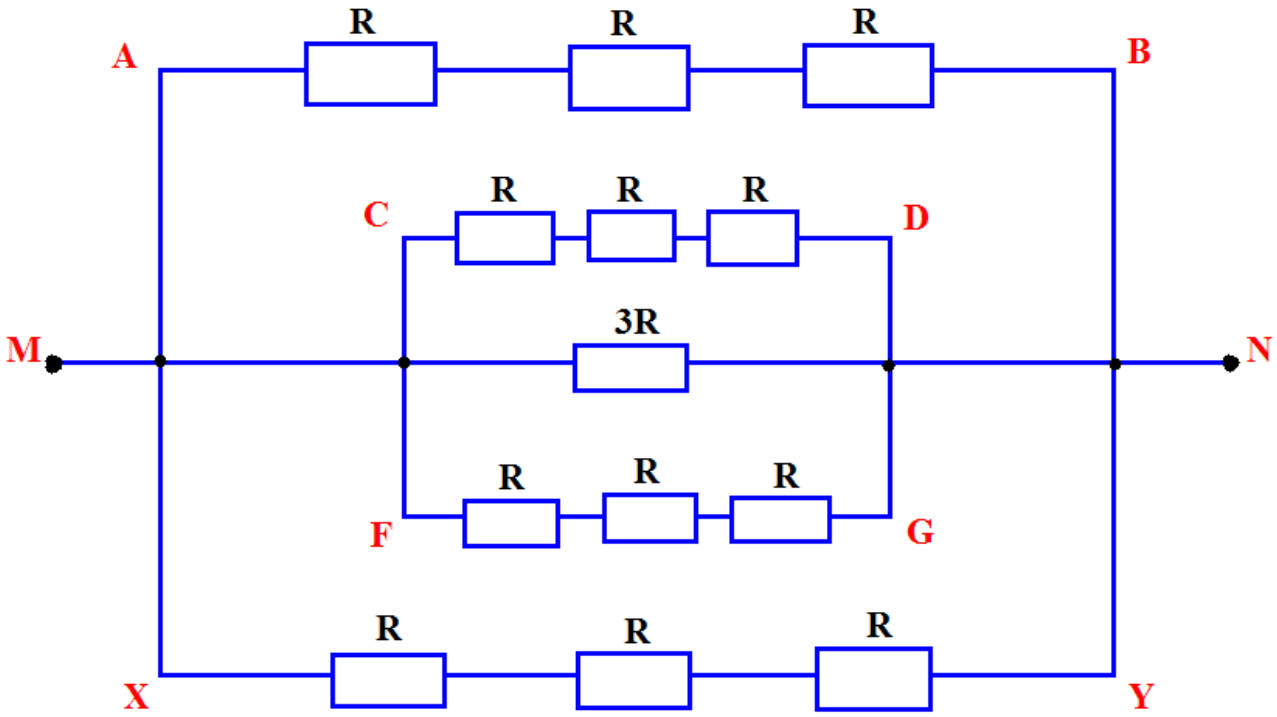


Berilgan:	Formula:	Yechilishi:
$R_1=R_2=R_3=$ $=R_5=R_6=R_7=2R;$ $R_4=6R$	*Qarshiliklarni parallel ulash*: $\frac{1}{R_P}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}+\ldots\frac{1}{R_n}$ *Qarshiliklarni ketma – ket ulash*: $R_K=R_1+R_2+R_3+\ldots R_n$	$\frac{1}{R_{56}}=\frac{1}{R_5}+\frac{1}{R_6}:\ R_{56}=R$ $R_{567}=R_{56}+R_7=R+2R=3R$ $\frac{1}{R_{5674}}=\frac{1}{R_{567}}+\frac{1}{R_4};\ R_{5674}=2R$ $R_{56742}=R_{5674}+R_2=4R$ $R_{13}=R_1+R_3=2R+2R=4R$ $\frac{1}{R_{MN}}=\frac{1}{R_{13}}+\frac{1}{R_{56742}}=\frac{1}{4R}+\frac{1}{4R}$ $R_{MN}=2R$ *Javob*: $R_{MN}=2R$
R_{MN}-?		

34.Rasmda tasvirlangan elektr zanjirining M va N nuqtalari orasidagi qarshilikni aniqlang?



Bu zanjirni soddalashtiramiz (hisoblash oson bo’lishi uchun o’zimizdan belgilash kiritamiz)



Yechilishi:

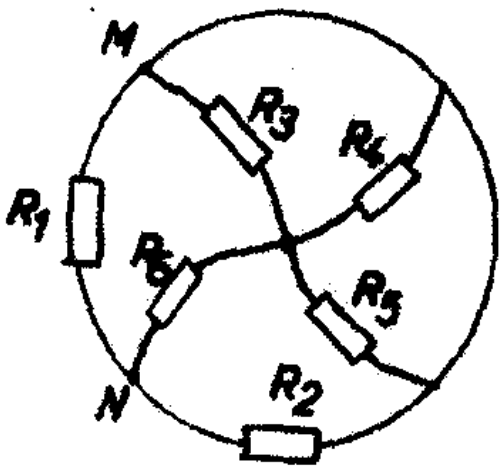
$$R_{AB} = R + R + R = 3R; \qquad R_{CD} = R + R + R = 3R$$

$$R_{FG} = R + R + R = 3R; \qquad R_{XY} = R + R + R = 3R$$

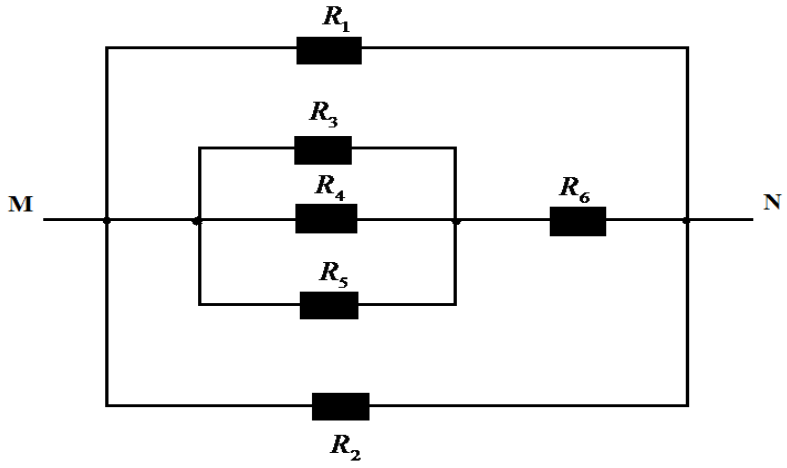
$$\frac{1}{R_{MN}} = \frac{1}{3R} + \frac{1}{3R} + \frac{1}{3R} + \frac{1}{3R} + \frac{1}{3R} = \frac{5}{3R}$$

$$R_{MN} = \frac{3}{5}R$$

35. Rasmda tasvirlangan elektr zanjirining M va N nuqtalari orasidagi qarshilikni aniqlang?



Bu zanjirni soddalashtiramiz



Berilgan:

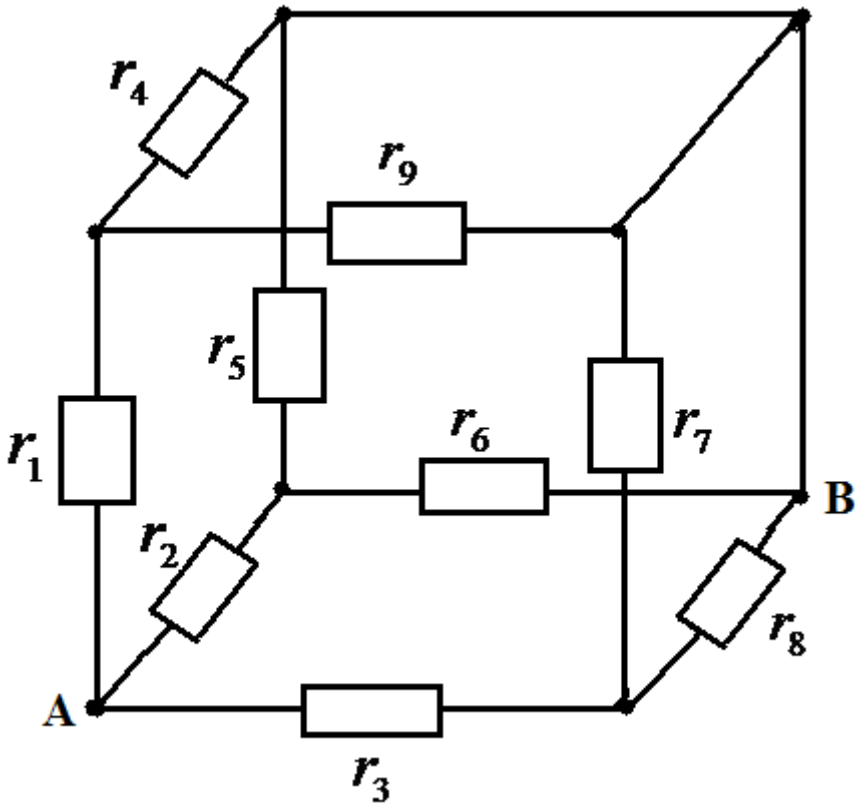
$$R_1 = R_3 = R_4 = R_5 = 3R;\ \ R_2 = 6R,\ R_6 = R$$

Yechilishi:

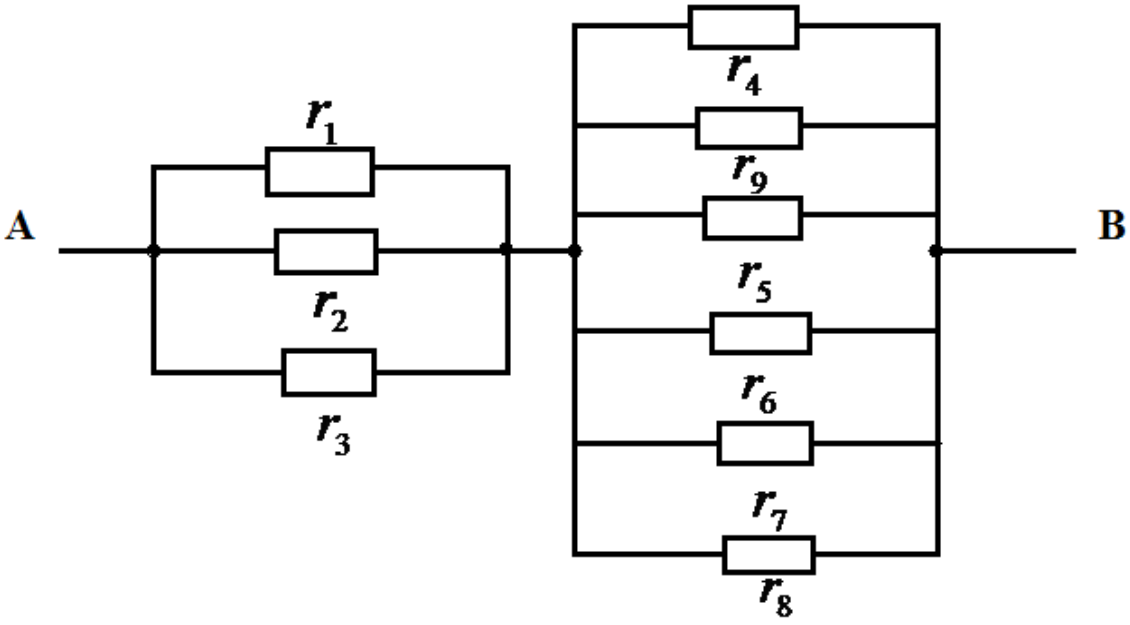
$$\frac{1}{R_{345}} = \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5}\ \ ;\ \ R_{345} = R;\ \ R_{3456} = R_{345} + R_6 = R + R = 2R$$

$$\frac{1}{R_{MN}} = \frac{1}{R_1} + \frac{1}{R_{3456}} + \frac{1}{R_2} = \frac{1}{3R} + \frac{1}{2R} + \frac{1}{6R};\ \ R_{MN} = R$$

36. Rasmda tasvirlangan elektr zanjirining A va B nuqtalari orasidagi qarshilikni aniqlang?



Bu zanjirni soddalashtiramiz:



Berilgan:

$$r_1 = r_2 = r_3 = r_4 = r_5 = r_6 = r_7 = r_8 = r_9 = R$$

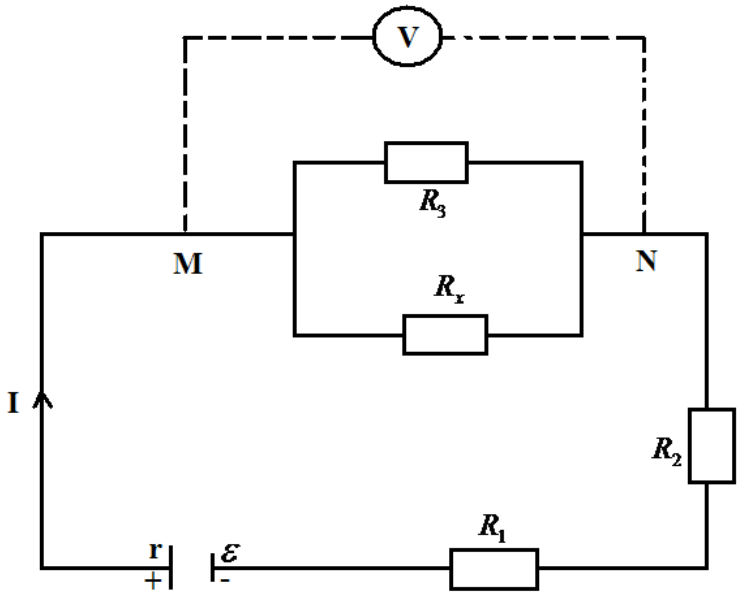
Yechilishi:

$$\frac{1}{R_1} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R};\ R_1 = \frac{R}{3}$$

$$\frac{1}{R_2} = \frac{1}{r_4} + \frac{1}{r_9} + \frac{1}{r_5} + \frac{1}{r_6} + \frac{1}{r_7} + \frac{1}{r_8} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{6}{R};\ R_2 = \frac{R}{6}$$

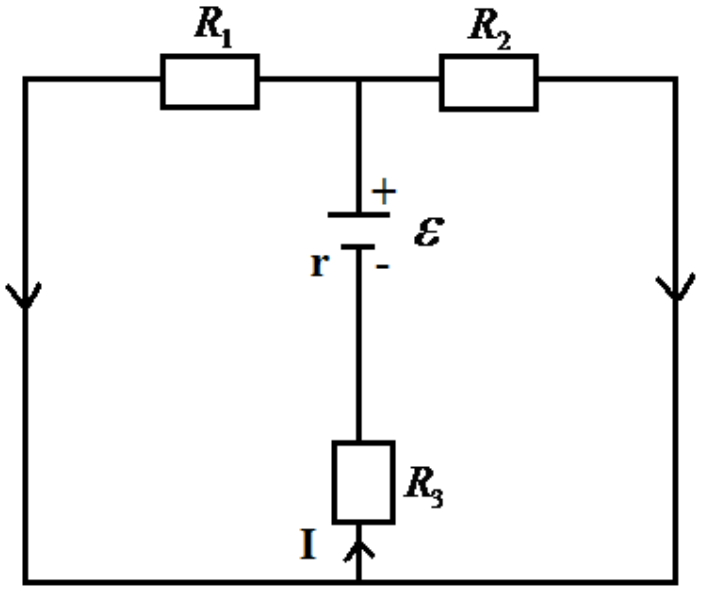
$$R_{AB} = R_1 + R_2 = \frac{R}{3} + \frac{R}{6} = \frac{3R}{6};\quad Javob:\ R_{AB} = \frac{1}{2}R$$

37. Elektr zanjir tarkibidagi R_x qarshilikni toping.



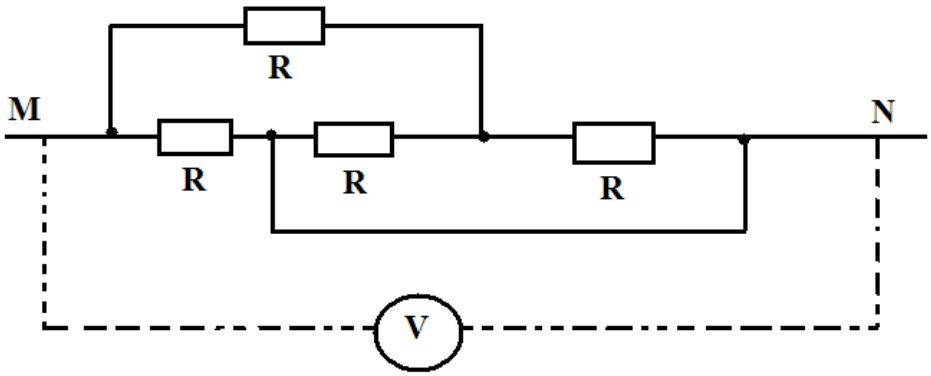
Berilgan:	Yechilishi:
$r = R_1 = 1\Omega;\quad R_2 = 2\Omega;$ $R_3 = 3\Omega$ $U_{MN} = 6V;\quad \varepsilon = 18V;$	*Zanjirning bir qismi uchun om qonuni va to'la zanjir uchun om qonunini tenglashtiramiz* $I = \frac{U}{R_{MN}}$ *va* $I = \frac{\varepsilon}{R_{umumiy} + r}$ $\frac{U}{R_{MN}} = \frac{\varepsilon}{R_{umumiy} + r};\quad R_{umumiy} = R_{MN} + R_1 + R_2$ $\frac{U}{R_{MN}} = \frac{\varepsilon}{R_{MN} + R_1 + R_2 + r};\ \frac{6}{R_{MN}} = \frac{18}{R_{MN} + 1 + 2 + 1};$ $18R_{MN} = 6R_{MN} + 24;\ 12R_{MN} = 24;\ R_{MN} = 2;\quad R_{MN} = \frac{R_3 \cdot R_x}{R_3 + R_x} = 2$ $3R_x = 6 + 2R_x;\quad R_x = 6\Omega$
R_x-?	

38. Elektr zanjiridagi tok kuchini aniqlang.

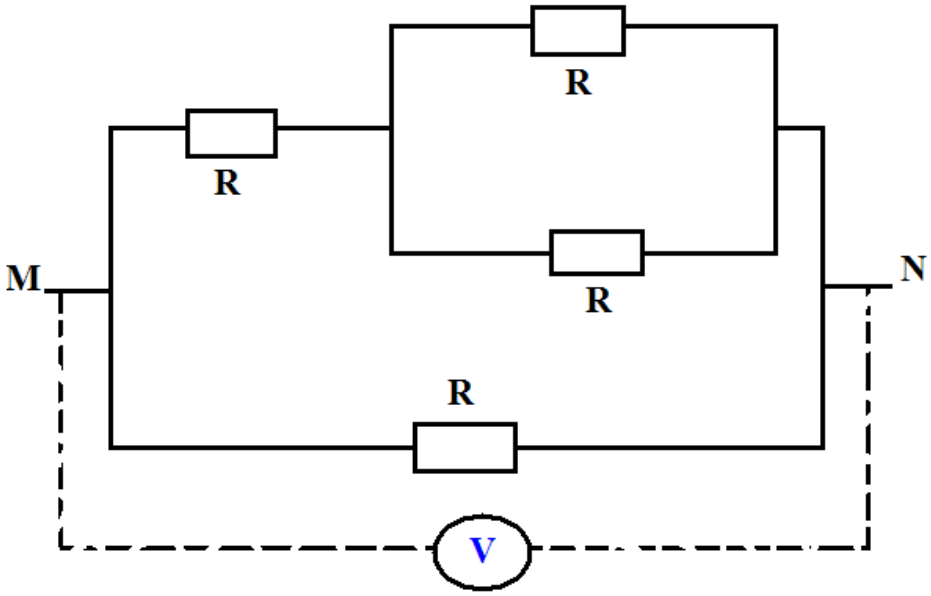


Berilgan:	Formula:	Yechilishi:
$R_1 = R_2 = 4\Omega;\ R_3 = 5\Omega$ $r = 1\Omega;\quad \varepsilon = 16V;$	*Qarshiliklarni parallel ulash:* $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_n}$ *Qarshiliklarni ketma – ket ulash:* $R_K = R_1 + R_2 + R_3 + \ldots R_n$	$I = \frac{\varepsilon}{R_{umum} + r}$; $\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2};\quad R_{12} = 2\Omega$ $R_{umum} = R_{12} + R_3 = 2 + 5 = 7\Omega$ $I = \frac{16}{7+1} = \frac{16}{8} = 2\,A$
I-?		

39. Elektr zanjiridagi tok kuchini aniqlang.

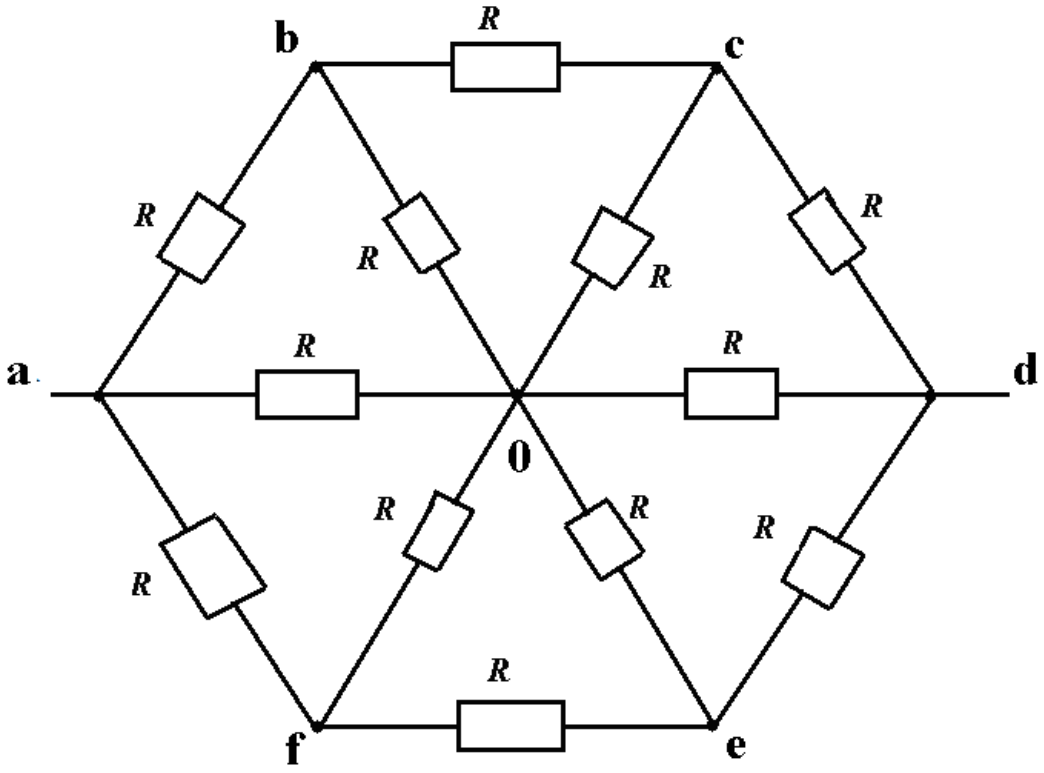


Bu zanjirni soddalashtiramiz :

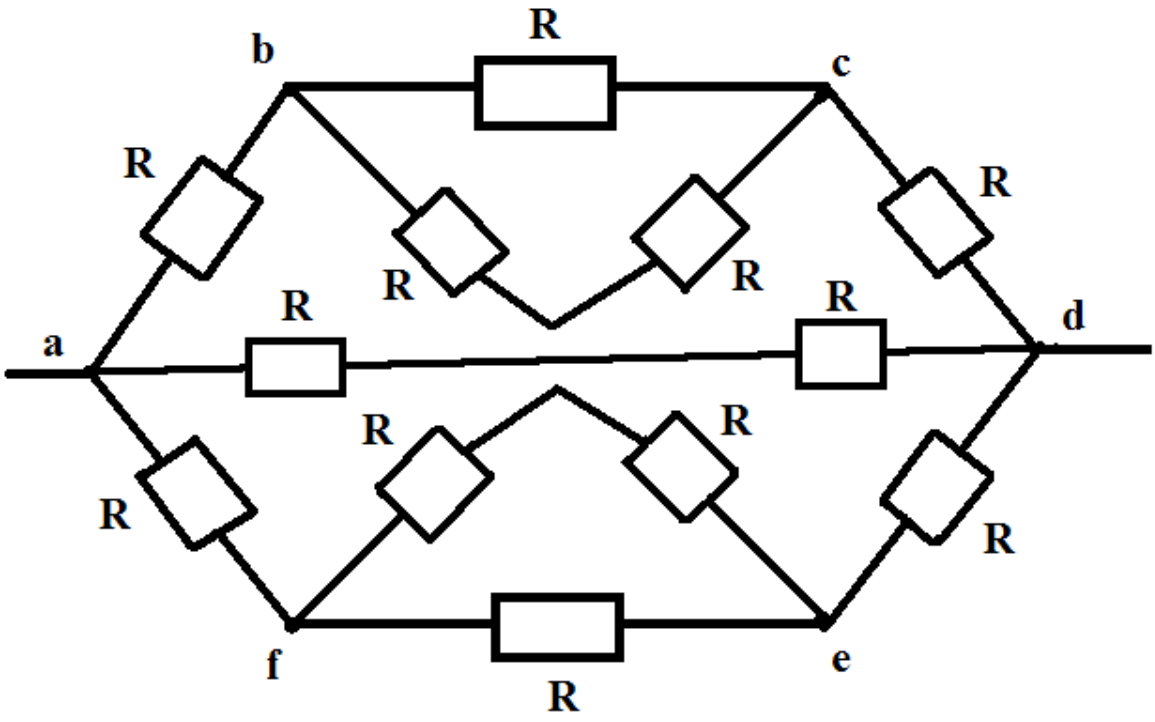


Berilgan:	Formula:	Yechilishi:
$R=1\Omega;\ U=3V$	*Qarshiliklarni parallel ulash*: $\frac{1}{R_P}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}+\ldots\frac{1}{R_n}$	$I=\frac{U}{R_{umum}}$; $\frac{1}{R_{12}}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R};\quad R_{12}=0{,}5\Omega$
I-?	*Qarshiliklarni ketma – ket ulash*: $R_K=R_1+R_2+R_3+\ldots R_n$	$R_{123}=R_{12}+R_3=0{,}5+1=1{,}5\Omega$ $\frac{1}{R_{MN}}=\frac{1}{R_{123}}+\frac{1}{R_4}=\frac{1}{1{,}5}+\frac{1}{0{,}5}$; $R_{MN}=\frac{1{,}5}{4}\Omega$ $I=\frac{3}{\frac{1{,}5}{4}}=\frac{12}{1{,}5}=8A$

40.Rasmda tasvirlangan zanjir R bo’lgan qarshilikdan tuzilgan zanjirning AB nuqta orasidagi qarshiligini hisoblab toping.

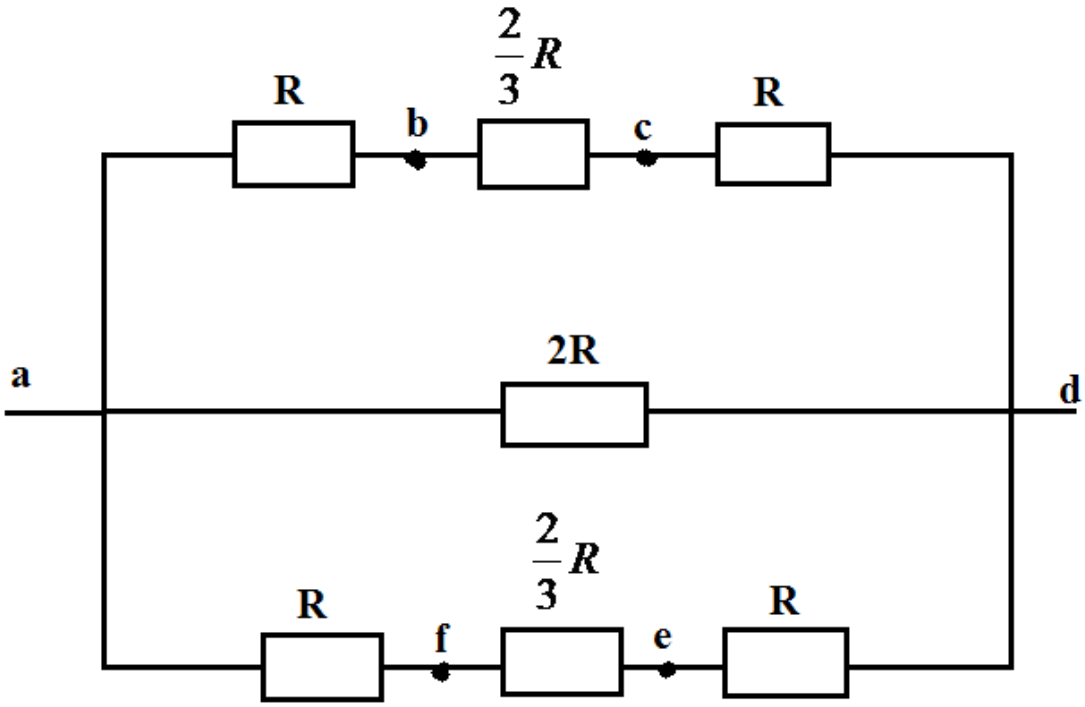


Barcha qarshiliklar bir xil bo’lganligi uchun o nuqtani uzamiz va quyidagicha chizmaga ega bo’lamiz.



$$\frac{1}{R_{bc}} = \frac{1}{R} + \frac{1}{2R} = \frac{3}{2R}; \qquad R_{bc} = \frac{2}{3}R$$

$$\frac{1}{R_{fe}} = \frac{1}{R} + \frac{1}{2R} = \frac{3}{2R}; \qquad R_{fe} = \frac{2}{3}R$$



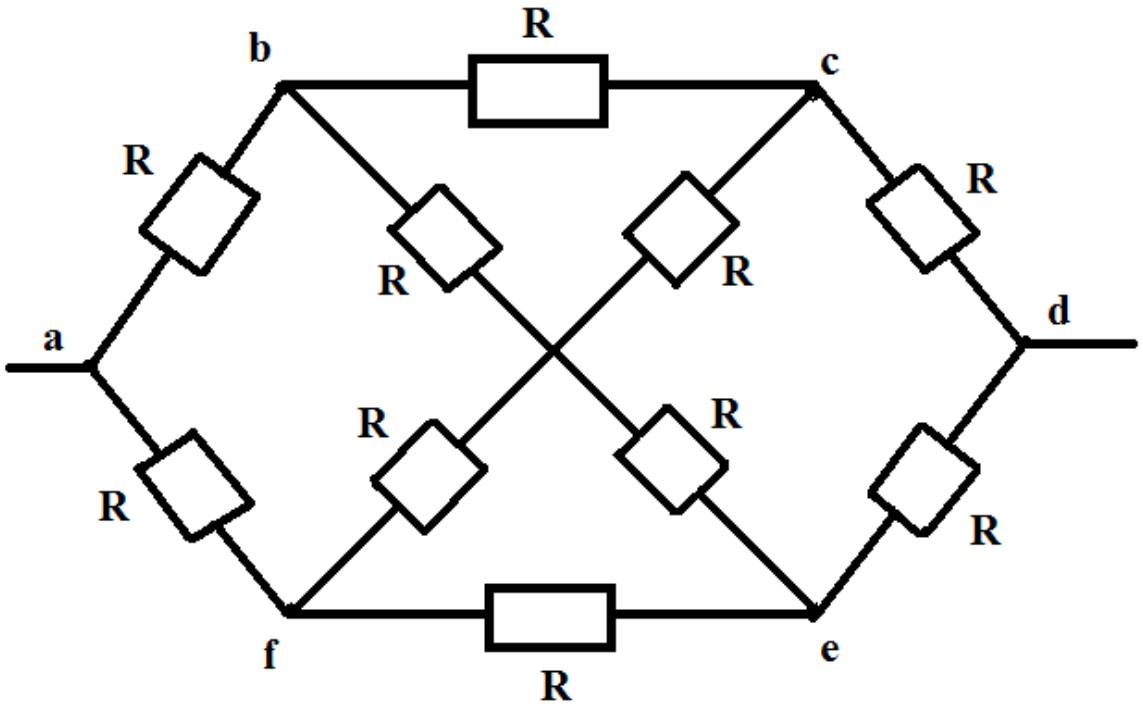
$$R_{abcd} = R + \frac{2}{3}R + R = \frac{8}{3}R$$

$$R_{afed} = R + \frac{2}{3}R + R = \frac{8}{3}R$$

$$\frac{1}{R_{ad}} = \frac{1}{R_{abcd}} + \frac{1}{R_{afed}} + \frac{1}{2R} = \frac{3}{8R} + \frac{3}{8R} + \frac{1}{2R} = \frac{10}{8R}$$

$$R_{ad} = \frac{8R}{10} = \frac{4R}{5}$$

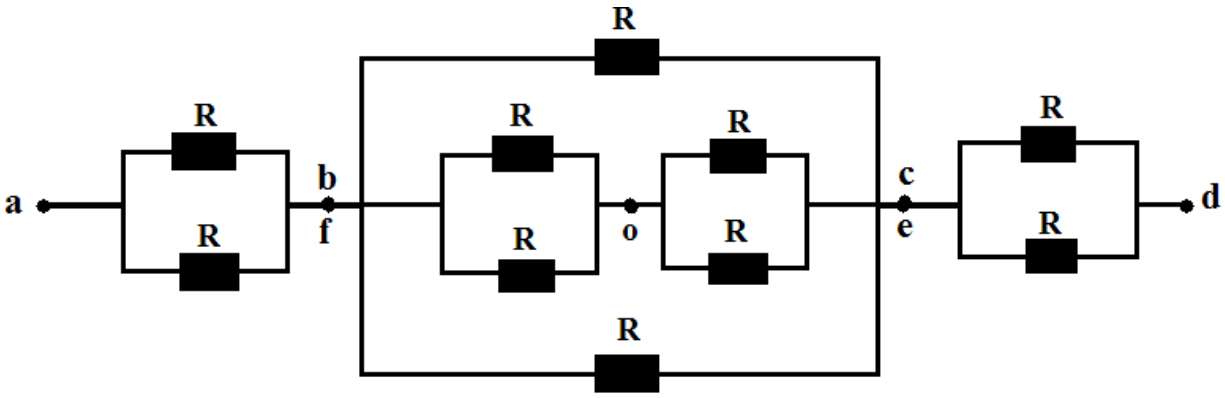
41.Rasmda tasvirlangan zanjir R bo’lgan qarshilikdan tuzilgan zanjir qismining umumiy qarshiligini hisoblab toping.



1-USUL

Zanjirning bf va ce nuqtalarini tutashtiramiz.

Bu zanjirni tushunarli bo’lishi uchun soddalashtiramiz:

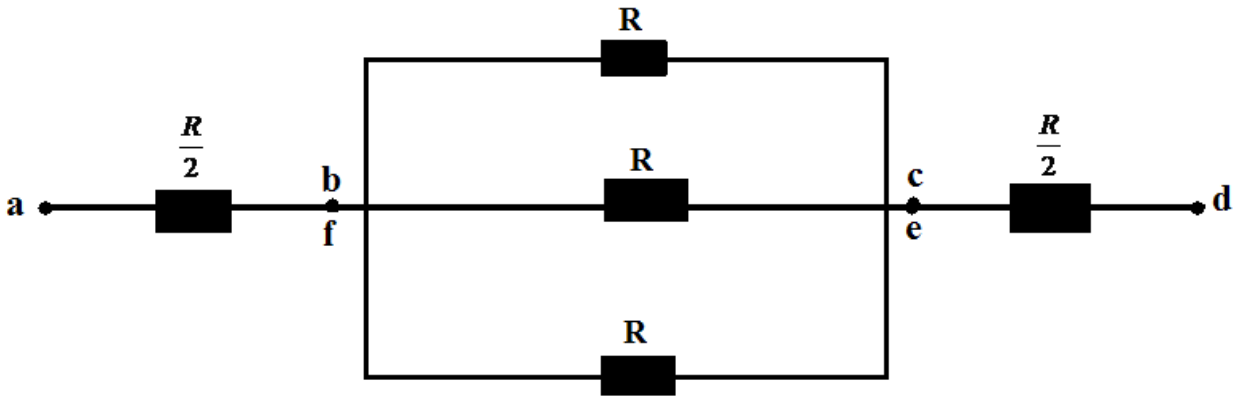


Berilganlardan foydalanib qarshiliklarni parallel ulash formulasidan foydalanib hisoblaymiz

$$\frac{1}{R_{ab}}=\frac{1}{R}+\frac{1}{R};\quad R_{ab}=\frac{R}{2};\quad \frac{1}{R_{bo}}=\frac{1}{R}+\frac{1}{R};\quad R_{bo}=\frac{R}{2}$$

$$\frac{1}{R_{oc}}=\frac{1}{R}+\frac{1}{R};\quad R_{oc}=\frac{R}{2};\quad R_{bc}=R_{bo}+R_{oc}=\frac{R}{2}+\frac{R}{2}=R$$

$$\frac{1}{R_{cd}}=\frac{1}{R}+\frac{1}{R};\quad R_{cd}=\frac{R}{2}$$

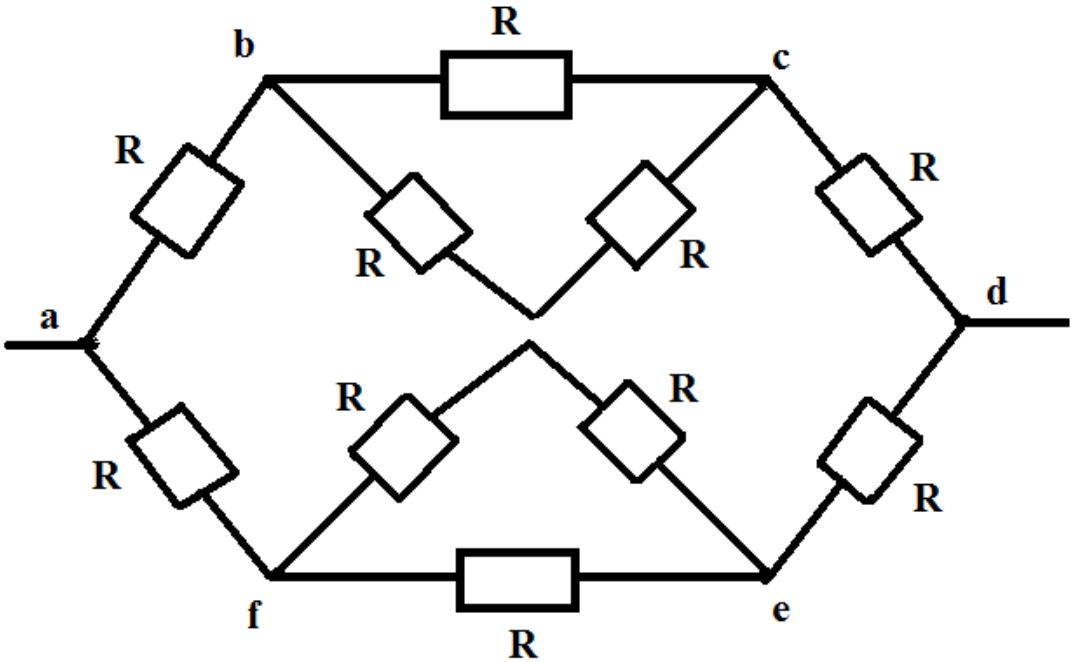


$$\frac{1}{R_{bc}}=\frac{1}{R}+\frac{1}{R}+\frac{1}{R};\quad R_{bc}=\frac{R}{3}$$

$$R_{ad}=\frac{R}{2}+\frac{R}{3}+\frac{R}{2}=\frac{4R}{3}$$

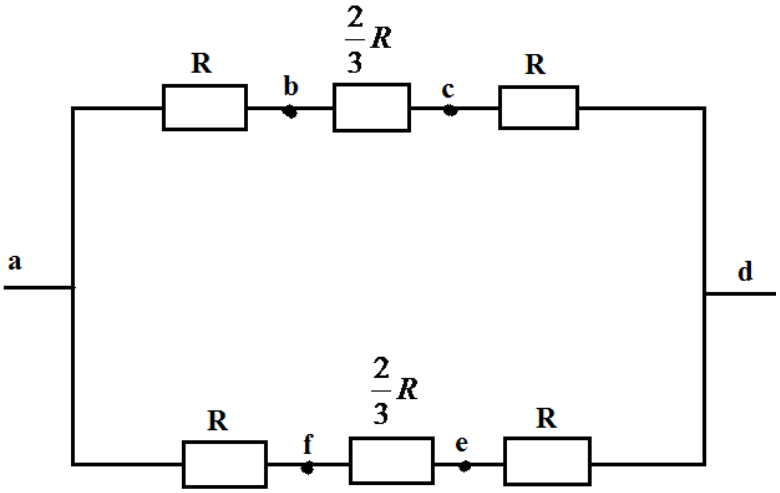
2-SUL

Barcha qarshiliklar bir xil bo'lganligi uchun o nuqtani uzamiz va quyidagicha chizmaga ega bo'lamiz.



$$\frac{1}{R_{bc}}=\frac{1}{R}+\frac{1}{2R}=\frac{3}{2R};\qquad R_{bc}=\frac{2}{3}R$$

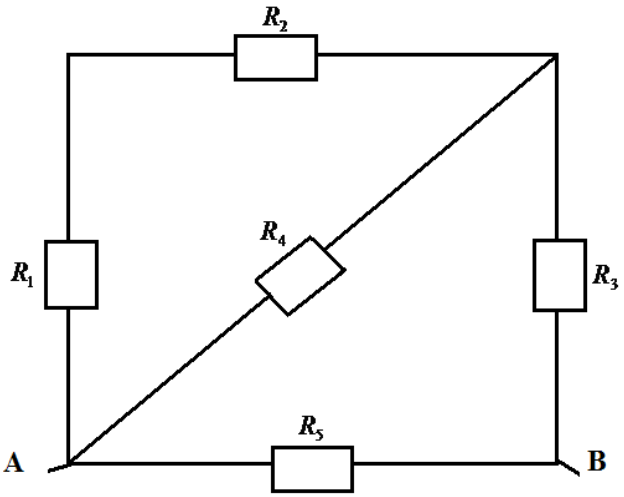
$$\frac{1}{R_{fe}}=\frac{1}{R}+\frac{1}{2R}=\frac{3}{2R};\qquad R_{fe}=\frac{2}{3}R$$



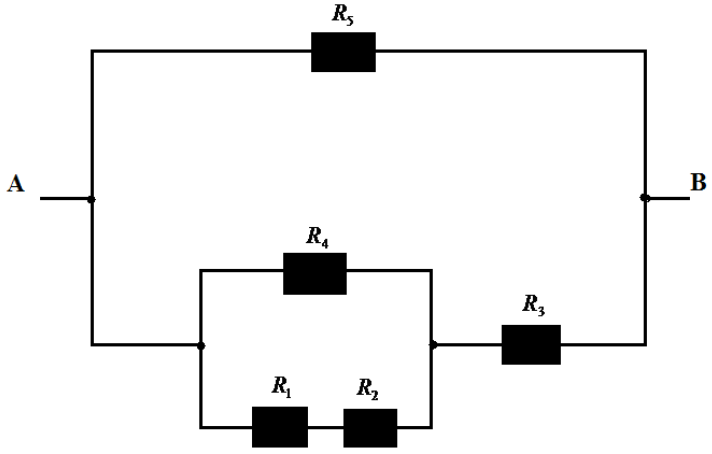
$$R_{abcd} = R + \frac{2}{3}R + R = \frac{8}{3}R: \qquad R_{afed} = R + \frac{2}{3}R + R = \frac{8}{3}R$$

$$\frac{1}{R_{ad}} = \frac{1}{R_{abcd}} + \frac{1}{R_{afed}} = \frac{3}{8R} + \frac{3}{8R} = \frac{6}{8R}: \qquad R_{ad} = \frac{8R}{6} = \frac{4R}{3}$$

42. Rasmda tasvirlangan qarshiliklari R ga teng. Zanjir umumiy qarshiligini hisoblab toping. $R_1 = R_2 = R_3 = R_4 = R_5 = R$



Bu zanjirni soddalashtiramiz



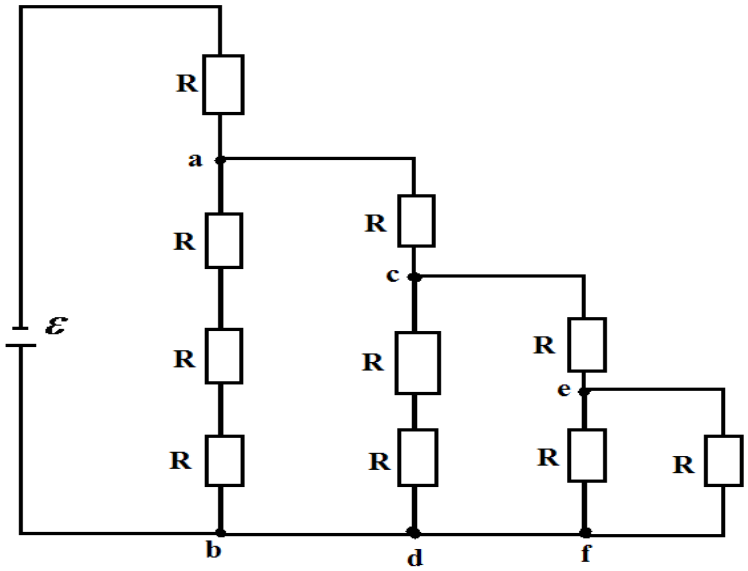
$$R_{12} = R_1 + R_2 = R + R = 2R$$

$$\frac{1}{R_{124}} = \frac{1}{R_{12}} + \frac{1}{R_4} = \frac{1}{2R} + \frac{1}{R} = \frac{3}{2R}; \quad R_{124} = \frac{2}{3}R$$

$$R_{1243} = R_{124} + R_3 = \frac{2}{3}R + R = \frac{5}{3}R$$

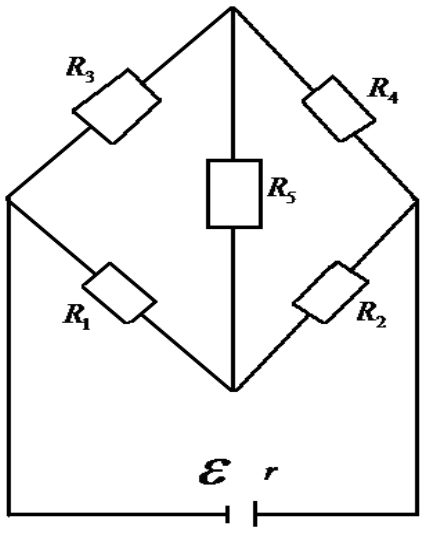
$$\frac{1}{R_{AB}} = \frac{1}{R_{1243}} + \frac{1}{R_5} = \frac{3}{5R} + \frac{1}{R} = \frac{8}{5R}; \quad R_{AB} = \frac{5}{8}R$$

43.Rasmda tasvirlangan sxemadagi tok manbai orqali o'tuvchi I tokni aniqlang. Sxemaning hamma qarshiligi birday va $R = 68\Omega$ ga teng. Manbaning E.Yu.K esa $\varepsilon = 14V$ .Manbaning ichki qarshiligini hisobga olmang.



Berilgan:	Formula:	Yechilishi:
$R = 68\Omega;$ $\varepsilon = 14v;$ $r = 0$ I-?	*To'liq zanjir uchun Om qonunu formulasidan foydalanamiz* $I = \frac{\varepsilon}{R+r};$ $r = 0$ *bo'lganligi uchun* $I = \frac{\varepsilon}{R};$	$\frac{1}{R_{ef}} = \frac{1}{R} + \frac{1}{R}; \quad R_{ef} = \frac{R}{2}$ $R_{efc} = \frac{R}{2} + R = \frac{3}{2}R; \quad R_{cd} = R + R = 2R$ $\frac{1}{R_{fecd}} = \frac{2}{3R} + \frac{1}{2r}; \; R_{fecd} = \frac{6}{7}R$ $R_{fecda} = R_{fecd} + R = \frac{6}{7}R + R = \frac{13}{7}R$ $R_{ba} = R + R + R = 3R$ $\frac{1}{R_{fecdab}} = \frac{1}{R_{fecda}} + \frac{1}{R_{ba}} = \frac{7}{13R} + \frac{1}{3R}; \; R_{fecdab} = \frac{39}{34}R$ $R_{Umumiy} = R_{fecdab} + R = \frac{39}{34}R + R = \frac{73}{34}R$ $R_{Umumiy} = \frac{73}{34}R = \frac{73}{34} \cdot 68 = 146\Omega; \quad I = \frac{\varepsilon}{R} = \frac{14}{146} \approx 0{,}1A$

44.Rasmda tasvirlangan sxema qanday quvvat istemol qiladi.
Elementning E.Yu.K $\varepsilon = 12V$, ichki qarshiligi
$r = 0{,}4\Omega;\ R_1 = 2\Omega;\ R_2 = 4\Omega;\ R_3 = 3\Omega;\ R_4 = 6\Omega;\ R_5 = 10\Omega;$



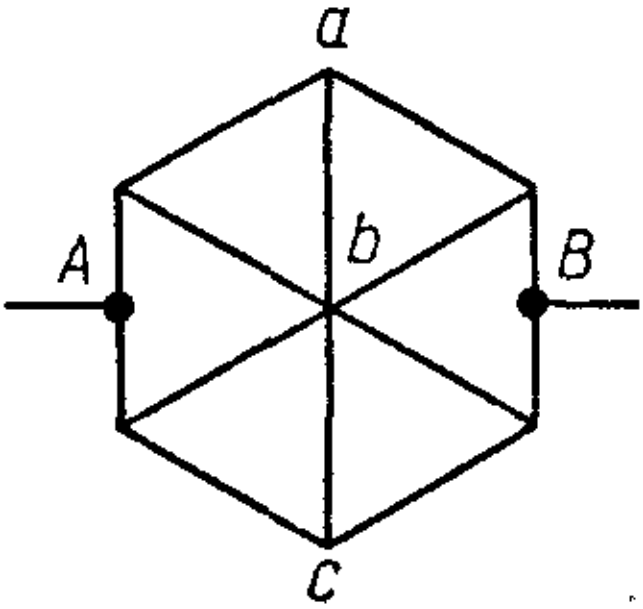
$R_1/R_3 = R_2/R_4$ shartdan R_5 qarshilikdan tok o'tmaganligi kelib chiqadi.Shuning uchun zanjirning to'la R qarshiligi R_1, R_2 va R_3, R_4 qarshiliklardan tuzilgan ikki parallel zanjir qarshiligi va elementning r ichki qarshiligi yig'indisidan iborat bo'ladi.

$$R_{34} = R_3 + R_4 = 3 + 6 = 9\Omega; \qquad R_{12} = R_1 + R_2 = 2 + 4 = 6\Omega$$

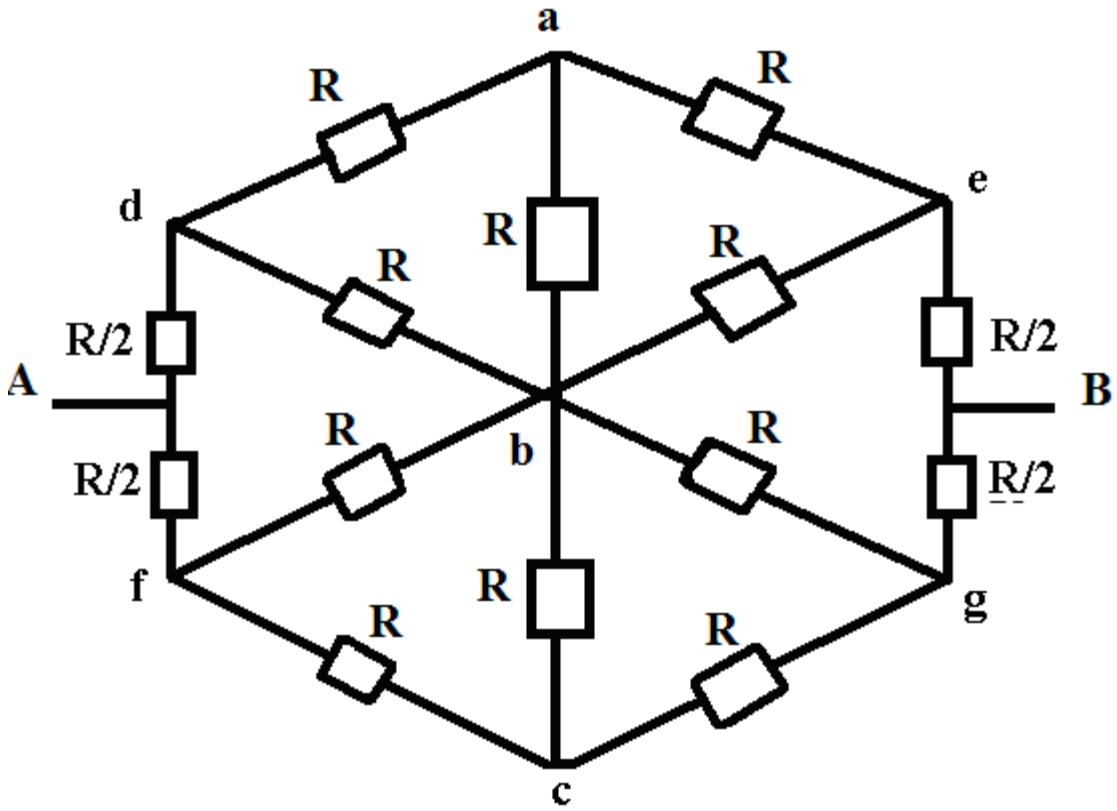
$$\frac{1}{R} = \frac{1}{R_{34}} + \frac{1}{R_{12}} = \frac{1}{9} + \frac{1}{6} = \frac{2}{18} + \frac{3}{18} = \frac{5}{18}; \quad R = \frac{18}{5} = 3{,}6\Omega$$

$$N = I \cdot \varepsilon = \frac{\varepsilon}{R+r} \cdot \varepsilon = \frac{\varepsilon^2}{R+r} = \frac{144}{3{,}6+0{,}4} = \frac{144}{4} = 36W$$

45.Rasmda tasvirlangan sxemaning har bir qismining qarshiligi R bo'lgan qarshilikdan tuzilgan zanjir qismining umumiy qarshiligini hisoblab toping.



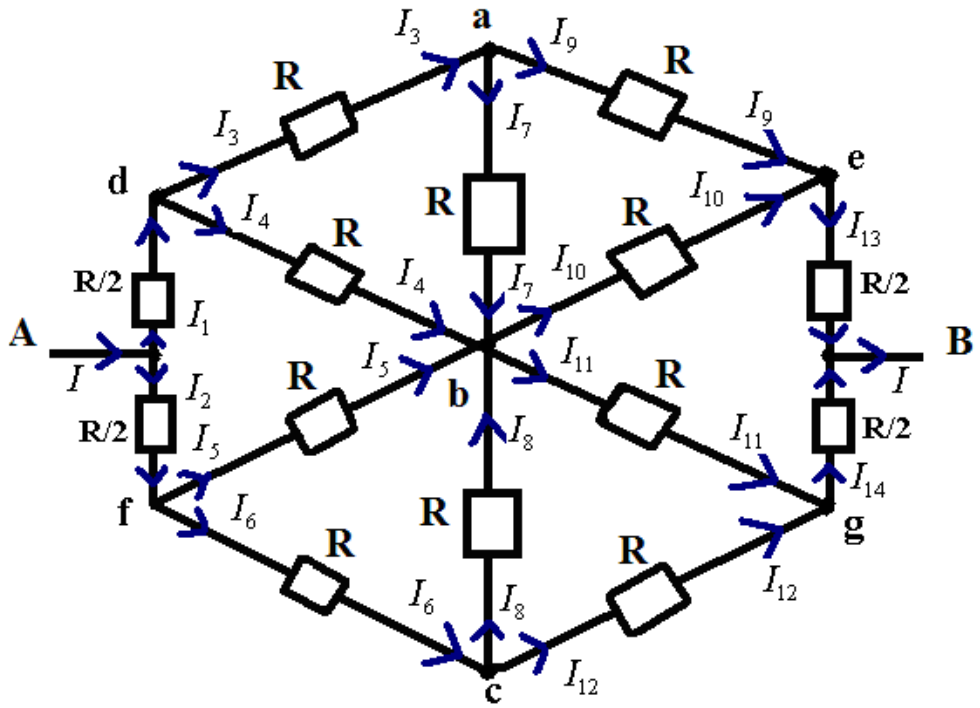
Sxemadagi qarshiliklarni tushunarli bo’lishi uchun uni quyidagicha tasvirlaymiz



Sxemaning qarshiliklari bir-xil bo’lganligi uchun a ,b,c nuqtalarda potensial bir xil $\varphi_a = \varphi_b = \varphi_c$ bo’lganligi uchun potensiallar farqi $\varphi_a - \varphi_b = \varphi_c - \varphi_b = 0$ ga teng bo’ladi va ab va bc qismdan tok o’tmaydi.

ISBOT

Har bir qarshilikdan oqayotgan toklarni taqsimlab chiqamiz



Kirxgofning 1-qoidasiga ko’ra tugunlarga kiruvchi va chiquvchi toklar tengligidan quyidagi natijaga ega bo’lamiz.

$$I = I_1 + I_2 \qquad I_3 = I_7 + I_9$$

$$I_1 = I_3 + I_4 \qquad I_6 = I_8 + I_{12}$$

$$I_2 = I_5 + I_6 \qquad I_{13} = I_{10} + I_9$$

$$I_4 + I_5 + I_7 + I_8 = I_{10} + I_{11} \qquad I_{14} = I_{11} + I_{12}$$

Bu natijalardan foydalanib rezistorlarni parallel ulangan qismlaridagi toklar tengligidan quyidagilarni hisoblaymiz.

Ad va Af rezistorlar paralelligidan $U_1 = U_2$ va Krixgov 1-qoidasidan chiqqan natijalardan toklarni topamiz

$$U_1 = U_2$$

$$I_1 \cdot \frac{R}{2} = I_2 \cdot \frac{R}{2}$$

$$I_1 = I_2 = \frac{I}{2}$$

Be va Bg rezistorlar paralelligidan $U_{13} = U_{14}$ va Krixgov 1-qoidasidan chiqqan natijalardan toklarni topamiz

$$U_{13} = U_{14}$$

$$I_{13} \cdot \frac{R}{2} = I_{14} \cdot \frac{R}{2}$$

$$I_{13} = I_{14} = \frac{I}{2}$$

da va db rezistorlar paralelligidan $U_3 = U_4$ va Krixgov 1-qoidasidan chiqqan natijalardan toklarni topamiz

$$U_3 = U_4$$

$$I_3 \cdot R = I_4 \cdot R$$

$$I_3 = I_4 = \frac{I_1}{2} = \frac{I}{4}$$

fb va fc rezistorlar paralelligidan $U_5 = U_6$ va Krixgov 1-qoidasidan chiqqan natijalardan toklarni topamiz

$$U_5 = U_6$$

$$I_5 \cdot R = I_6 \cdot R$$

$$I_5 = I_6 = \frac{I_2}{2} = \frac{I}{4}$$

ae va be rezistorlar paralelligidan $U_9 = U_{10}$ va Krixgov 1-qoidasidan chiqqan natijalardan toklarni topamiz

$$U_9 = U_{10}$$

$$I_9 \cdot R = I_{10} \cdot R$$

$$I_9 = I_{10} = \frac{I_{13}}{2} = \frac{I}{4}$$

gb va gc rezistorlar paralelligidan $U_{11} = U_{12}$ va Krixgov 1-qoidasidan chiqqan natijalardan toklarni topamiz

$$U_{11} = U_{12}$$

$$I_{11} \cdot R = I_{12} \cdot R$$

$$I_{11} = I_{12} = \frac{I_{14}}{2} = \frac{I}{4}$$

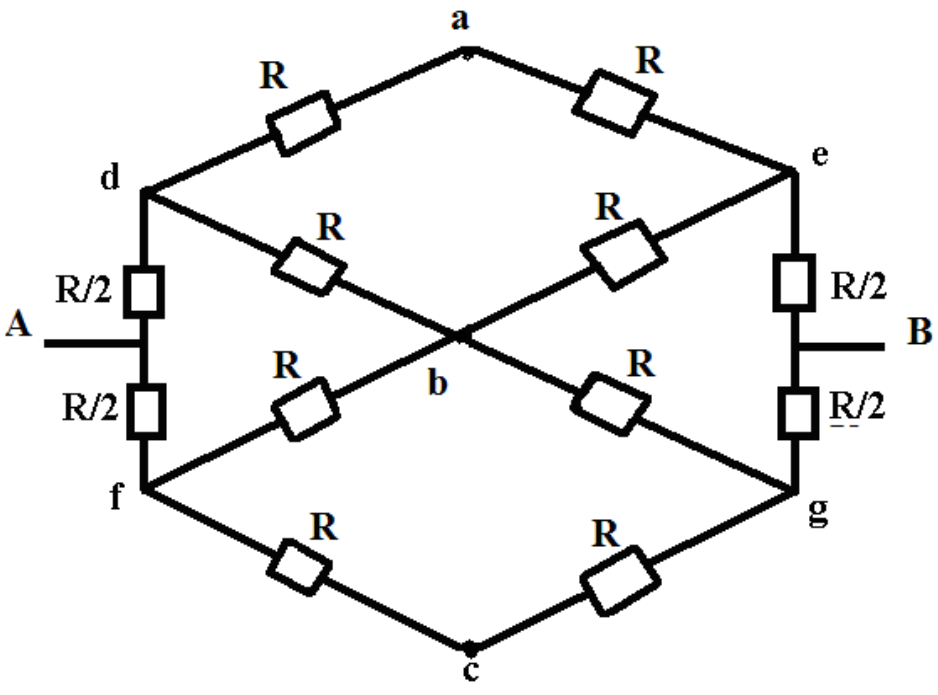
Krixgovning 1-qoidasidan chiqqan oxirgi natijadan

$$I_4 + I_5 + I_7 + I_8 = I_{10} + I_{11}$$

$$\frac{I}{4} + \frac{I}{4} + I_7 + I_8 = \frac{I}{4} + \frac{I}{4}$$

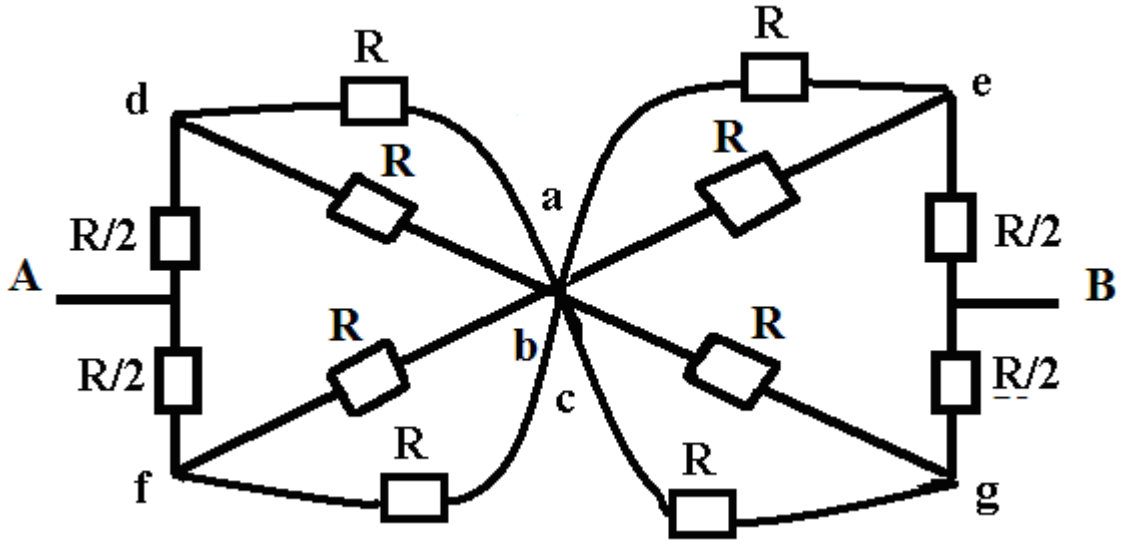
$$I_7 + I_8 = 0; \qquad I_7 = 0; \; I_8 = 0$$

natijaga ega bo'lamiz. Demak 7 chi va 8 chi rezistorlardan tok o'tmaydi. Shuning uchun ab va bc qarshiliklarni olib tashlaymiz va sxema quyudagi ko'rinishga ega bo'ladi.



Bu sxema 2-xil usulda bajariladi

1-Usul: abc nuqtalarni tutashtiramiz



abc birlashgan nuqtani o’z holicha qoldirib hisoblaymiz.
Buning uchun Ab va bB qismlardagi qarshiliklarni hisoblaymiz.

$$\frac{1}{R_{da}}=\frac{1}{R}+\frac{1}{R};\quad R_{da}=\frac{R}{2}$$

$$R_{Aa}=\frac{R}{2}+R_{da}=\frac{R}{2}+\frac{R}{2}=R$$

$$\frac{1}{R_{fc}}=\frac{1}{R}+\frac{1}{R};\quad R_{fc}=\frac{R}{2}$$

$$R_{Ac}=\frac{R}{2}+R_{dc}=\frac{R}{2}+\frac{R}{2}=R$$

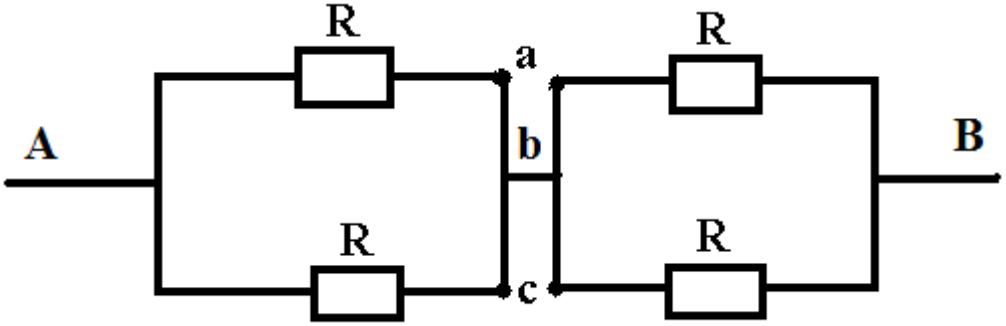
$$\frac{1}{R_{ea}}=\frac{1}{R}+\frac{1}{R};\quad R_{ea}=\frac{R}{2}$$

$$R_{Ba}=\frac{R}{2}+R_{ea}=\frac{R}{2}+\frac{R}{2}=R$$

$$\frac{1}{R_{gc}}=\frac{1}{R}+\frac{1}{R};\quad R_{gc}=\frac{R}{2}$$

$$R_{Bc}=\frac{R}{2}+R_{gc}=\frac{R}{2}+\frac{R}{2}=R$$

Ushbu hisoblashdan so’ng sxema quyidagicha sodda holga keladi.



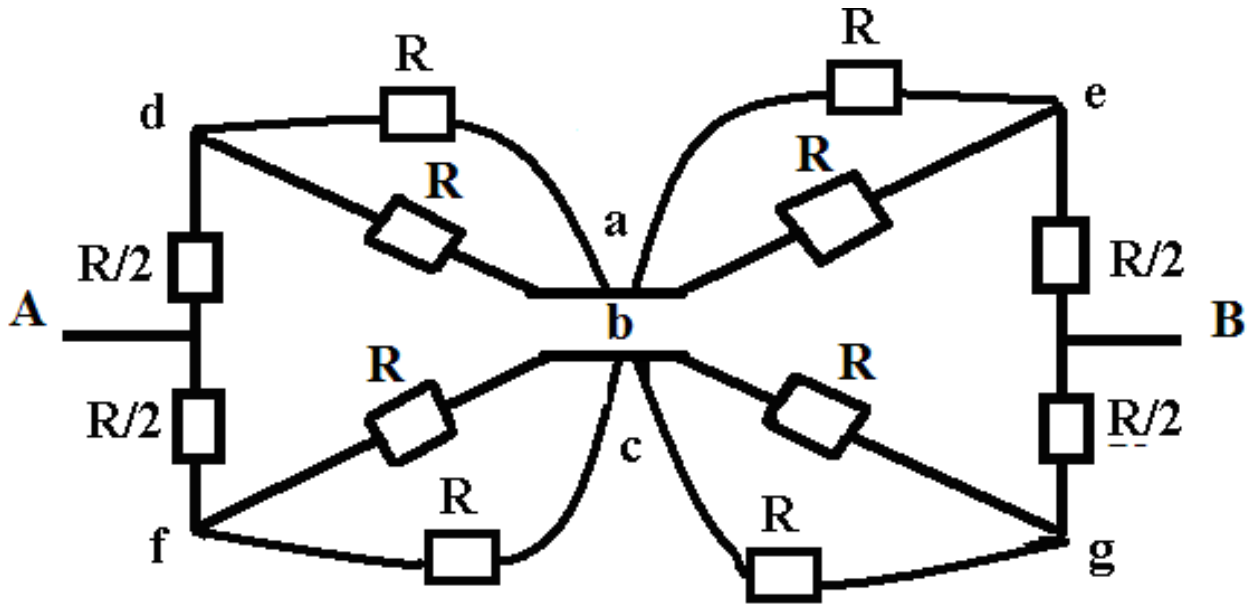
$$\frac{1}{R_{Ab}}=\frac{1}{R}+\frac{1}{R};\qquad R_{Ab}=\frac{R}{2}$$

$$\frac{1}{R_{bB}}=\frac{1}{R}+\frac{1}{R};\qquad R_{bB}=\frac{R}{2}$$

$$R_{AB}=R_{Ab}+R_{bB}=\frac{R}{2}+\frac{R}{2}=R$$

$$Javob: R_{AB}=R$$

2-Usul sxemani b nuqtadan kesamiz va sxema quyidagi holga keladi



$$\frac{1}{R_{da}}=\frac{1}{R}+\frac{1}{R};\quad R_{da}=\frac{R}{2}$$

$$R_{Aa}=\frac{R}{2}+R_{da}=\frac{R}{2}+\frac{R}{2}=R$$

$$\frac{1}{R_{fc}}=\frac{1}{R}+\frac{1}{R};\quad R_{fc}=\frac{R}{2}$$

$$R_{Ac}=\frac{R}{2}+R_{dc}=\frac{R}{2}+\frac{R}{2}=R$$

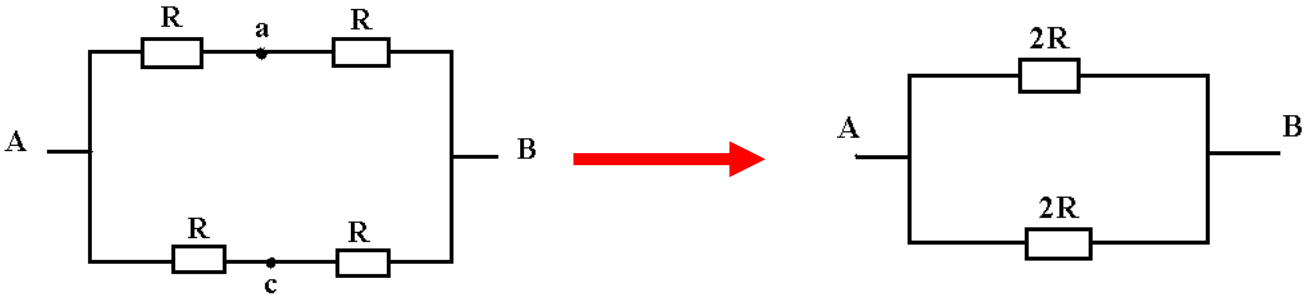
$$\frac{1}{R_{ea}}=\frac{1}{R}+\frac{1}{R};\quad R_{ea}=\frac{R}{2}$$

$$R_{Ba}=\frac{R}{2}+R_{ea}=\frac{R}{2}+\frac{R}{2}=R$$

$$\frac{1}{R_{gc}}=\frac{1}{R}+\frac{1}{R};\quad R_{gc}=\frac{R}{2}$$

$$R_{Bc}=\frac{R}{2}+R_{gc}=\frac{R}{2}+\frac{R}{2}=R$$

Ushbu hisoblashdan so'ng sxema quyidagicha sodda holga keladi.

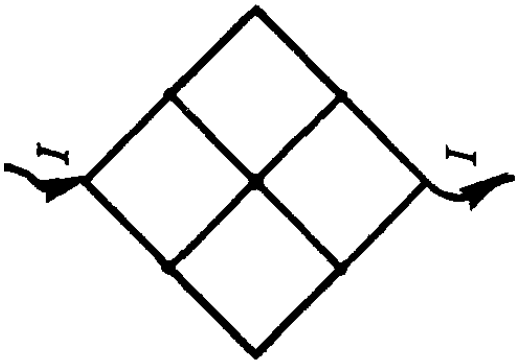


$$\frac{1}{R_{AB}}=\frac{1}{2R}+\frac{1}{2R};$$

$$R_{AB}=R$$

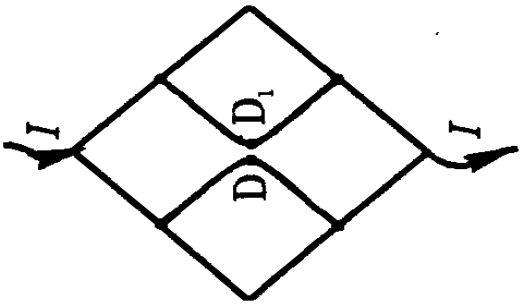
$$Javob: R_{AB}=R$$

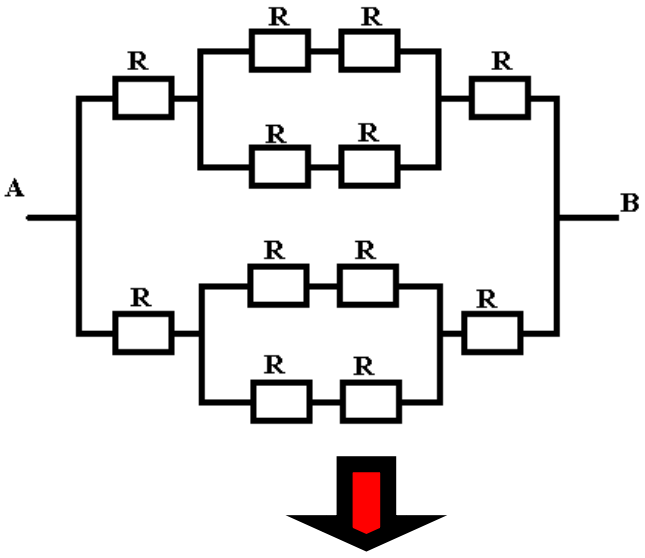
46.Quyidagi sxemaning umumiy qarshiligini toping

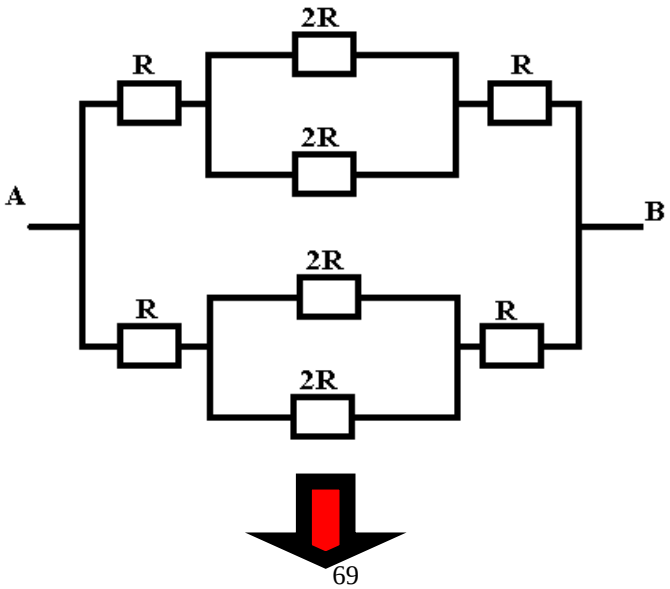


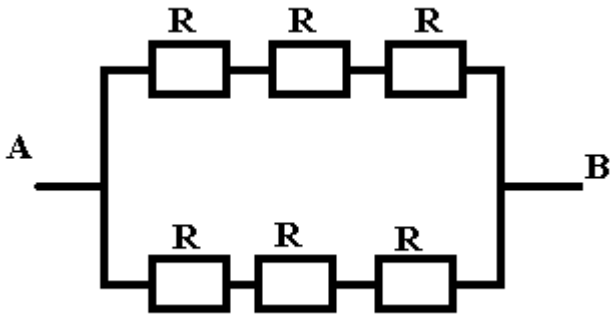
Bu masala ikki usulda yechiladi

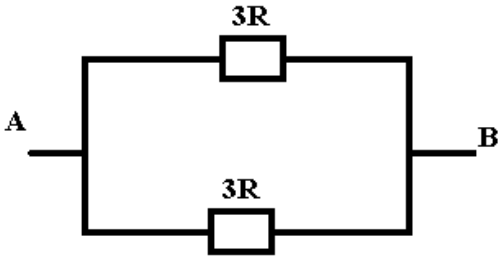
1-Usul:Markazidan ikkiga ayiramiz







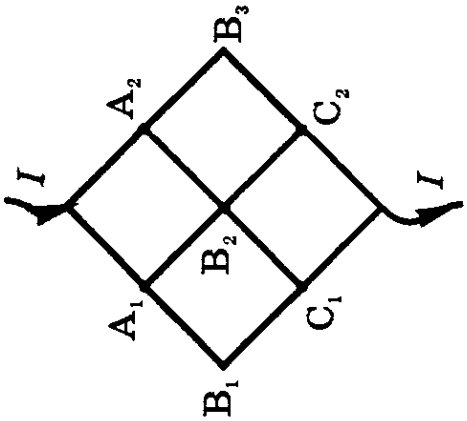




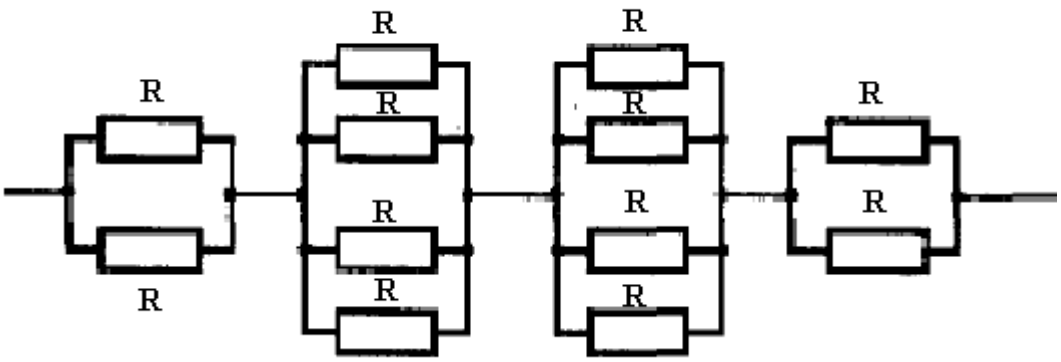
$$\frac{1}{R_{AB}}=\frac{1}{3R}+\frac{1}{3R};$$

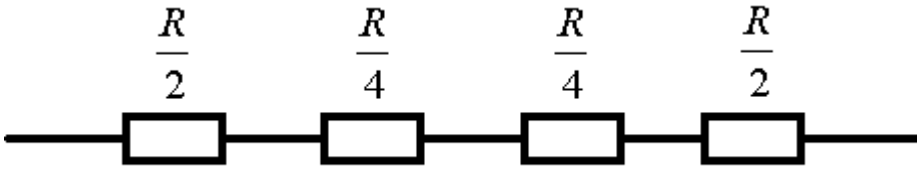
$$R_{AB}=\frac{3}{2}R$$

2-Usul: Sxemaning A_1 va A_2; B_1 ,B_2 va B_3; C_1 va C_2 nuqtalarni tutashtiramiz.



Sxemani quyidagicha soddalashadi.

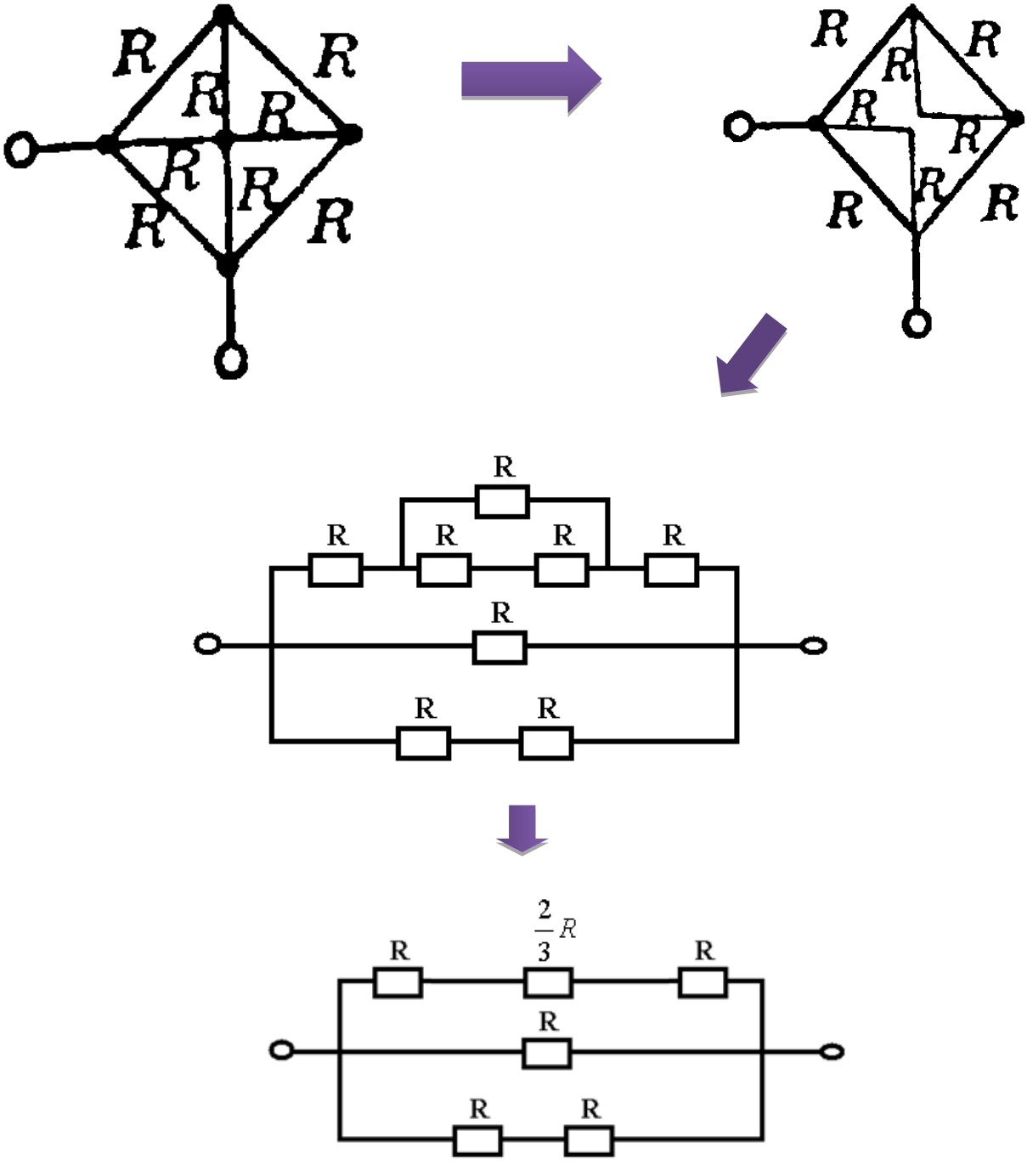


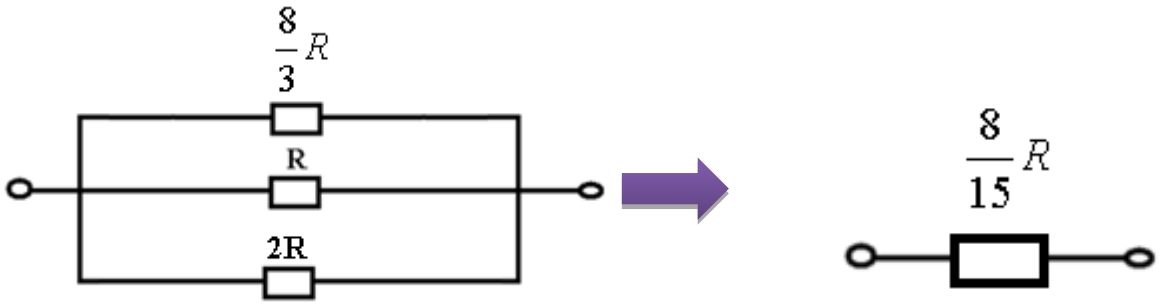


$$R_{Um} = \frac{R}{2} + \frac{R}{4} + \frac{R}{4} + \frac{R}{2} = \frac{3R}{2};$$

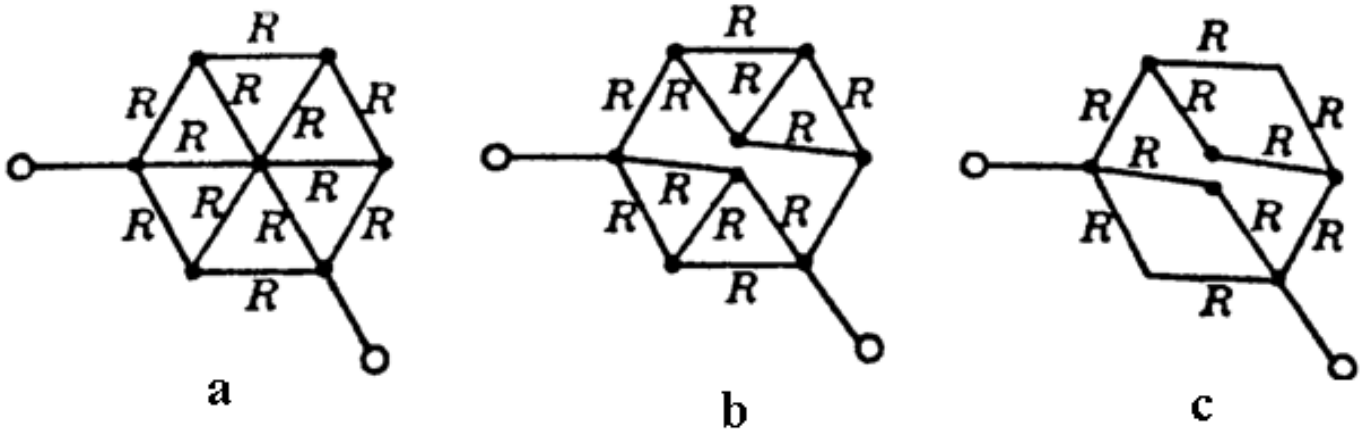
$$R_{Um} = \frac{3R}{2};$$

47. Zarjirning umumiy qarshiligini toping

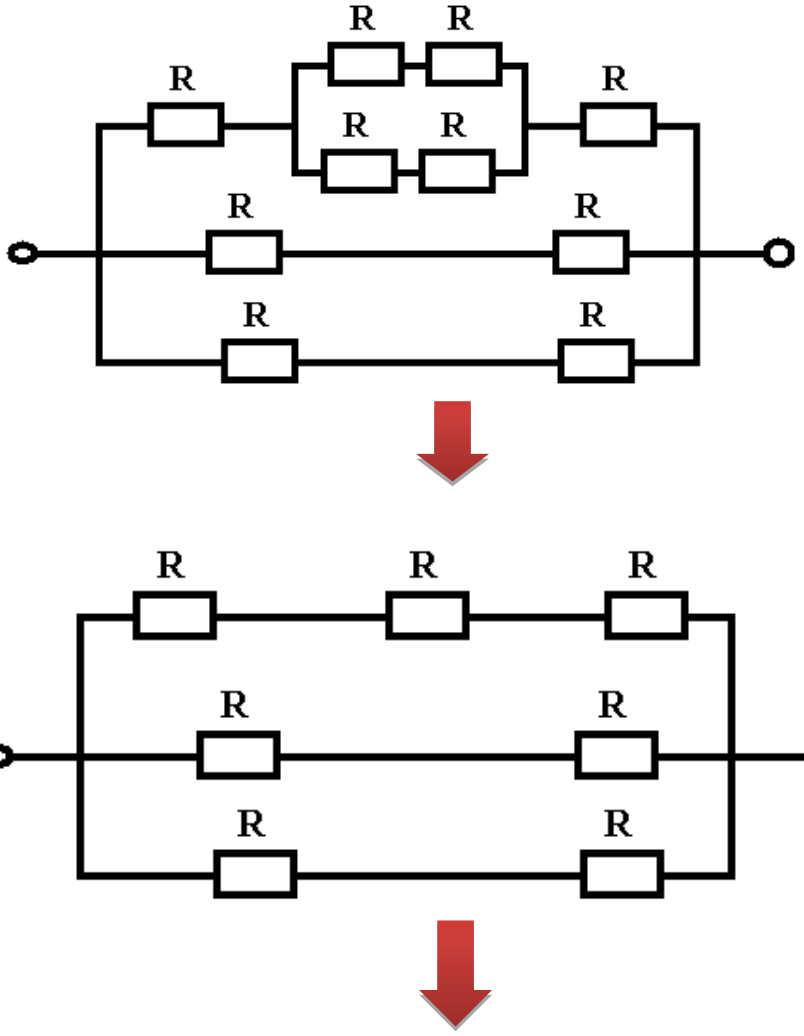


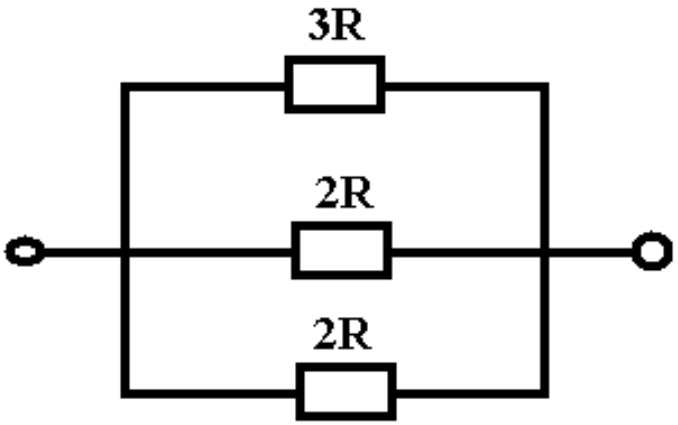


48.Zarjirning umumiy qarshiligini toping(a-rasm)



Zanjirning umumiy qarshiligini topish uchun b va c rasmlar kabi soddalashtiramiz.B va c rasmni quyidagi tushunarli holga keltiramiz.

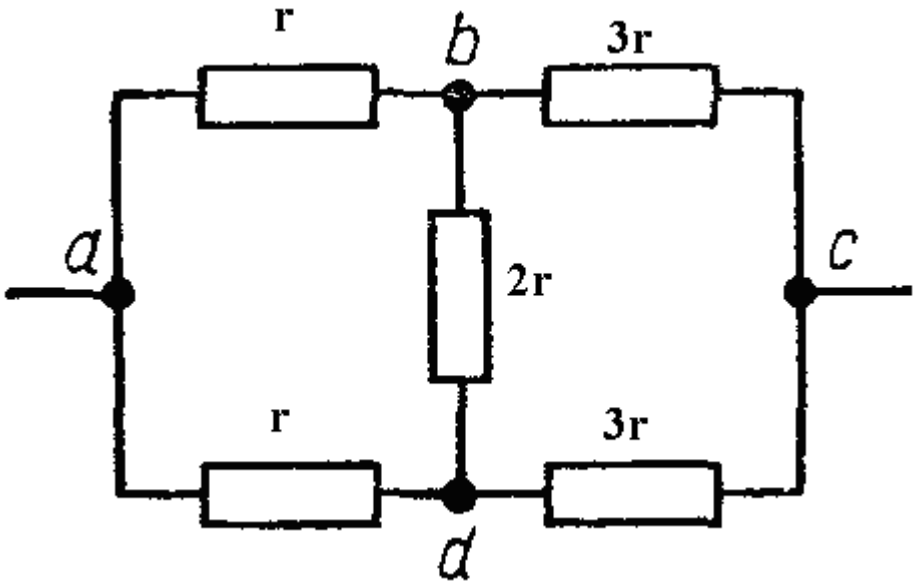




$$\frac{1}{R_{Um}} = \frac{1}{3R} + \frac{1}{2R} + \frac{1}{2R} = \frac{8}{6R};$$

$$R_{Um} = \frac{3R}{4}$$

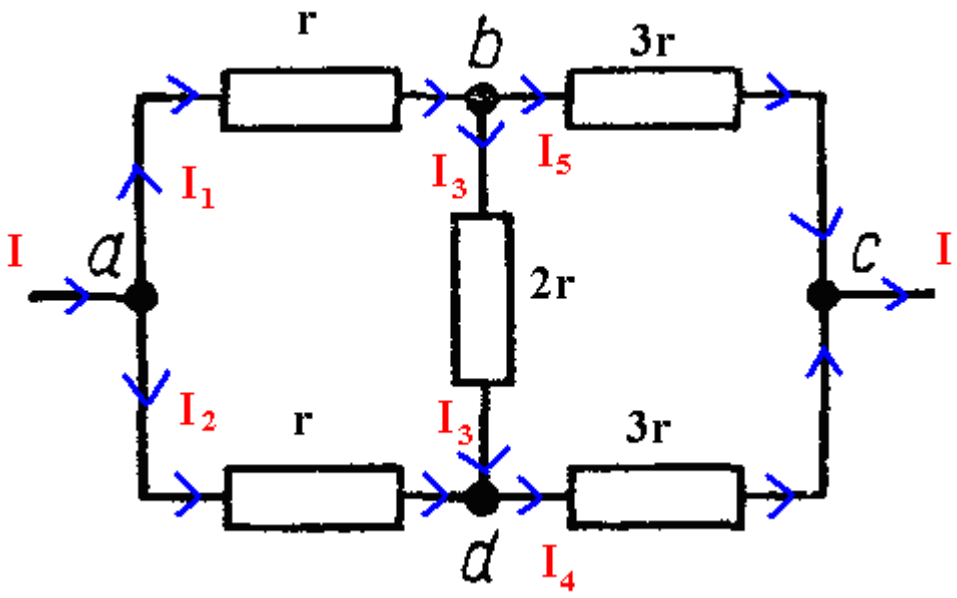
48. Sxemaning ac nuqtalari orasidagi umumiy qarshilikni toping.



b va d nuqtalarda potensiallari bir xil $\varphi_b = \varphi_d$ bo'lganligi uchun bd nuqtalar orasidagi potensiallar farqi $U_{bd} = \varphi_b - \varphi_d = 0$ ga teng ya'ni bd yo'nalishda tok o'tmaydi

ISBOT

Tok o'tmasligini isbotlash uchun sxemaga tok kuchini joylashtirib chiqamiz yo'nalishi bilan



Kirxgofning 1-qoidasiga ko’ra tugunlarga kiruvchi va chiquvchi toklar tengligidan quyidagi natijaga ega bo’lamiz.

$$I = I_1 + I_2$$

$$I_4 + I_5 = I$$

$$I_3 + I_5 = I_1$$

$$I_3 + I_2 = I_4$$

ab,ad rezistorlar va bc, dc rezistorlar o’zaro parallel ulanganligi uchun bu yo’nalishlardagi kuchlanishlari teng bo’ladi:

$$U_1 = U_2$$

$$I_1 \cdot r = I_2 \cdot r$$

$$I_1 = I_2$$

$$U_5 = U_4$$

$$I_5 \cdot 3r = I_4 \cdot 3r$$

$$I_5 = I_4$$

Bu natijalarni Kirxgofning 1-qoidasidan chiqqan natijalarga etib qo’ysak:

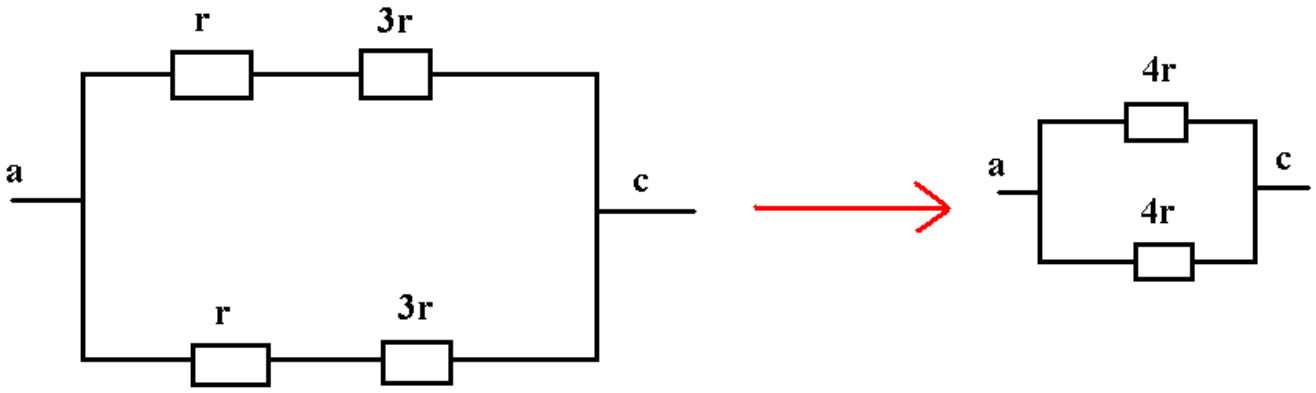
$$I_1 = I_2 = \frac{I}{2}$$

$$I_4 = I_5 = \frac{I}{2}$$

$$I_3 + I_5 = I_1 \Rightarrow I_3 + \frac{I}{2} = \frac{I}{2} \Rightarrow I_3 = 0$$

$$I_3 + I_2 = I_4 \Rightarrow I_3 + \frac{I}{2} = \frac{I}{2} \Rightarrow I_3 = 0$$

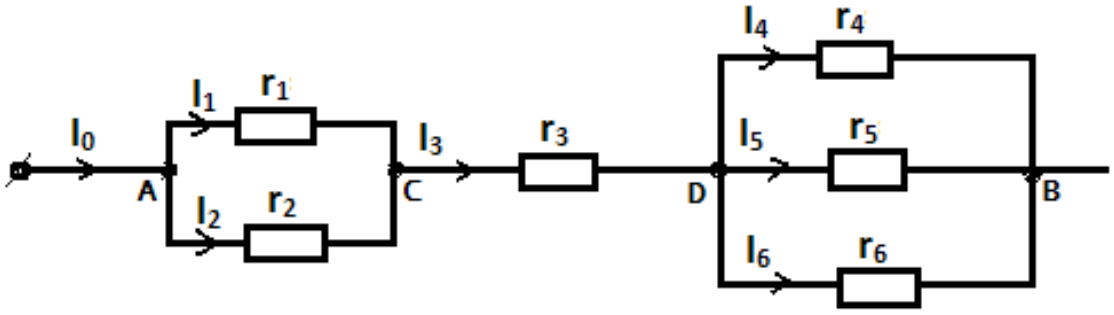
Bu natijalardan ko’rinadiki 3-rezistordan tok o’tmas ekan. Shuning uchun bd rezistorni olib tashlaymiz va sxemamiz quyidagi soda holga keladi:



$$\frac{1}{R_{ab}} = \frac{1}{4r} + \frac{1}{4r};$$

$$R_{ab} = 2r$$

49. Agar kuchlanish U_{AB} =48V, r_1= r_3=3Ω; r_2=6 Ω, r_4=5 Ω ,r_5=10 Ω va r_6=30 Ω bo’lsa, zanjirdagi har bir qarshilikdan o’tayotgan tok kuchlari $I_1;I_2;I_3;I_4;I_5;I_6$ va zanjirning AC ,CD, DB qismlaridagi U_{AC} ,U_{CD} ,U_{DB} kuchlanishlarni toping.



Berilgan:

U_{AB} =48v, r_1= r_3=3Ω; r_2=6 Ω, r_4=5 Ω ,r_5=10 Ω va r_6=30 Ω

Topish kerak:

$I_1;I_2;I_3;I_4;I_5;I_6$? U_{AC} =? U_{CD} =? U_{DB}?

Yechish:

Berilgan sxemada tarmoqlangan nuqtalari A,C,D va B harflar bilan belgilaymiz.Zanjirning AC,CD,DB qismlari o’zaro ketma-ket ulanganligi sababli zanjirning umumiy qarshiligi $R_{AB} = R_{AC} + R_{CD} + R_{DB}$ bo’ladi.

Zanjirning AC va DB qismlarida rezistorlar o’zaro parallel ulangan.

Shuning uchun:

$$\frac{1}{R_{AC}}=\frac{1}{r_1}+\frac{1}{r_2};\qquad R_{AC}=\frac{r_1 r_2}{r_1+r_2}=\frac{3\cdot 6}{3+6}=\frac{18}{9}=2\Omega$$

$$\frac{1}{R_{DB}}=\frac{1}{r_4}+\frac{1}{r_5}+\frac{1}{r_6};\quad \frac{1}{R_{DB}}=\frac{1}{5}+\frac{1}{10}+\frac{1}{30};\quad R_{DB}=3\Omega$$

$$R_{CD}=r_3=3\Omega$$

Endi R_{AB} ni topamiz

$$R_{AB}=2+3+3=8\Omega$$

Om qonuniga binoan zanjirdan o'tayotgan tokning kuchi I_0 resistor r_3 dan o'tayotgan tokning kuchi I_3 ga teng,ya'ni

$$I_0=I_3=\frac{U_{AB}}{R_{AB}}=\frac{48}{8}=6A$$

Zanjir qismlaridagi $U_{AC};U_{CD};U_{DB}$ kuchlanishlar quyidagiga teng bo'ladi:

$$U_{AC}=I_0\cdot R_{AC}=6\cdot 2=12V$$

$$U_{CD}=I_0\cdot R_{CD}=6\cdot 3=18V$$

$$U_{DB}=I_0\cdot R_{BD}=6\cdot 3=18V$$

Parallel ulashda kuchlanishlar tengligidan

$$U_1=U_2=U_{AC}=12V$$

$$U_3=U_{CD}=18V$$

$$U_4=U_5=U_6=U_{DB}=18V$$

Har bir rezistorlardagi kuchlanishlarni topdik.

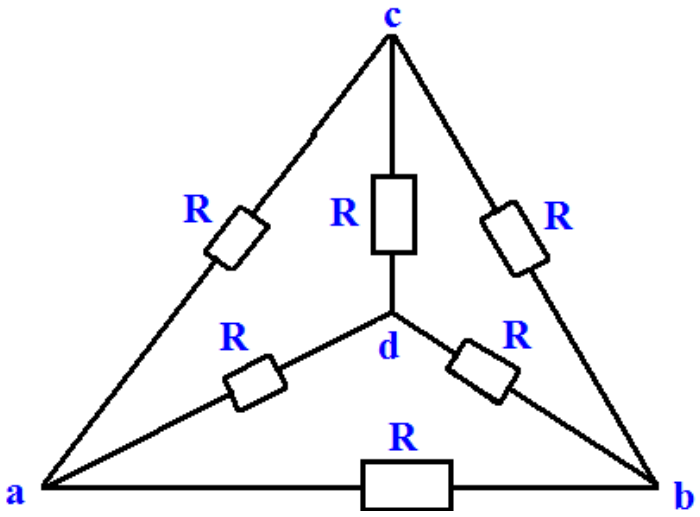
Endi rezistorlardan o'tayotgan tok kuchi quyidagilarga teng bo'ladi:

$$I_1=\frac{U_1}{r_1}=\frac{12}{3}=4A;I_2=\frac{U_2}{r_2}=\frac{12}{6}=2A;I_4=\frac{U_4}{r_4}=\frac{18}{5}=3{,}6A;$$

$$I_5=\frac{U_5}{r_5}=\frac{18}{10}=1{,}8A;I_6=\frac{U_6}{r_6}=\frac{18}{30}=0{,}6A$$

LABORATORIYA YO'LI BILAN YECHILGAN MASALALAR

1.Uchburchakli piramidaning barcha tomonlarining qarshiligi R ga teng sxemaning ab;ad va ac nuqtalar orasidagi umumiy qarshiligini toping.

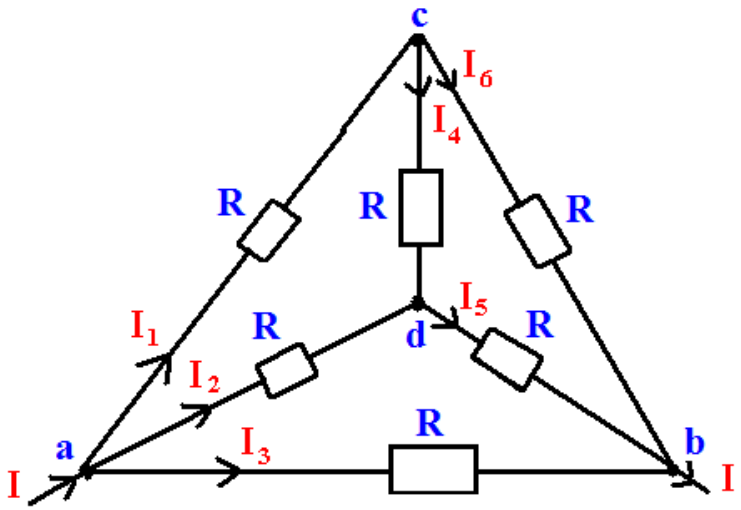


1-HOL $(R_{ab} = ?)$

Sxemaning qarshiliklari bir-xil bo'lganligi uchun va c va d nuqtalarga potensiallar tengligi uchun, potensiallar farqi cd rezistorda 0 ga teng. Shuning uchun cd rezistordan tok o'tmaydi.

ISBOT

ab nuqtalar orasidagi umumiy qarshilikni hisoblash uchun biz avval har bir rezistor orqali o'tadigan toklarni topib chiqamiz. Demak a va b nuqta orasidagi qarshilikni toping deyilgani uchun a nuqtadan I tok kirib b nuqtadan chiqib ketadi deb faraz qilib masalani ishlaymiz.Va sxemada tok kuchlarini quyidagicha joylashtirib chiqamiz.



Kirxgofning 1-qoidasiga ko'ra tugunlarga kiruvchi va chiquvchi toklar tengligidan quyidagi natijaga ega bo'lamiz.

$$I = I_1 + I_2 + I_3$$

$$I_6 + I_5 + I_3 = I$$

$$I_4 + I_2 = I_5$$

ac; ad; ab rezistorlar va cb; db; ab rezistorlar o'zaro parallel ulanganligi uchun bu yo'nalishlardagi kuchlanishlar teng bo'ladi:

$$U_1 = U_2 = U_3$$

$$I_1 \cdot R = I_2 \cdot R = I_3 \cdot R$$

$$I_1 = I_2 = I_3$$

$$U_3 = U_5 = U_6$$

$$I_3 \cdot R = I_5 \cdot R = I_6 \cdot R$$

$$I_3 = I_5 = I_6$$

Bu natijalarni Kirxgofning 1-qoidasidan chiqqan natijalarga etib qo'ysak

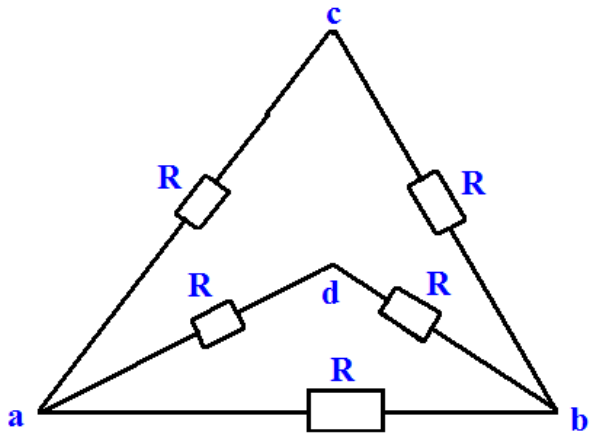
$$I_1 = I_2 = I_3 = \frac{I}{3}$$

$$I_3 = I_5 = I_6 = \frac{I}{3}$$

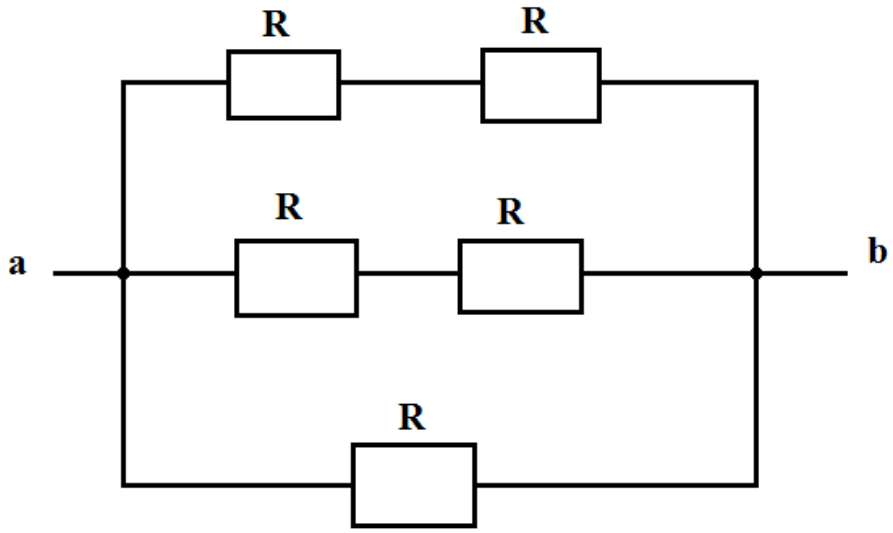
$$I_4 + I_2 = I_5 \Rightarrow I_4 + \frac{I}{3} = \frac{I}{3} \Rightarrow I_4 = 0$$

Bulardan cd rezistordagi tok kuchi $I_4 = 0$ ga teng bo'lib chiqdi. Demak cd yo'nalishda tok harakatlanmaydi potensiallar farqi $\varphi_c - \varphi_d = U_{cd} = 0$ bo'ladi

Shuning uchun cd rezistorni sxemadan olib tashlaymiz va sxemamiz quyidagi ko'rinishga keladi.

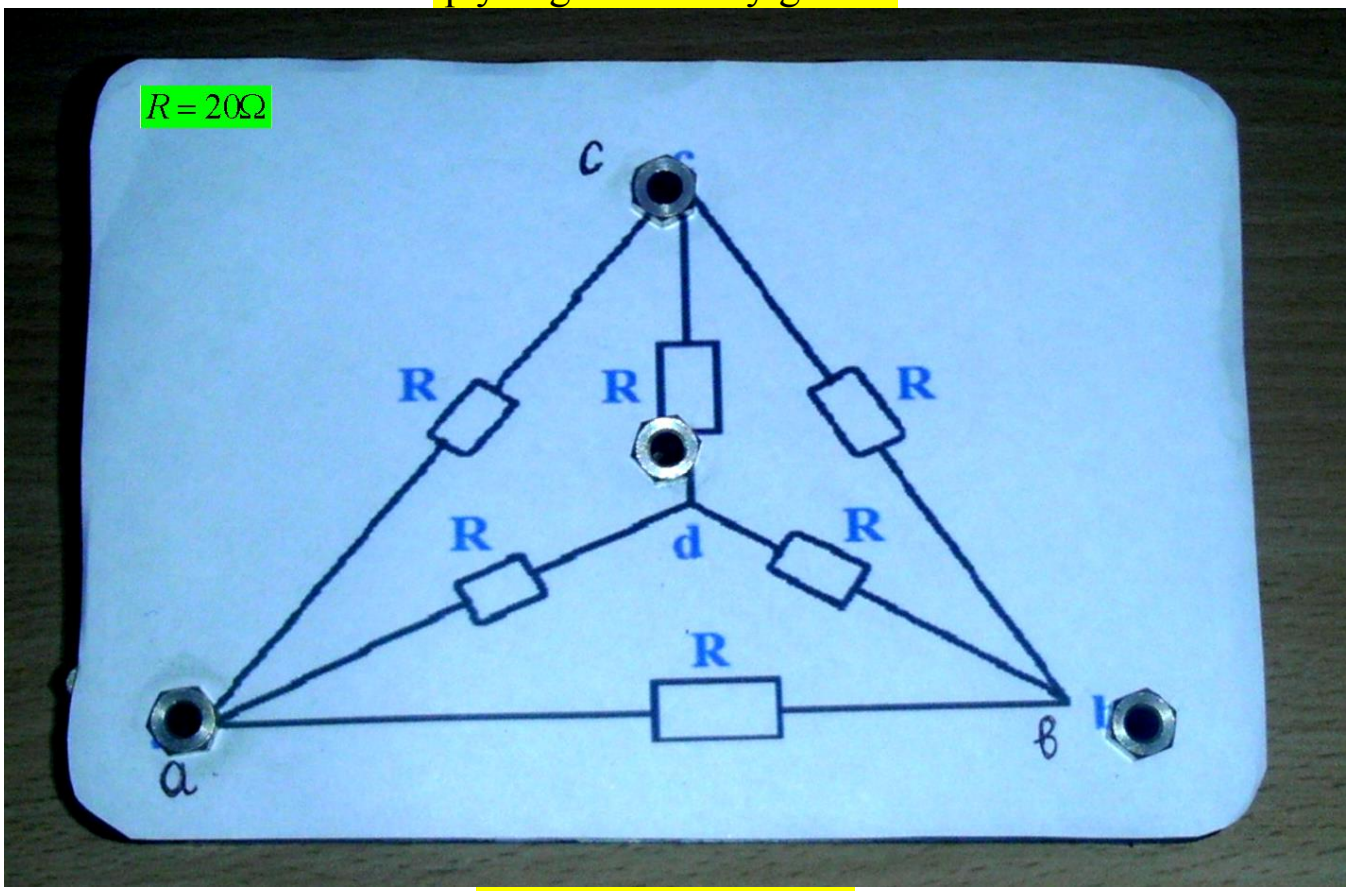


Buni sxemani quyidagi soda ko'rinishda yozamiz

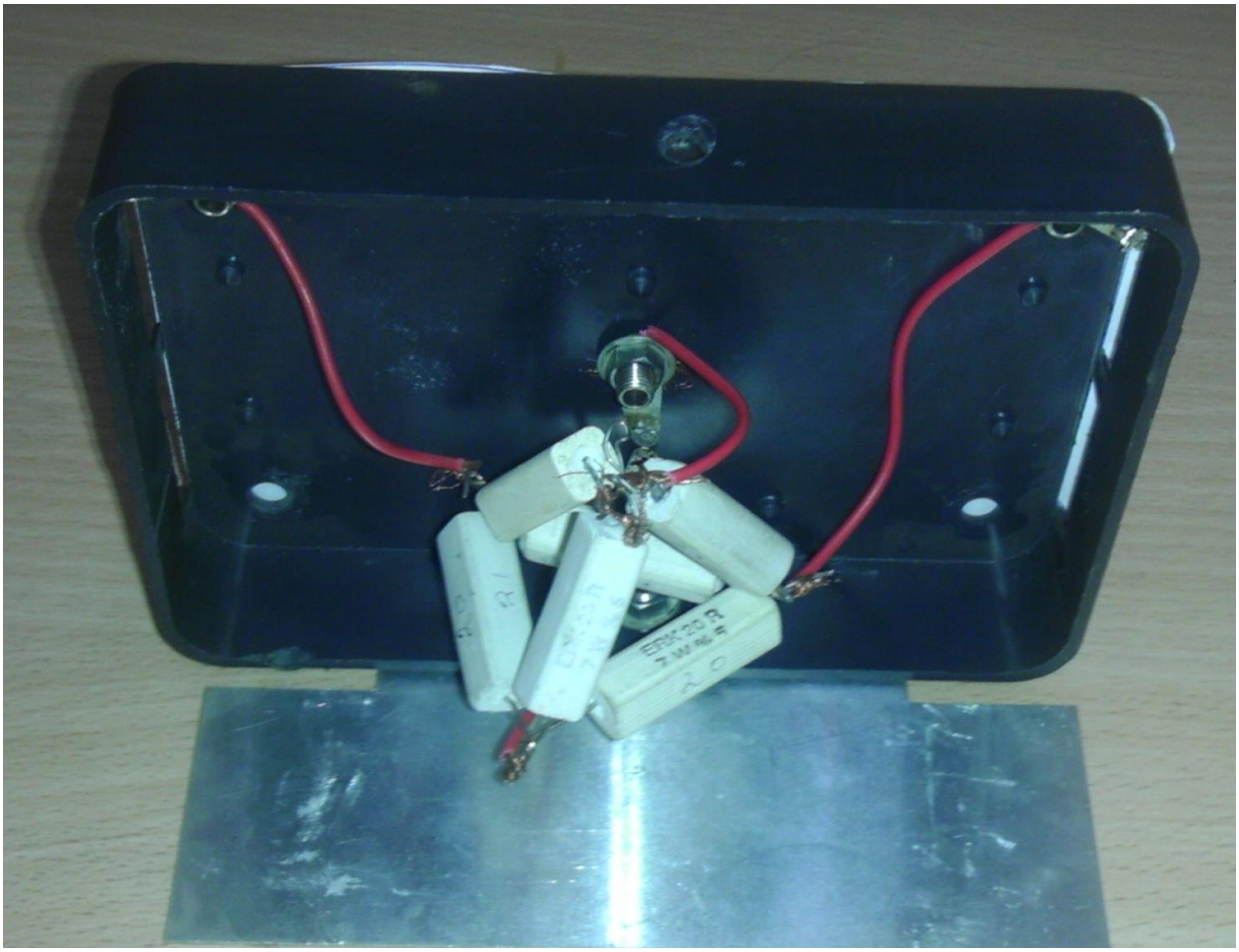


$$\frac{1}{R_{ab}}=\frac{1}{2R}+\frac{1}{2R}+\frac{1}{R};\qquad R_{ab}=0{,}5R$$

Endi nazariy yo’ bilan isbotlab chiqarilgan natijalarni amalda bajaramiz. $\varphi_c-\varphi_d=U_{cd}=0$ ushbu formulani amalda isbotlaymiz .Buning uchun quyidagi sxemani yig’amiz



Old va orqa ko’rinishi



Sxemamizni a va b nuqtasini tok manbaiga ketma-ket, c va d nuqtalarni esa voltimetrga parallel ulaymiz.

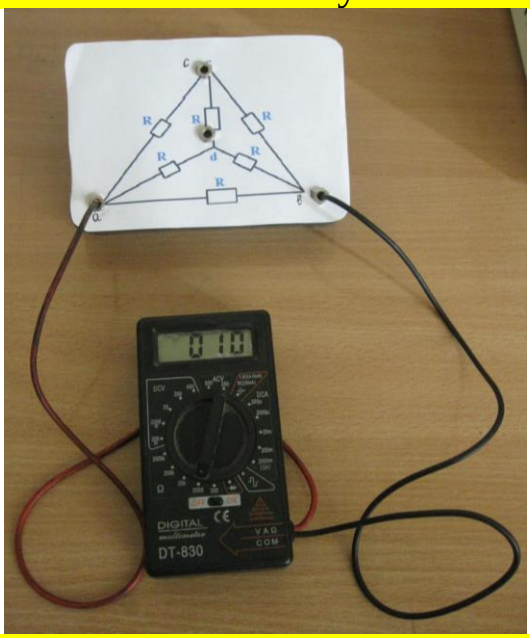


Tajribadan ko’rinib turibdiki voltimetr 0 ni ko’rsatmoqda demak c va d nuqtalarda potensiallar farqi o ga teng ekan.

Tajribadan ya’na shunday xulosa chiqarish mumkinki Kirxgofning

1-qoidasida: Tugunlarga kiruvchi va chiquvchi toklar teng deyilgan edi yaqiqatdan ham a nuqtaga ulangan ampermetr ham b nuqtaga ulangan ampermetrham birxil tokni ko’rsatmoqda.

Endi umumiy qarshilikni hisoblaymiz buning uchun qarshilik o’lchaydigan asbob ommetrdan foydalanamiz.



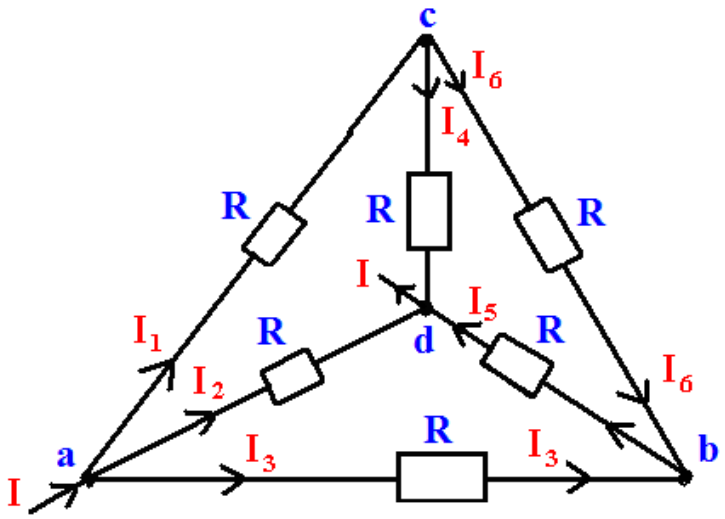
Bizga nazariy yo’l bilan $R_{ab} = 0{,}5R$ javob chiqqan edi, biz qarshiligi $R = 20\Omega$ bo’lgan qarshiliklardan foydalanganimiz uchun javob $R_{ab} = 0{,}5 \cdot 20 = 10\Omega$ chiqdi.

2-HOL $(R_{ad} = ?)$

Sxemaning qarshiliklari bir-xil bo’lganligi uchun va c va b nuqtalarga potensiallar tengligi uchun, potensiallar farqi cb rezistorda 0 ga teng shuning uchun cb rezistordan tok o’tmaydi.

ISBOT

ad nuqtalar orasidagi umumiy qarshilikni hisoblash uchun biz avval har bir rezistor orqali o’tadigan toklarni topib chiqamiz. Demak a va d nuqta orasidagi qarshilikni toping deyilgani uchun a nuqtadan I tok kirib d nuqtadan chiqib ketadi deb faraz qilib masalani ishlaymiz.Va sxemada tok kuchlarini quyidagicha joylashtirib chiqamiz.



Kirxgofning 1-qoidasiga ko’ra tugunlarga kiruvchi va chiquvchi toklar tengligidan quyidagi natijaga ega bo’lamiz.

$$I = I_1 + I_2 + I_3$$

$$I_2 + I_4 + I_5 = I$$

$$I_3 + I_6 = I_5$$

ac;ad;ab rezistorlar va cd;ad;bd rezistorlar o’zaro parallel ulanganligi uchun bu yo’nalishlardagi Kuchlanishlari teng bo’ladi

$$U_1 = U_2 = U_3$$

$$I_1 \cdot R = I_2 \cdot R = I_3 \cdot R$$

$$I_1 = I_2 = I_3$$

$$U_2 = U_4 = U_5$$

$$I_2 \cdot R = I_4 \cdot R = I_5 \cdot R$$

$$I_2 = I_4 = I_5$$

Bu natijalarni Kirxgofning 1-qoidasidan chiqqan natijalarga etib qo’ysak

$$I_1 = I_2 = I_3 = \frac{I}{3}$$

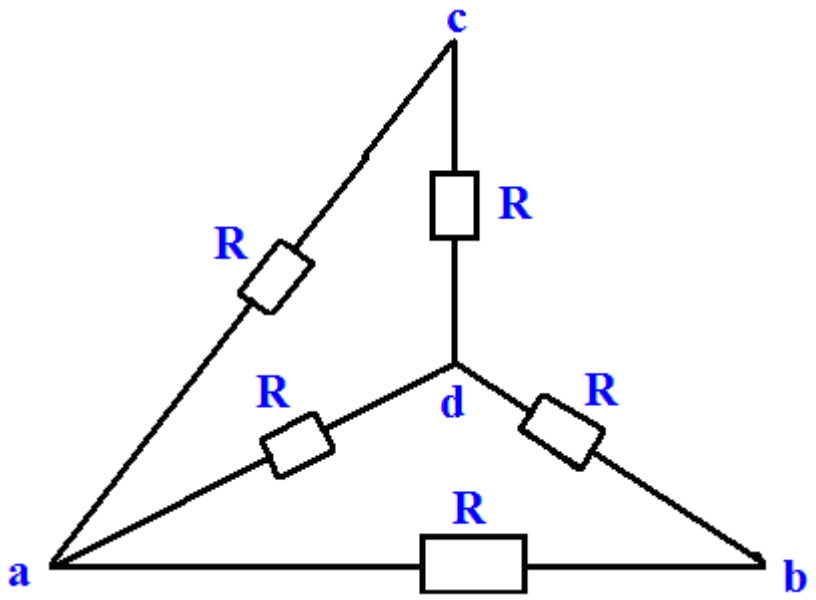
$$I_2 = I_4 = I_5 = \frac{I}{3}$$

$$I_3 + I_6 = I_5 \Rightarrow I_6 + \frac{I}{3} = \frac{I}{3} \Rightarrow I_6 = 0$$

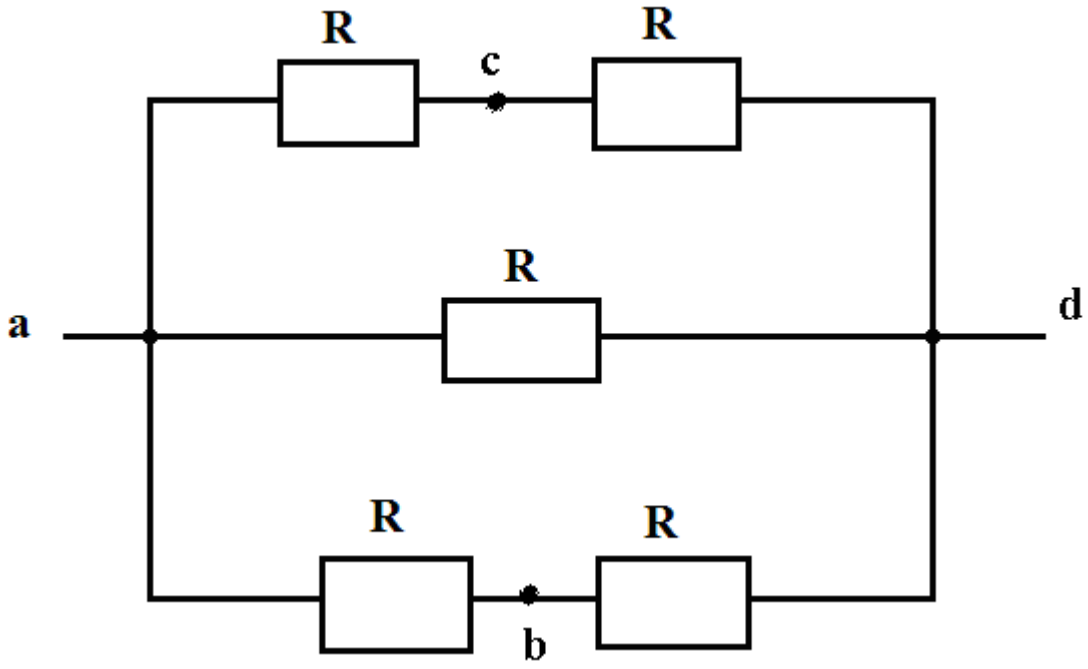
Bulardan cb rezistordagi tok kuchi $I_6 = 0$ ga teng bo’lib chiqdi. Demak cb yo’nalishda tok harakatlanmaydi potensiallar farqi

$\varphi_c - \varphi_b = U_{cb} = 0$ bo'ladi

Shuning uchun cb rezistorni sxemadan olib tashlaymiz va sxemamiz quyidagi ko'rinishga keladi.



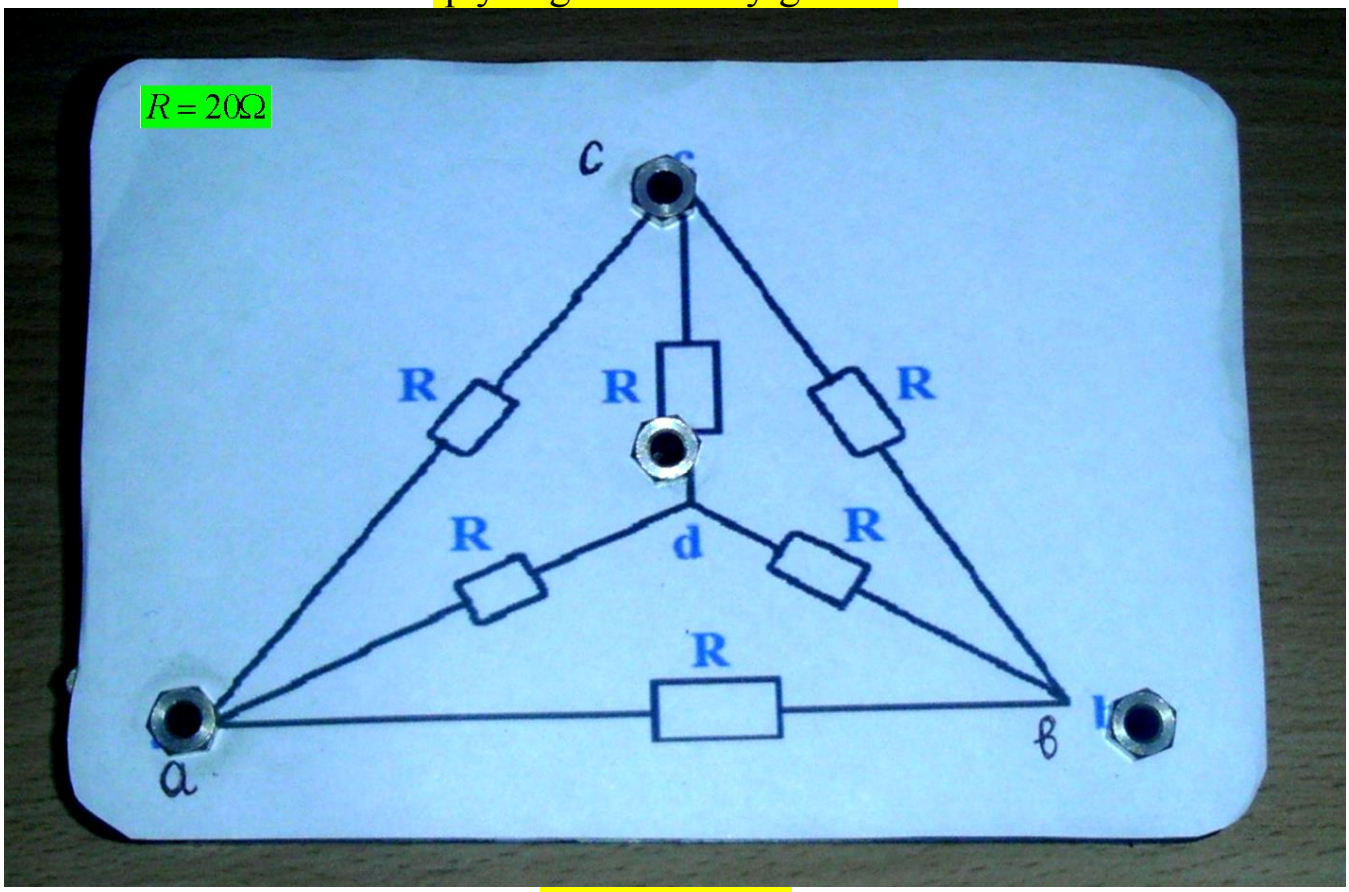
Buni sxemani quyidagi soda ko'rinishda yozamiz



$$\frac{1}{R_{ad}} = \frac{1}{2R} + \frac{1}{2R} + \frac{1}{R}; \qquad R_{ad} = 0{,}5R$$

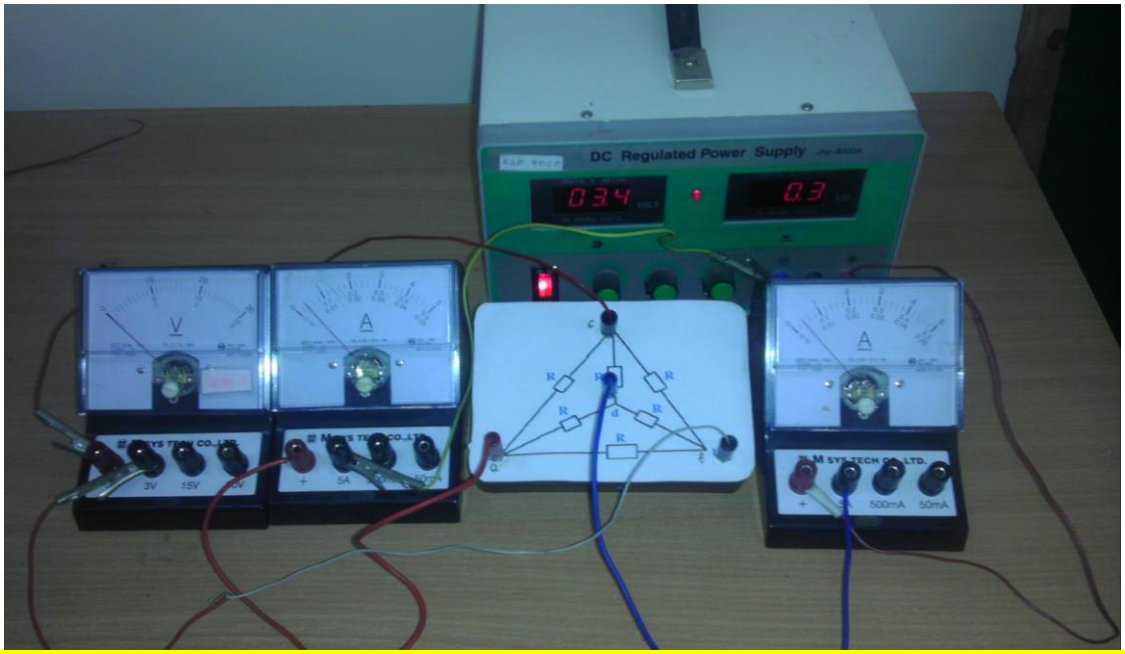
Endi nazariy yo' bilan isbotlab chiqarilgan natijalarni amalda bajaramiz.

$\varphi_c - \varphi_b = U_{cb} = 0$ ushbu formulani amalda isbotlaymiz .Buning uchun quyidagi sxemani yig'amiz



Old ko'rinishi

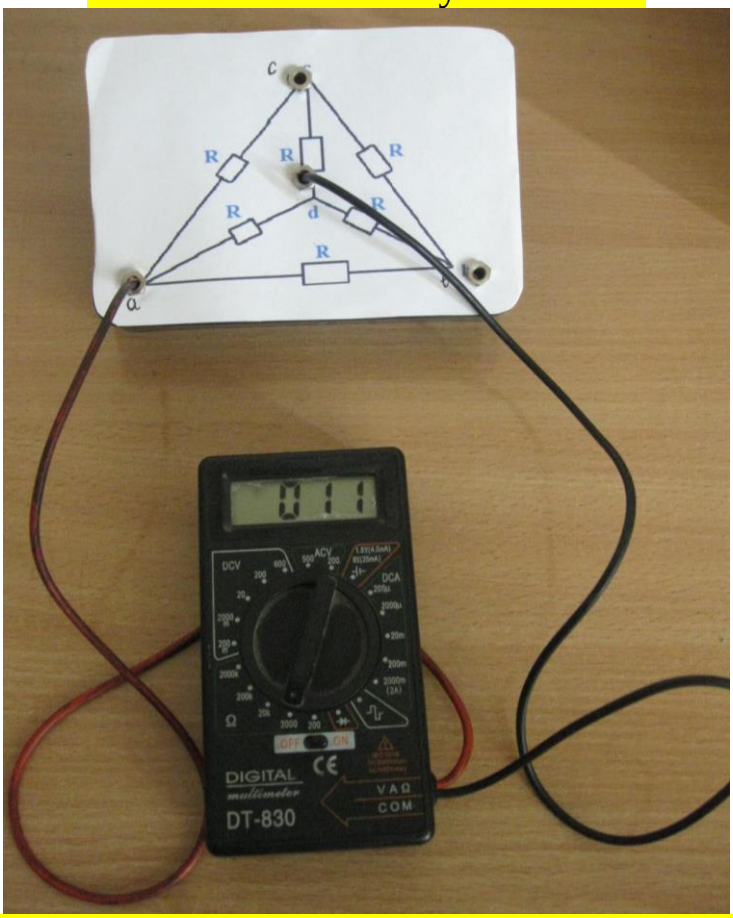
Sxemamizni a va d nuqtasini tok manbaiga ketma-ket, c va b nuqtalarni esa voltimetrga parallel ulaymiz.



Tajribadan ko'rinib turibdiki voltimetr 0 ni ko'rsatmoqda demak c va b nuqtalarda potensiallar farqi o ga teng ekan.

Tajribadan ya'na shunday xulosa chiqarish mumkinki Kirxgofning

1-qoidasida: Tugunlarga kiruvchi va chiquvchi toklar teng deyilgan edi yaqiqatdan ham a nuqtaga ulangan ampermetr ham d nuqtaga ulangan ampermetrham bir xil tokni ko’rsatmoqda.
Endi umumiy qarshilikni hisoblaymiz buning uchun qarshilik o’lchaydigan asbob ommetrdan foydalanamiz.



Bizga nazariy yo’l bilan $R_{ad} = 0{,}5R$ javob chiqqan edi, biz qarshiligi $R = 20\Omega$ bo’lgan qarshiliklardan foydalanganimiz uchun javob $R_{ad} = 0{,}5 \cdot 20 = 10\Omega$ chiqishi kerak edi lekin ulash simlarining qarshiligi borligi uchun $R_{ad} = 11\Omega$ chiqdi.

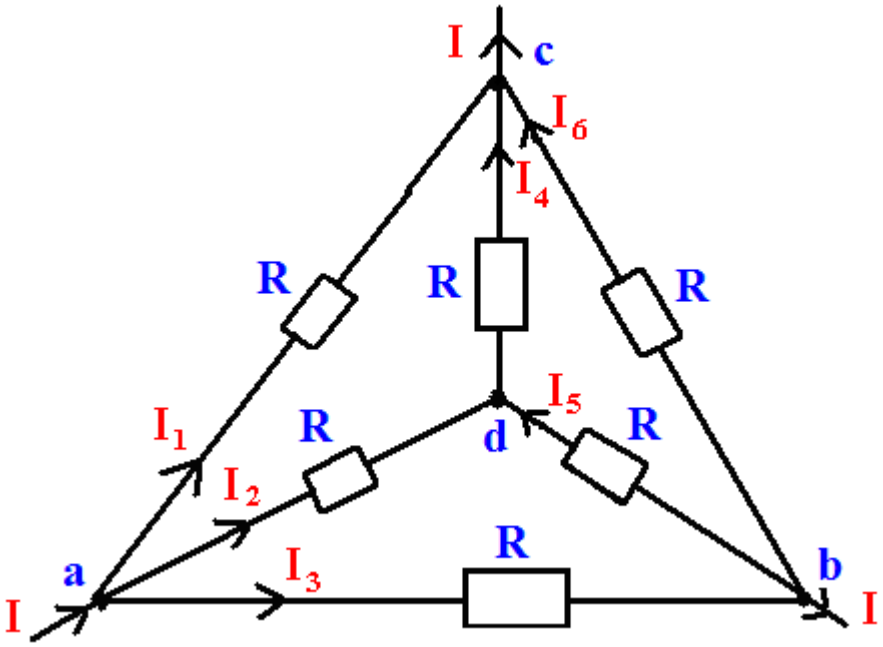
3-HOL $(R_{ac} = ?)$

Sxemaning qarshiliklari bir-xil bo’lganligi uchun va b va d nuqtalarga potensiallar tengligi uchun,potensiallar farqi bd rezistorda 0 ga teng shuning uchun bd rezistordan tok o’tmaydi.

ISBOT

ac nuqtalar orasidagi umumiy qarshilikni hisoblash uchun biz avval har bir rezistor orqali o’tadigan toklarni topib chiqamiz. Demak a va c nuqta orasidagi qarshilikni toping deyilgani uchun a nuqtadan I tok kirib c

nuqtadan chiqib ketadi deb faraz qilib masalani ishlaymiz.Va sxemada tok kuchlarini quyidagicha joylashtirib chiqamiz.



Kirxgofning 1-qoidasiga ko’ra tugunlarga kiruvchi va chiquvchi toklar tengligidan quyidagi natijaga ega bo’lamiz.

$$I = I_1 + I_2 + I_3$$

$$I_1 + I_4 + I_6 = I$$

$$I_2 + I_5 = I_4$$

ac;ad;ab rezistorlar va ac;dc;bc rezistorlar o’zaro parallel ulanganligi uchun bu yo’nalishlardagi kuchlanishlari teng bo’ladi

$$U_1 = U_2 = U_3$$

$$I_1 \cdot R = I_2 \cdot R = I_3 \cdot R$$

$$I_1 = I_2 = I_3$$

$$U_1 = U_4 = U_6$$

$$I_1 \cdot R = I_4 \cdot R = I_6 \cdot R$$

$$I_1 = I_4 = I_6$$

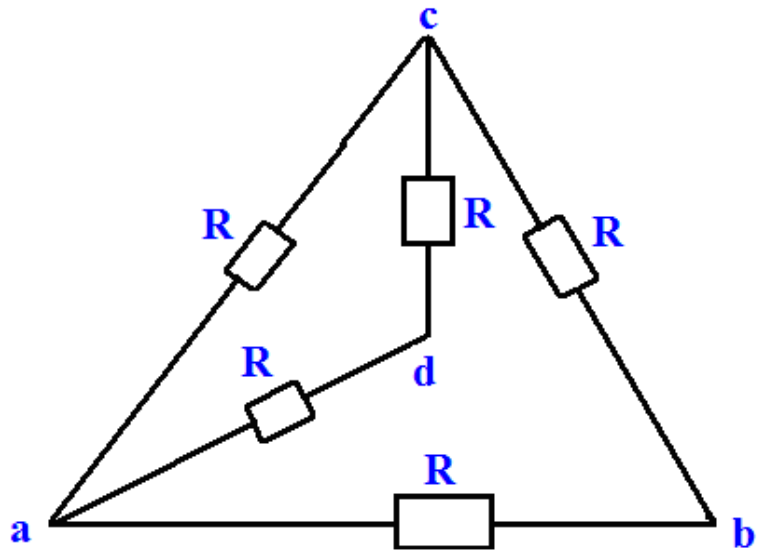
Bu natijalarni Kirxgofning 1-qoidasidan chiqqan natijalarga etib qo’ysak

$$I_1 = I_2 = I_3 = \frac{I}{3};\ I_1 = I_4 = I_6 = \frac{I}{3}$$

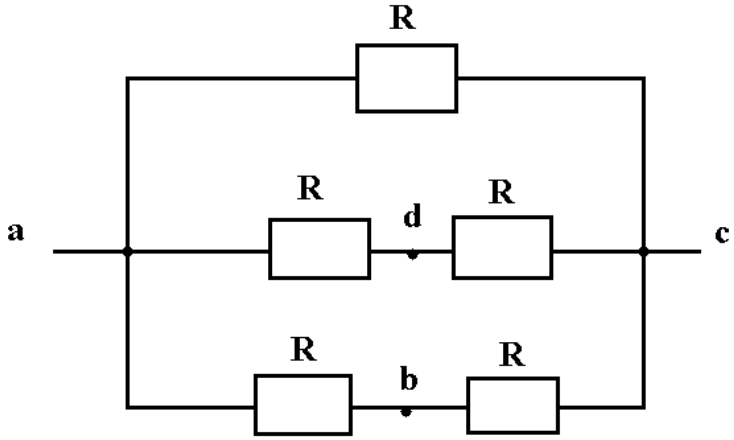
$$I_2 + I_5 = I_4 \ \Rightarrow \frac{I}{3} + I_5 = \frac{I}{3} \ \Rightarrow I_5 = 0$$

Bulardan db rezistordagi tok kuchi $I_5 = 0$ ga teng bo'lib chiqdi. Demak db yo'nalishda tok harakatlanmaydi potensiallar farqi $\varphi_b - \varphi_d = U_{bd} = 0$ bo'ladi

Shuning uchun db rezistorni sxemadan olib tashlaymiz va sxemamiz quyidagi ko'rinishga keladi.

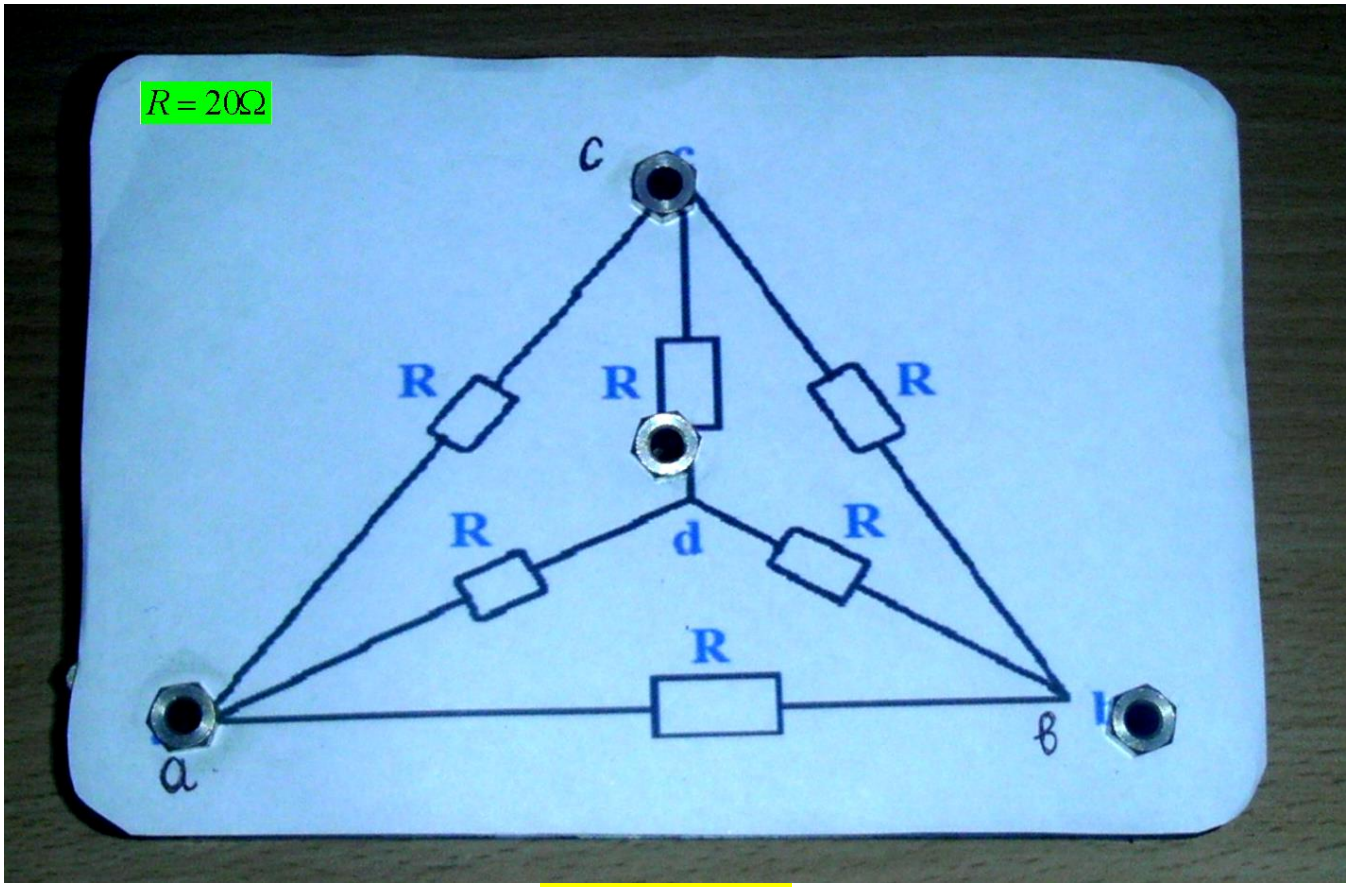


Buni sxemani quyidagi soda ko'rinishda yozamiz



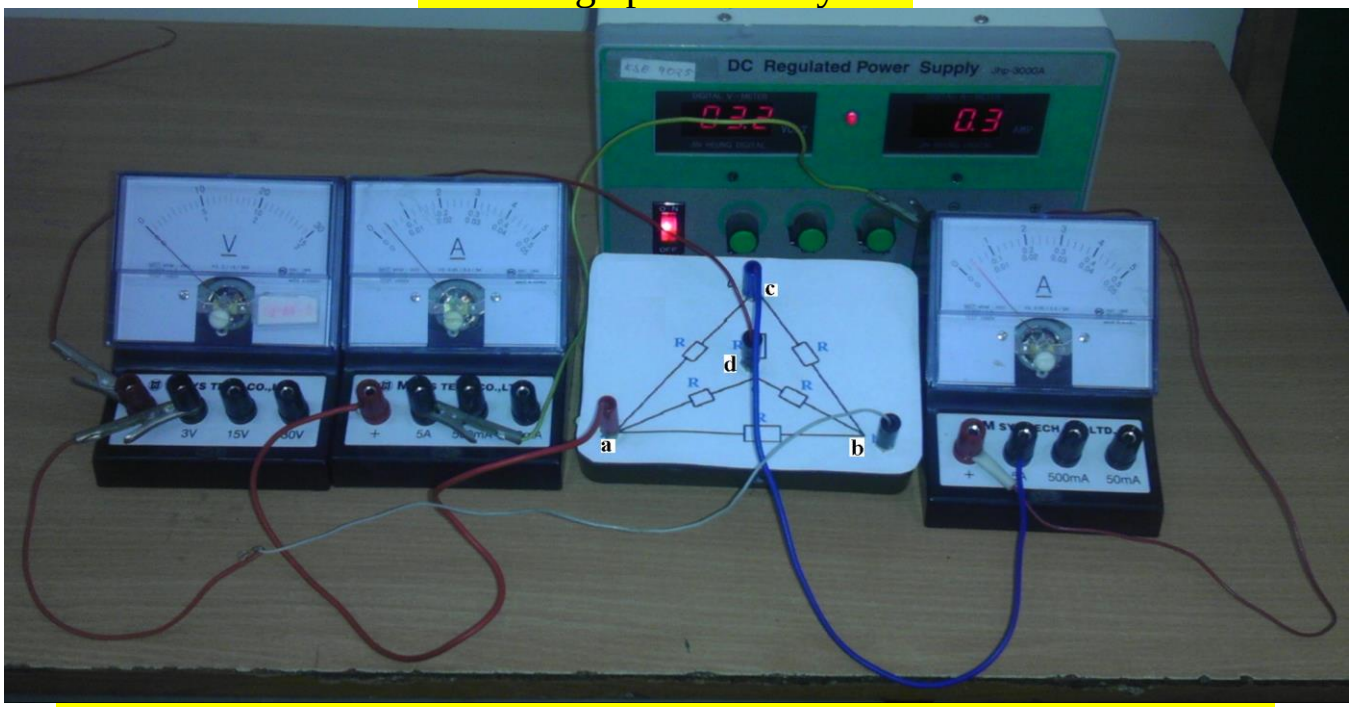
$$\frac{1}{R_{ac}} = \frac{1}{2R} + \frac{1}{2R} + \frac{1}{R}; \qquad R_{ac} = 0{,}5R$$

Endi nazariy yo' bilan isbotlab chiqarilgan natijalarni amalda bajaramiz. $\varphi_d - \varphi_b = U_{db} = 0$ ushbu formulani amalda isbotlaymiz .Buning uchun quyidagi sxemani yig'amiz



Old ko’rinishi

Sxemamizni a va c nuqtasini tok manbaiga ketma-ket, d va b nuqtalarni esa voltimetrga parallel ulaymiz.

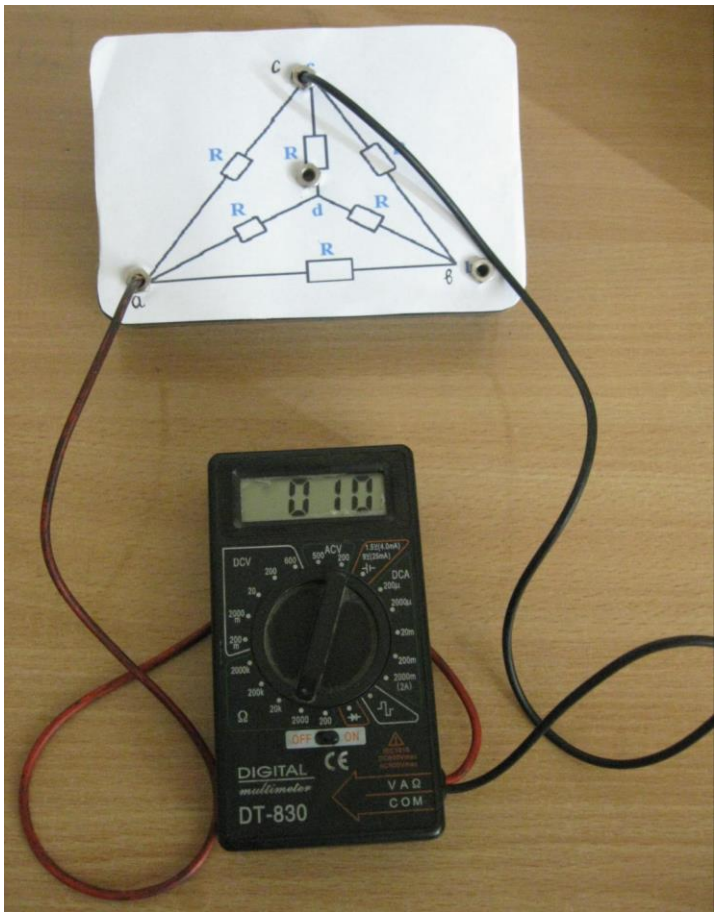


Tajribadan ko’rinib turibdiki voltimetr 0 ni ko’rsatmoqda demak d va b nuqtalarda potensiallar farqi o ga teng ekan.

Tajribadan ya’na shunday xulosa chiqarish mumkinki Kirxgofning

1-qoidasida: Tugunlarga kiruvchi va chiquvchi toklar teng deyilgan edi yaqiqatdan ham a nuqtaga ulangan ampermetr ham c nuqtaga ulangan ampermetrham bir xil tokni ko’rsatmoqda.

Endi umumiy qarshilikni hisoblaymiz buning uchun qarshilik o’lchaydigan asbob ommetrdan foydalanamiz.



Bizga nazariy yo’l bilan $R_{ac} = 0{,}5R$ javob chiqqan edi, biz qarshiligi $R = 20\Omega$ bo’lgan qarshiliklardan foydalanganimiz uchun javob $R_{ac} = 0{,}5 \cdot 20 = 10\Omega$ chiqdi.

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4.O.I.Ahmadjonov, Sh.O.Ahmadjonov “Fizikadan testlar”namunalari va izohlari.
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